Spark
POLYTECHNIC

قسم
هندسة الطاقة
الكهربائية
فريق سبارك

ملخص و شرح الطالب ليث عواد

ماشين 1

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جروب طلاب البور والاتصالات - البوليتكنك

ELECTRICAL MACHINES 1

Laith abdulsalam awwad

Spark POLYTECHNIC
11 عاماً من العطاء

مقدمة

بسم الله الرحمن الرحيم

والحمد لله رب العالمين نحمده ونستعين، ونشهد أن لا أله إلا الله وأن سيدنا محمد عبده ورسوله الأمين، فصلّ اللهم وسلم تسليمًا كثيرًا على سيدنا محمد وعلى آله وصحبه أجمعين، أقدم لكم تلخيصي عن مادة ماشين 1، الذي يمثل موضوعًا مهمًا في تخخص الكهرباء والميكاترونكس، وإنني بذلت فيه قصارى جهدي ونمقت الأحرف والكلمات متمنيًا أن يكون عند حسن ظنكم .

ونظرًا لأهمية هذا الموضوع فإنني قد تناولت كل ما يحيط به من جوانب بشكل مختصر ، ثم قمت بالتفصيل ببعض المواضيع والعناصر التي كان لا بد لي من أن أعرج عليها وأنا في صدد الحديث عن هذا الموضوع نظرًا لأهميتها البالغة ، لكنني حرصت على تجنب الإسهاب والإطالة غير المرغوب فيها، وعملت على ذكر النقاط الأساسية والمهمة .

هذا العمل مجهود بشري لايخلو من الاخطاء

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chapter 1 (Introduction to machinery principle)

Rotation motion, Newton's law and power relationships

- **Clockwise (CW) and Counterclockwise (CCW)**

– CCW is assumed as the positive direction(+), CW is assumed as the negative direction (-).

تحدد اتجاه حركة الدوران اما مع عقارب الساعة او عكس عقارب الساعة .

- **Linear and rotation motion**

– Position and angular (الخطي وا الزاوي)

(meter) (degree or radian)

– Speed and angular speed(السرعة الخطية وا السرعة الزاوية)

$$v = \frac{dr}{dt} \qquad \omega = \frac{d\theta}{dt}$$

ω_m	angular velocity expressed in radians per second
f_m	angular velocity expressed in revolutions per second
n_m	angular velocity expressed in revolutions per minute

$$n_m = 60 f_m$$

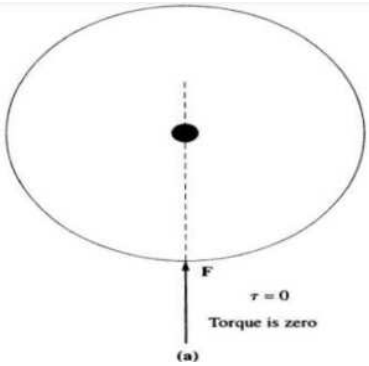
$$f_m = \frac{\omega_m}{2\pi}$$

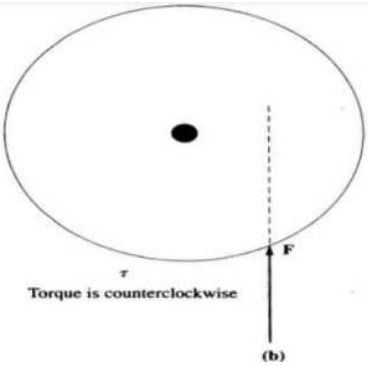
– Acceleration and angular acceleration

$$a = \frac{dv}{dt} \qquad \alpha = \frac{d\omega}{dt}$$

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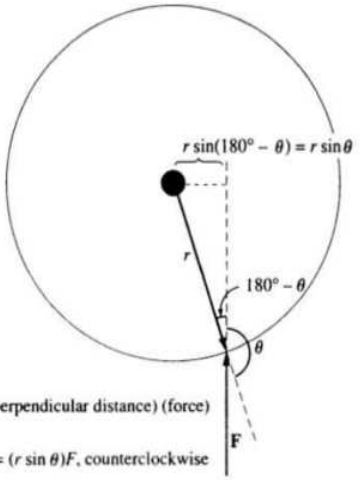
- **Torque (العزم)**





τ = (force applied)(perpendicular distance)
= $(F)\,(r \sin \theta)$
= $rF \sin \theta$

where θ is the angle between the vector **r** and the vector **F**.



- **Newton's law of rotation**

1. Force $F = ma$

F = net force applied to an object
m = mass of the object
a = resulting acceleration

2. Torque $\tau = J\alpha$

1-1. A motor's shaft is spinning at a speed of 3000 r/min. What is the shaft speed in radians per second?

SOLUTION The speed in radians per second is

$$\omega = (3000 \text{ r/min})\left(\frac{1 \text{ min}}{60 \text{ s}}\right)\left(\frac{2\pi \text{ rad}}{1 \text{ r}}\right) = 314.2 \text{ rad/s}$$

1-2. A flywheel with a moment of inertia of 2 $\text{kg} \cdot \text{m}^2$ is initially at rest. If a torque of 5 $\text{N} \cdot \text{m}$ (counterclockwise) is suddenly applied to the flywheel, what will be the speed of the flywheel after 5 s? Express that speed in both radians per second and revolutions per minute.

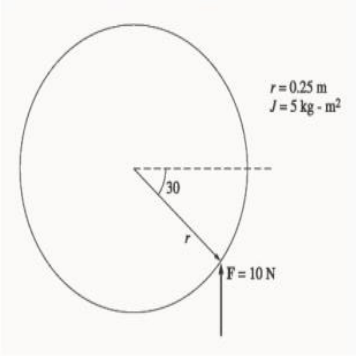
SOLUTION The speed in radians per second is:

$$\omega = \alpha t = \left(\frac{\tau}{J}\right) t = \frac{5 \text{ N} \cdot \text{m}}{2 \text{ kg} \cdot \text{m}^2}(5 \text{ s}) = 12.5 \text{ rad/s}$$

The speed in revolutions per minute is:

$$n = (12.5 \text{ rad/s})\left(\frac{1 \text{ r}}{2\pi \text{ rad}}\right)\left(\frac{60 \text{ s}}{1 \text{ min}}\right) = 119.4 \text{ r/min}$$

1-3. A force of 5 N is applied to a cylinder, as shown in Figure P1-1. What are the magnitude and direction of the torque produced on the cylinder? What is the angular acceleration α of the cylinder?



SOLUTION The magnitude and the direction of the torque on this cylinder is:

$$\tau_{\text{ind}} = rF \sin\theta, \text{CCW}$$

$$\tau_{\text{ind}} = (0.25 \text{ m})(10 \text{ N}) \sin 30° = 1.25 \text{ N} \cdot \text{m}, \text{ CCW}$$

The resulting angular acceleration is:

$$\alpha = \frac{\tau}{J} = \frac{1.25 \text{ N} \cdot \text{m}}{5 \text{ kg} \cdot \text{m}^2} = 0.25 \text{ rad/s}^2$$

- **Work W(الشغل)**

For linear motion, work is defined as the application of a force through a distance. In equation form,

بالنسبة للحركة الخطية ، يُعرَّف الشغل بأنه تطبيق قوة عبر مسافة. في شكل معادلة ،

$$W = \int F\,dr$$

where it is assumed that the force is coilinear with the direction of motion. For the special case of a constant force applied collinearly with the direction of motion, this equation becomes just

حيث يُفترض أن تكون القوة على شكل ملفوف مع اتجاه الحركة. بالنسبة للحالة الخاصة للقوة الثابتة المطبقة بشكل خطي متواصل مع اتجاه الحركة ، تصبح هذه المعادلة عادلة

$$W = Fr$$

The units of work are joules in SI and foot-pounds in the English system. For rotational motion, work is the application of a torque through an angle. Here the equation for work is

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وحدات العمل هي الجول في النظام الدولي للوحدات ، والقدم بالجنيه في نظام الإنجليزي. بالنسبة للحركة الدورانية ، الشغل هو تطبيق عزم من خلال زاوية. هنا معادلة الشغل

$$W = \int \tau \, d\theta$$

and if the torque is constant,

وإذا كان عزم الدوران ثابتًا ،

$$W = \tau\theta$$

- **Power P**

Power is the rate of doing work, or the increase in work per unit time. The equation for power is

القوة هي معدل القيام بالعمل ، أو الزيادة في العمل لكل وحدة زمنية. معادلة القوة هي

$$P = \frac{dW}{dt}$$

It is usually measured in joules per second (watts), but also can be measured in foot-pounds per second or in horsepower. By this definition, and assuming that force is constant and collinear with the direction of motion, power is given by

يقاس عادةً بالجول في الثانية (واط) ، ولكن يمكن أيضًا قياسه بالقدم / الرطل في الثانية أو بالقدرة الحصانية. من خلال هذا التعريف ، وبافتراض أن القوة ثابتة ومترابطة مع اتجاه الحركة ، تُعطى القوة بواسطة

$$P = \frac{dW}{dt} = \frac{d}{dt}(Fr) = F\left(\frac{dr}{dt}\right) = Fv$$

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Similarly, assuming constant torque, power in rotational motion is given by

وبالمثل ، بافتراض عزم ثابت ، يتم إعطاء القدرة في الحركة الدورانية بواسطة

$$P = \frac{dW}{dt} = \frac{d}{dt}(\tau\theta) = \tau\left(\frac{d\theta}{dt}\right) = \tau\omega$$

$$P = \tau\omega$$

تعتبر هذه المعادلة مهمة جدًا في دراسة الآلات الكهربائية ، لأنها يمكن أن تصف القوة الميكانيكية على عمود المحرك أو المولد.

- **Conversion between watts and horsepower**

1. Watts and horsepower

$$P\ (\text{watts}) = \frac{\tau\ (\text{lb-ft})\ n\ (\text{r/min})}{7.04}$$

$$P\ (\text{horsepower}) = \frac{\tau\ (\text{lb-ft})\ n\ (\text{r/min})}{5252}$$

2. Conversion between two units

$5252 / 7.04 = 746.02$

$1\text{hp} = 746\text{W} = 0.746\text{kW}$

1-4. A motor is supplying 60 N · m of torque to its load. If the motor's shaft is turning at 1800 r/min, what is the mechanical power supplied to the load in watts? In horsepower?

SOLUTION The mechanical power supplied to the load is

$$P = \tau\omega = (60\ \text{N}\cdot\text{m})(1800\ \text{r/min})\ \frac{1\ \text{min}}{60\ \text{s}}\ \frac{2\pi\ \text{rad}}{1\ \text{r}} = 11{,}310\ \text{W}$$

$$P = (11{,}310\ \text{W})\ \frac{1\ \text{hp}}{746\ \text{W}} = 15.2\ \text{hp}$$

- **The magnetic field**

I. A current-carrying wire produces a magnetic field in the area around it.

2. A time-changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of transfonner action.)

3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of motor action.)

4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of generator action.)

1. ينتج السلك الحامل للتيار مجالًا مغناطيسيًا في المنطقة المحيطة به.

2. يحث المجال المغناطيسي المتغير بمرور الوقت جهدًا في ملف من الأسلاك إذا كان يمر عبر هذا الملف. (هذا هو أساس عمل المتحول.)

3. السلك الحامل للتيار في وجود مجال مغناطيسي له قوة مستحثة عليه. (هذا هو أساس العمل الحركي.)

4. السلك المتحرك في وجود مجال مغناطيسي به جهد مستحث فيه. (هذا هو أساس عمل المولد.)

- **Produce a magnetic field – Ampere's law**

1. The magnetic field is produced by ampere's law

2. The core is a ferromagnetic material

1. يتم إنتاج المجال المغناطيسي بواسطة قانون الأمبير

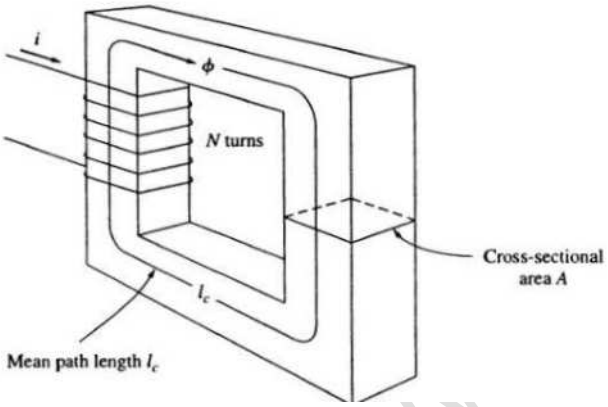
2. اللب مادة مغناطيسية

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$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{net}$$

$$Hl_c = Ni$$

$$H = \frac{Ni}{l_c}$$



- **From the magnetic field to magnetic flux density**

1. **When the magnetic field is applied on a ferromagnetic material, the magnetic flux density B will be produced**

$$\mathbf{B} = \mu \mathbf{H}$$

where

$\mathbf{H}$ = magnetic field intensity

μ = magnetic *permeability* of material

$\mathbf{B}$ = resulting magnetic flux density produced

The permeability of free space is called μ_0, and its value is $\mu_o = 4\pi \times 10^{-7}$ H/m

The permeability of any other material compared to the permeability of free space is called its relative permeability:

$$\mu_r = \frac{\mu}{\mu_0}$$

التدفق المغناطيسي (magnetic flux) : هو عدد الخطوط المجال المغناطيسي الموجودة في الدائرة المغناطيسية ويتم قياسه بالويبر(waber) .

يحافظ التدفق المغناطيسي على قيمته خلال اي جزء مستقيم في الدائرة المغناطيسية فهو مماثل للتيار في الدوائر الكهربائية.

شدة المجال المنغاطيسي (*H*)(magnetic field intensity) : يمكن حسابها في الدوائر المغناطيسية البسيطة بتطبيق نظرية امبير والتي تقول بأن التكامل الخطي لشدة المجال المغناطيسي *H* حول مسار مغلق يساوي التيار الكلي داخل المسار .

كثافة التدفق المغناطيسي (Magnetic flux density): هي كمية التدفق المغناطيسي التي تعبر مساحة معينة وتكون عمودية على خطوط المجال المغناطيسي ، كما يمكن القول ايضاً ان الشدة للمجال المغناطيسي التي تمثل التأثيرات الحساسة ، غالبا ماتكون على شكل ميكانيكي كقوة لابلاس والعزم الكهرومغناطيسي .

- **Magnetic flux density and magnetic flux**

1. Magnetic flux density

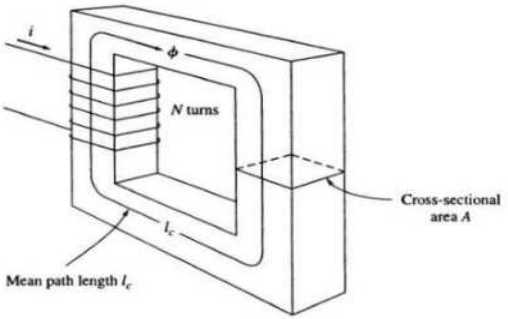
$$B = \mu H = \frac{\mu N i}{l_c}$$

2. Magnetic flux

$$\phi = \int_A \mathbf{B} \cdot d\mathbf{A}$$

$$\phi = BA$$

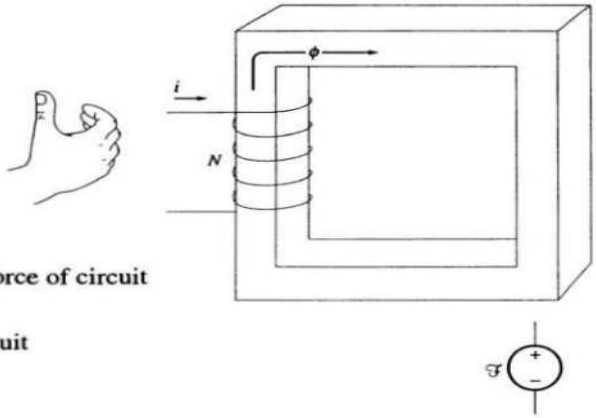
$$\phi = BA = \frac{\mu N i A}{l_c}$$



- **Magnetic Circuit – magnetomotive force**

$\mathcal{F} = Ni$

$\mathcal{F} = \phi \mathcal{R}$



$\mathcal{F}$ = magnetomotive force of circuit
ϕ = flux of circuit
$\mathcal{R}$ = *reluctance* of circuit

1. Magnetic circuit

$$\phi = BA = \frac{\mu NiA}{l_c}$$

$$= Ni\left(\frac{\mu A}{l_c}\right)$$

$$\phi = \mathcal{F}\left(\frac{\mu A}{l_c}\right)$$

$$\mathcal{R} = \frac{l_c}{\mu A}$$

$$\mathcal{F} = \phi \mathcal{R}$$

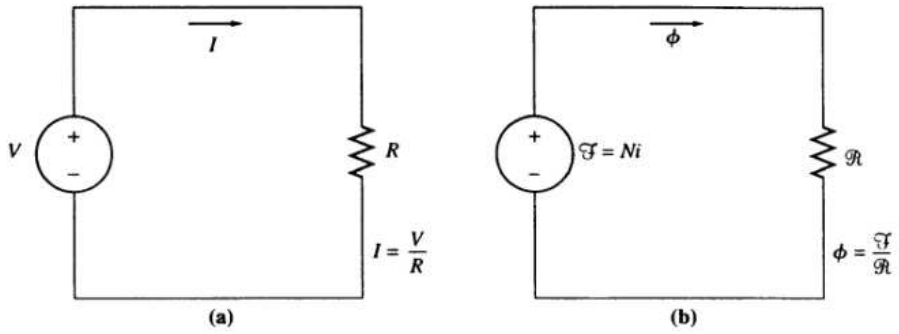


FIGURE 1–4
(a) A simple electric circuit. (b) The magnetic circuit analog to a transformer core.

الممانعة أو (المقاومة المغناطيسية) (R_m ،$\Re$): تعرف بأنها النسبة ما بين القوة الدافعة المغناطيسية والتدفق المغناطيسي ، وتعبر هذه العلاقة عن قانون اوم للدائرة المغناطيسية ولمسار مغناطيسي طوله L ويمكن كتابتها على الشكل التالي .

$$R_m = \frac{NI}{\Phi} = \frac{NI}{BH} = \frac{NI}{\mu HA} = \frac{l}{\mu A}$$

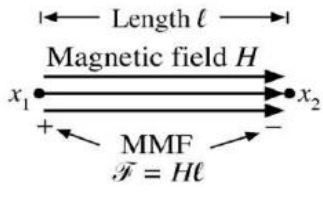
$$R_m = \frac{l}{\mu A}$$

- **Magnetic field *H* and magnetomotive force $\mathcal{F}$**

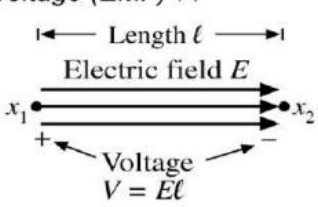
Magnetomotive force (MMF) $\mathcal{F}$ between points x_1 and x_2 is related to the magnetic field H according to

$$\mathcal{F} = \int_{x_1}^{x_2} H \cdot d\ell$$

Example: uniform magnetic field of magnitude H



Analogous to electric field of strength E, which induces voltage (EMF) V:



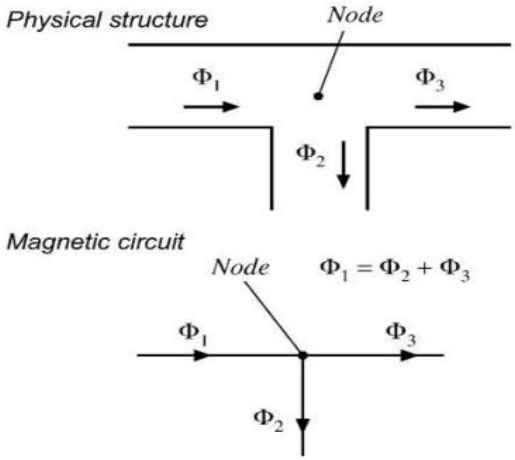
- **Magnetic analog of kirchoff's current law**

التدفق يشبه التيار ف أذا كان عندي قلب معدني كما موضح في الشكل فأنه سوف يتشعب مثل ما بتشعب التيار عند العقدة .

Divergence of $\boldsymbol{B} = 0$

Flux lines are continuous and cannot end

Total flux entering a node must be zero



- **Reluctance in magnetic circuit**

1. Series connection

$$\mathcal{R}_{eq} = \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \cdots$$

2. Parallel connection

$$\frac{1}{\mathcal{R}_{eq}} = \frac{1}{\mathcal{R}_1} + \frac{1}{\mathcal{R}_2} + \frac{1}{\mathcal{R}_3} + \cdots$$

- **The errors in magnetic circuit computation**
- **الأخطاء في حساب الدائرة المغناطيسية**

I. The magnetic circuit concept assumes that all flux is confined within a magnetic core . Unfortunately, this is not quite true. The permeability of a ferromagnetic core is 2000 to 6000 times that of air, but a small fraction of the flux escapes from the core into the surrounding low-permeability air. This flux outside the core is called leakage flux, and it plays a very important role in electric machine design.

2. The calculation of reluctance assumes a certain mean path length and cross sectional area for the core . These assumptions are not really very good, especially at corners.

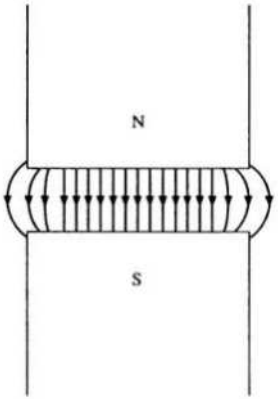
3. In ferromagnetic materials, the permeability varies with the amount of flux already in the material. This nonlinear effect is described in detail. It adds yet another source of error to magnetic circuit analysis, since the reluctances used in magnetic circuit calculations depend on the permeability of the material.

1. يفترض مفهوم الدائرة المغناطيسية أن كل التدفق محصور داخل قلب مغناطيسي. لسوء الحظ ، هذا ليس صحيحًا تمامًا. نفاذية النواة المغناطيسية هي 2000 إلى 6000 مرة من الهواء ، لكن جزءًا صغيرًا من التدفق يتسرب من اللب إلى الهواء المحيط منخفض النفاذية. يسمى هذا التدفق خارج القلب بتدفق التسرب ، ويلعب دورًا مهمًا جدًا في تصميم الماكينة الكهربائية.

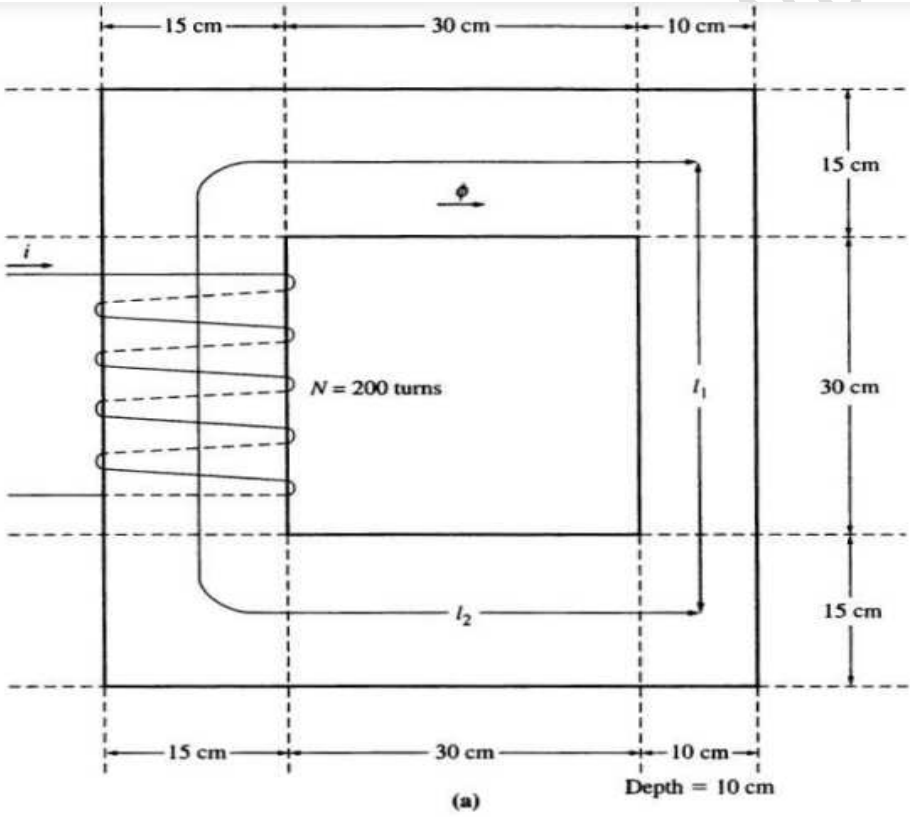
2. يفترض حساب الممانعة وجود متوسط معين لطول المسار ومنطقة المقطع العرضي للب. هذه الافتراضات ليست جيدة حقًا ، خاصة في الزوايا.

3. في المواد المغناطيسية ، تختلف النفاذية مع كمية التدفق الموجودة بالفعل في المادة. تم وصف هذا التأثير غير الخطي بالتفصيل. إنه يضيف مصدرًا آخر للخطأ لتحليل الدائرة المغناطيسية ، لأن الترددات المستخدمة في حسابات الدائرة المغناطيسية تعتمد على نفاذية المادة.

4. Air gap "fringing effect"



Example 1-1. A ferromagnetic core is shown in Figure 1–7a. Three sides of this core are of uniform width, while the fourth side is somewhat thinner. The depth of the core (into the page) is 10 cm, and the other dimensions are shown in the figure. There is a 200-turn coil wrapped around the left side of the core. Assuming relative permeability μ_r of 2500, how much flux will be produced by a 1-A input current?



The mean path length of region 1 is 45 cm, and the cross-sectional area is 10 × 10 cm = 100 cm^2. Therefore, the reluctance in the first region is

$$\mathcal{R}_1 = \frac{l_1}{\mu A_1} = \frac{l_1}{\mu_r \mu_0 A_1} \tag{1–32}$$

$$= \frac{0.45 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.01 \text{ m}^2)}$$

$$= 14{,}300 \text{ A} \cdot \text{turns/Wb}$$

The mean path length of region 2 is 130 cm, and the cross-sectional area is 15 × 10 cm = 150 cm^2. Therefore, the reluctance in the second region is

$$\mathcal{R}_2 = \frac{l_2}{\mu A_2} = \frac{l_2}{\mu_r \mu_0 A_2} \tag{1–32}$$

$$= \frac{1.3 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.015 \text{ m}^2)}$$

$$= 27{,}600 \text{ A} \cdot \text{turns/Wb}$$

Therefore, the total reluctance in the core is

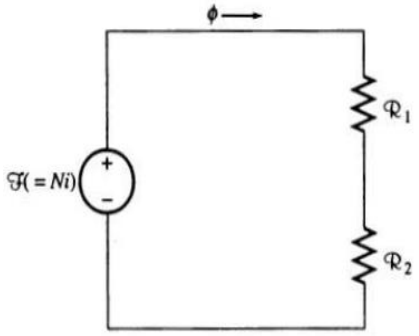
$$\mathcal{R}_{eq} = \mathcal{R}_1 + \mathcal{R}_2$$
$$= 14{,}300 \text{ A} \cdot \text{turns/Wb} + 27{,}600 \text{ A} \cdot \text{turns/Wb}$$
$$= 41{,}900 \text{ A} \cdot \text{turns/Wb}$$

The total magnetomotive force is

$$\mathcal{F} = Ni = (200 \text{ turns})(1.0 \text{ A}) = 200 \text{ A} \cdot \text{turns}$$

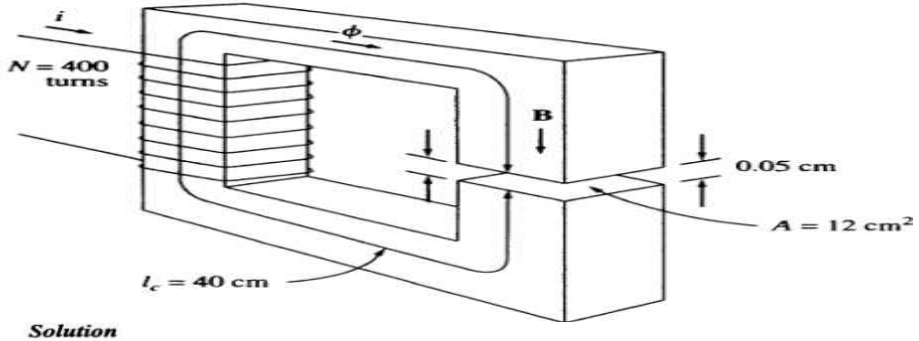
The total flux in the core is given by

$$\phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{200 \text{ A} \cdot \text{turns}}{41{,}900 \text{ A} \cdot \text{turns/Wb}}$$
$$= 0.0048 \text{ Wb}$$



(b)

Example 1–2. Figure 1–8a shows a ferromagnetic core whose mean path length is 40 cm. There is a small gap of 0.05 cm in the structure of the otherwise whole core. The cross-sectional area of the core is 12 cm², the relative permeability of the core is 4000, and the coil of wire on the core has 400 turns. Assume that fringing in the air gap increases the effective cross-sectional area of the air gap by 5 percent. Given this information, find (a) the total reluctance of the flux path (iron plus air gap) and (b) the current required to produce a flux density of 0.5 T in the air gap.



Solution

The magnetic circuit corresponding to this core is shown in Figure 1–8b.

(a) The reluctance of the core is

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_r \mu_0 A_c} \qquad (1\text{–}32)$$

$$= \frac{0.4 \text{ m}}{(4000)(4\pi \times 10^{-7})(0.002 \text{ m}^2)}$$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb}$$

The effective area of the air gap is $1.05 \times 12 \text{ cm}^2 = 12.6 \text{ cm}^2$, so the reluctance of the air gap is

$$\mathcal{R}_a = \frac{l_a}{\mu_0 A_a} \qquad (1\text{–}32)$$

$$= \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7})(0.00126 \text{ m}^2)}$$

$$= 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

(a) The ferromagnetic core of Example 1–2. (b) The magnetic circuit corresponding to (a).

Therefore, the total reluctance of the flux path is

$$\mathcal{R}_{eq} = \mathcal{R}_c + \mathcal{R}_a$$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb} + 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

$$= 382{,}300 \text{ A} \cdot \text{turns/Wb}$$

Note that the air gap contributes most of the reluctance even though it is 800 times shorter than the core.

(b) Equation (1–28) states that

$$\mathcal{F} = \phi \mathcal{R} \qquad (1\text{–}28)$$

Since the flux $\phi = BA$ and $\mathcal{F} = Ni$, this equation becomes

$$Ni = BA\mathcal{R}$$

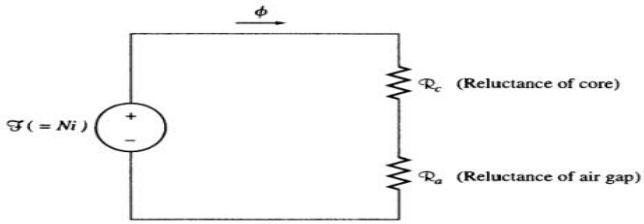
so

$$i = \frac{BA\mathcal{R}}{N}$$

$$= \frac{(0.5\ \text{T})(0.00126\ \text{m}^2)(383{,}200\ \text{A} \cdot \text{turns/Wb})}{400\ \text{turns}}$$

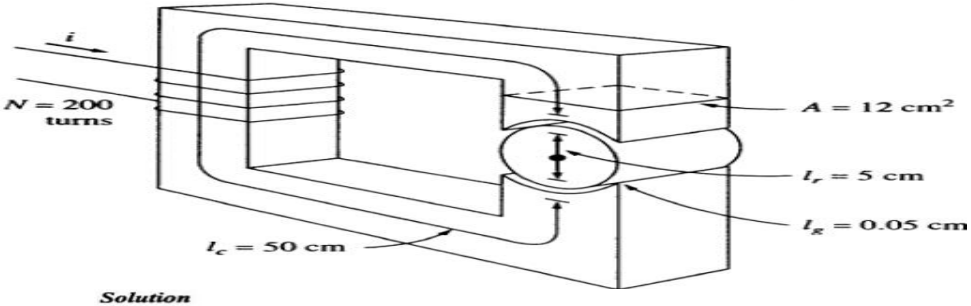
$$= 0.602\ \text{A}$$

Notice that, since the *air-gap* flux was required, the effective air-gap area was used in the above equation.



(b)

Example 1–3. Figure 1–9a shows a simplified rotor and stator for a dc motor. The mean path length of the stator is 50 cm, and its cross-sectional area is 12 cm^2. The mean path length of the rotor is 5 cm, and its cross-sectional area also may be assumed to be 12 cm^2. Each air gap between the rotor and the stator is 0.05 cm wide, and the cross-sectional area of each air gap (including fringing) is 14 cm^2. The iron of the core has a relative permeability of 2000, and there are 200 turns of wire on the core. If the current in the wire is adjusted to be 1 A, what will the resulting flux density in the air gaps be?



Solution

To determine the flux density in the air gap, it is necessary to first calculate the magnetomotive force applied to the core and the total reluctance of the flux path. With this information, the total flux in the core can be found. Finally, knowing the cross-sectional area of the air gaps enables the flux density to be calculated.

The reluctance of the stator is

$$\mathcal{R}_s = \frac{l_s}{\mu_r \mu_0 A_s}$$

$$= \frac{0.5\ \text{m}}{(2000)(4\pi \times 10^{-7})(0.0012\ \text{m}^2)}$$

$$= 166{,}000\ \text{A} \cdot \text{turns/Wb}$$

The reluctance of the rotor is

$$\mathcal{R}_r = \frac{l_r}{\mu_r \mu_0 A_r}$$

$$= \frac{0.05\ \text{m}}{(2000)(4\pi \times 10^{-7})(0.0012\ \text{m}^2)}$$

$$= 16{,}600\ \text{A} \cdot \text{turns/Wb}$$

The reluctance of the air gaps is

$$\mathcal{R}_a = \frac{l_a}{\mu_r \mu_0 A_a}$$

$$= \frac{0.0005 \text{ m}}{(1)(4\pi \times 10^{-7})(0.0014 \text{ m}^2)}$$

$$= 284{,}000 \text{ A} \cdot \text{turns/Wb}$$

The magnetic circuit corresponding to this machine is shown in Figure 1–9b. The total reluctance of the flux path is thus

$$\mathcal{R}_{eq} = \mathcal{R}_s + \mathcal{R}_{a1} + \mathcal{R}_r + \mathcal{R}_{a2}$$

$$= 166{,}000 + 284{,}000 + 16{,}600 + 284{,}000 \text{ A} \cdot \text{turns/Wb}$$

$$= 751{,}000 \text{ A} \cdot \text{turns/Wb}$$

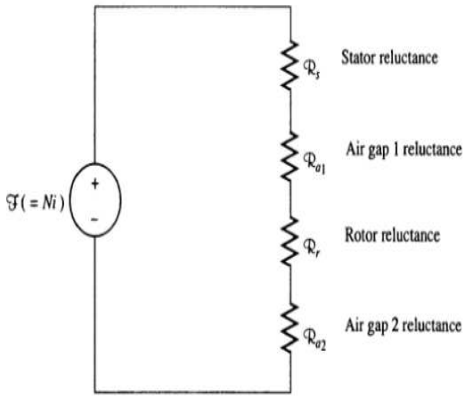
The net magnetomotive force applied to the core is

$$\mathcal{F} = Ni = (200 \text{ turns})(1.0 \text{ A}) = 200 \text{ A} \cdot \text{turns}$$

Therefore, the total flux in the core is

$$\phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{200 \text{ A} \cdot \text{turns}}{751{,}000 \text{ A} \cdot \text{turns/Wb}}$$

$$= 0.00266 \text{ Wb}$$

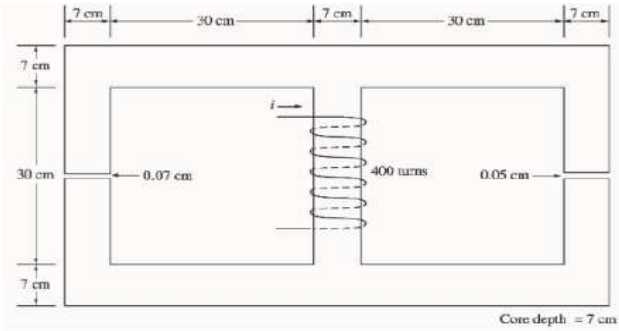


(b)

Finally, the magnetic flux density in the motor's air gap is

$$B = \frac{\phi}{A} = \frac{0.000266 \text{ Wb}}{0.0014 \text{ m}^2} = 0.19 \text{ T}$$

1-6. A ferromagnetic core with a relative permeability of 1500 is shown in Figure P1-3. The dimensions are as shown in the diagram, and the depth of the core is 7 cm. The air gaps on the left and right sides of the core are 0.070 and 0.020 cm, respectively. Because of fringing effects, the effective area of the air gaps is 5 percent larger than their physical size. If there are 400[1] turns in the coil wrapped around the center leg of the core and if the current in the coil is 1.0 A, what is the flux in each of the left, center, and right legs of the core? What is the flux density in each air gap?



SOLUTION This core can be divided up into five regions. Let $\mathcal{R}_1$ be the reluctance of the left-hand portion of the core, $\mathcal{R}_2$ be the reluctance of the left-hand air gap, $\mathcal{R}_3$ be the reluctance of the right-hand portion of the core, $\mathcal{R}_4$ be the reluctance of the right-hand air gap, and $\mathcal{R}_5$ be the reluctance of the center leg of the core. Then the total reluctance of the core is

$$\mathcal{R}_{TOT} = \mathcal{R}_5 + \frac{(\mathcal{R}_1 + \mathcal{R}_2)(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4}$$

$$\mathcal{R}_1 = \frac{l_1}{\mu_r \mu_0 A_1} = \frac{1.11 \text{ m}}{(2000)(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.07 \text{ m})} = 90.1 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_2 = \frac{l_2}{\mu_0 A_2} = \frac{0.0007 \text{ m}}{(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.07 \text{ m})(1.05)} = 108.3 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_3 = \frac{l_3}{\mu_r \mu_0 A_3} = \frac{1.11 \text{ m}}{(2000)(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.07 \text{ m})} = 90.1 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_4 = \frac{l_4}{\mu_0 A_4} = \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.07 \text{ m})(1.05)} = 77.3 \text{ kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_5 = \frac{l_5}{\mu_r \mu_0 A_5} = \frac{0.37 \text{ m}}{(2000)(4\pi \times 10^{-7} \text{ H/m})(0.07 \text{ m})(0.07 \text{ m})} = 30.0 \text{ kA} \cdot \text{t/Wb}$$

The total reluctance is

$$\mathcal{R}_{TOT} = \mathcal{R}_5 + \frac{(\mathcal{R}_1 + \mathcal{R}_2)(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} = 30.0 + \frac{(90.1 + 108.3)(90.1 + 77.3)}{90.1 + 108.3 + 90.1 + 77.3} = 120.8 \text{ kA} \cdot \text{t/Wb}$$

The total flux in the core is equal to the flux in the center leg:

$$\phi_{center} = \phi_{TOT} = \frac{\mathcal{F}}{\mathcal{R}_{TOT}} = \frac{(400 \text{ t})(1.0 \text{ A})}{120.8 \text{ kA} \cdot \text{t/Wb}} = 0.0033 \text{ Wb}$$

The fluxes in the left and right legs can be found by the "flux divider rule", which is analogous to the current divider rule.

$$\phi_{left} = \frac{(\mathcal{R}_3 + \mathcal{R}_4)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} \phi_{TOT} = \frac{(90.1 + 77.3)}{90.1 + 108.3 + 90.1 + 77.3} (0.0033 \text{ Wb}) = 0.00193 \text{ Wb}$$

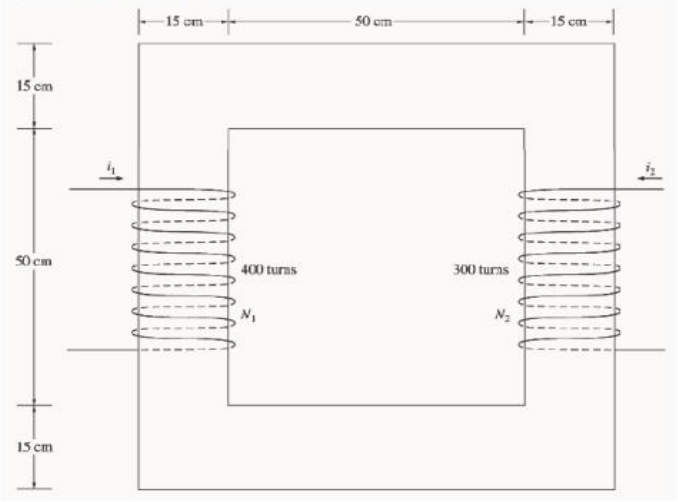
$$\phi_{right} = \frac{(\mathcal{R}_1 + \mathcal{R}_2)}{\mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} \phi_{TOT} = \frac{(90.1 + 108.3)}{90.1 + 108.3 + 90.1 + 77.3} (0.0033 \text{ Wb}) = 0.00229 \text{ Wb}$$

The flux density in the air gaps can be determined from the equation $\phi = BA$:

$$B_{left} = \frac{\phi_{left}}{A_{eff}} = \frac{0.00193 \text{ Wb}}{(0.07 \text{ cm})(0.07 \text{ cm})(1.05)} = 0.375 \text{ T}$$

$$B_{right} = \frac{\phi_{right}}{A_{eff}} = \frac{0.00229 \text{ Wb}}{(0.07 \text{ cm})(0.07 \text{ cm})(1.05)} = 0.445 \text{ T}$$

1-7. A two-legged core is shown in Figure P1-4. The winding on the left leg of the core (N_1) has 400 turns, and the winding on the right (N_2) has 300 turns. The coils are wound in the directions shown in the figure. If the dimensions are as shown, then what flux would be produced by currents $i_1 = 0.5$ A and $i_2 = 0.75$ A? Assume $\mu_r = 1000$ and constant.



SOLUTION The two coils on this core are would so that their magnetomotive forces are additive, so the total magnetomotive force on this core is

$$\mathcal{F}_{TOT} = N_1 i_1 + N_2 i_2 = (400\text{ t})(0.5\text{ A}) + (300\text{ t})(0.75\text{ A}) = 425\text{ A}\cdot\text{t}$$

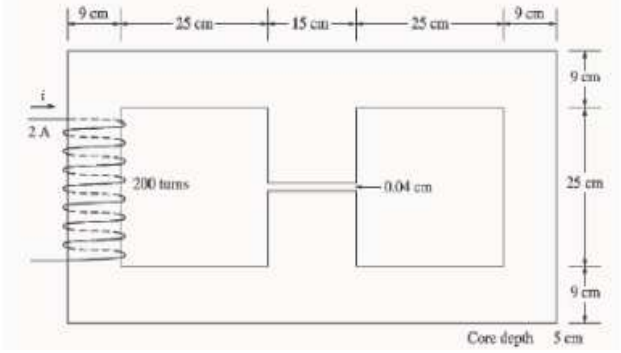
The total reluctance in the core is

$$\mathcal{R}_{TOT} = \frac{l}{\mu_r \mu_0 A} = \frac{2.60\text{ m}}{(1000)(4\pi\times10^{-7}\text{ H/m})(0.15\text{ m})(0.15\text{ m})} = 92.0\text{ kA}\cdot\text{t/Wb}$$

and the flux in the core is:

$$\phi = \frac{\mathcal{F}_{TOT}}{\mathcal{R}_{TOT}} = \frac{425\text{ A}\cdot\text{t}}{92.0\text{ kA}\cdot\text{t/Wb}} = 0.00462\text{ Wb}$$

1-8. A core with three legs is shown in Figure P1-5. Its depth is 5 cm, and there are 200 turns on the leftmost leg. The relative permeability of the core can be assumed to be 1500 and constant. What flux exists in each of the three legs of the core? What is the flux density in each of the legs? Assume a 4% increase in the effective area of the air gap due to fringing effects.



SOLUTION This core can be divided up into four regions. Let $\mathcal{R}_1$ be the reluctance of the left-hand portion of the core, $\mathcal{R}_2$ be the reluctance of the center leg of the core, $\mathcal{R}_3$ be the reluctance of the center air gap, and $\mathcal{R}_4$ be the reluctance of the right-hand portion of the core. Then the total reluctance of the core is

$$\mathcal{R}_{TOT} = \mathcal{R}_1 + \frac{(\mathcal{R}_2 + \mathcal{R}_3)\mathcal{R}_4}{\mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4}$$

$$\mathcal{R}_1 = \frac{l_1}{\mu_r \mu_0 A_1} = \frac{1.08\ \text{m}}{(1500)(4\pi \times 10^{-7}\ \text{H/m})(0.09\ \text{m})(0.05\ \text{m})} = 127.3\ \text{kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_2 = \frac{l_2}{\mu_r \mu_0 A_2} = \frac{0.34\ \text{m}}{(1500)(4\pi \times 10^{-7}\ \text{H/m})(0.15\ \text{m})(0.05\ \text{m})} = 24.0\ \text{kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_3 = \frac{l_3}{\mu_0 A_3} = \frac{0.0004\ \text{m}}{(4\pi \times 10^{-7}\ \text{H/m})(0.15\ \text{m})(0.05\ \text{m})(1.04)} = 40.8\ \text{kA} \cdot \text{t/Wb}$$

$$\mathcal{R}_4 = \frac{l_4}{\mu_r \mu_0 A_4} = \frac{1.08\ \text{m}}{(1500)(4\pi \times 10^{-7}\ \text{H/m})(0.09\ \text{m})(0.05\ \text{m})} = 127.3\ \text{kA} \cdot \text{t/Wb}$$

The total reluctance is

$$\mathcal{R}_{TOT} = \mathcal{R}_1 + \frac{(\mathcal{R}_2 + \mathcal{R}_3)\mathcal{R}_4}{\mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4} = 127.3 + \frac{(24.0 + 40.8)127.3}{24.0 + 40.8 + 127.3} = 170.2\ \text{kA} \cdot \text{t/Wb}$$

The total flux in the core is equal to the flux in the left leg:

$$\phi_{left} = \phi_{TOT} = \frac{\mathcal{F}}{\mathcal{R}_{TOT}} = \frac{(200\ \text{t})(2.0\ \text{A})}{170.2\ \text{kA} \cdot \text{t/Wb}} = 0.00235\ \text{Wb}$$

The fluxes in the center and right legs can be found by the "flux divider rule", which is analogous to the current divider rule.

$$\phi_{center} = \frac{\mathcal{R}_4}{\mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4}\phi_{TOT} = \frac{127.3}{24.0 + 40.8 + 127.3}(0.00235\ \text{Wb}) = 0.00156\ \text{Wb}$$

$$\phi_{right} = \frac{\mathcal{R}_2 + \mathcal{R}_3}{\mathcal{R}_2 + \mathcal{R}_3 + \mathcal{R}_4}\phi_{TOT} = \frac{24.0 + 40.8}{24.0 + 40.8 + 127.3}(0.00235\ \text{Wb}) = 0.00079\ \text{Wb}$$

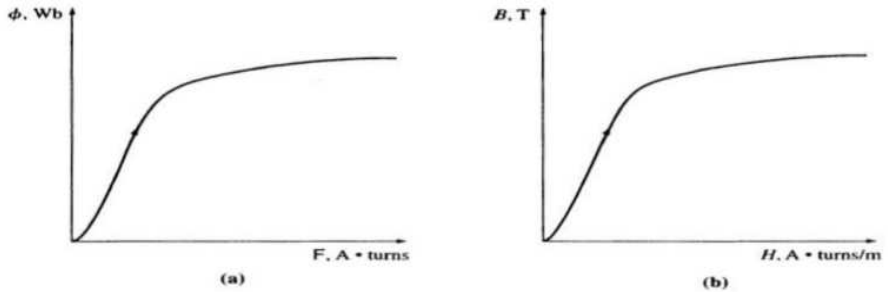
The flux density in the legs can be determined from the equation $\phi = BA$:

$$B_{left} = \frac{\phi_{left}}{A} = \frac{0.00235\ \text{Wb}}{(0.09\ \text{cm})(0.05\ \text{cm})} = 0.522\ \text{T}$$

$$B_{center} = \frac{\phi_{center}}{A} = \frac{0.00156\ \text{Wb}}{(0.15\ \text{cm})(0.05\ \text{cm})} = 0.208\ \text{T}$$

$$B_{right} = \frac{\phi_{left}}{A} = \frac{0.00079\ \text{Wb}}{(0.09\ \text{cm})(0.05\ \text{cm})} = 0.176\ \text{T}$$

- **Magnetic behavior of ferromagnetic material - Saturation**
- **السلوك المغناطيسي للمادة المغناطيسية – التشبع**



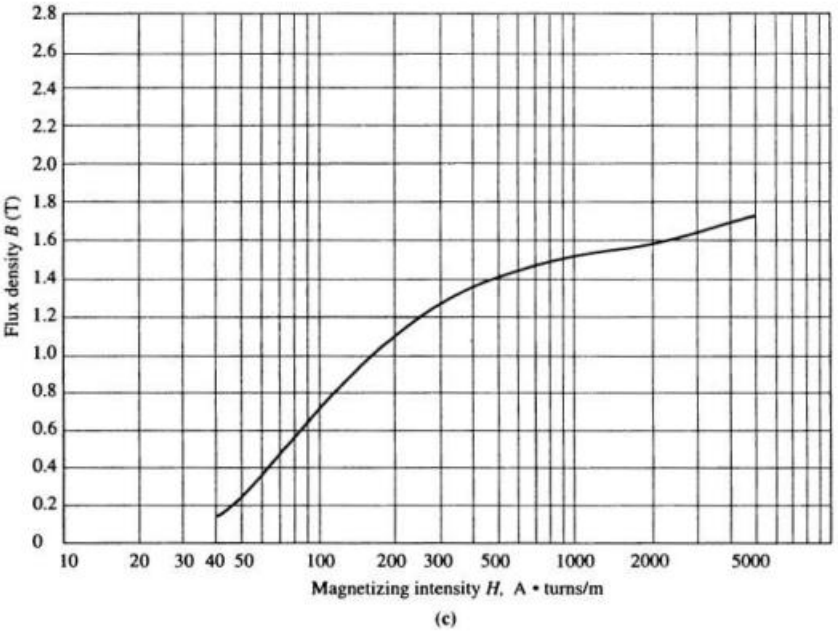
تكمن اهمية منحنى التمغنط (magnetization carve) في ان يحدد منطقة عمل الآلة الكهربائية اذا يفضل ان تعمل الآلة عند نقطة التشبع (saturation point)

اذا عملت الالة في منطقة عدم الاشباع فأن اي زيادة في الحمل (اي زيادة التيار على الحمل) يؤدي الى هبوط عالي في الجهد

اما اذا عملت غي منطقة فوق الاشباع فأن أي زيادة في التيار لا يؤدي الى احداث أي تغير في الالة ولكنه يؤدي الى حرق الملفات

- **Magnetic curve for a typical steel**

laith abdulsalam



(c)

- **A plot of relative permeability μ_r for typical steel**

Example 1–4. Find the relative permeability of the typical ferromagnetic material whose magnetization curve is shown in Figure 1–10c at (a) $H = 50$, (b) $H = 100$, (c) $H = 500$, and (d) $H = 1000$ A • turns/m.

Solution

The permeability of a material is given by

$$\mu = \frac{B}{H}$$

and the relative permeability is given by

Example 1–5. A square magnetic core has a mean path length of 55 cm and a cross-sectional area of 150 cm^2. A 200-turn coil of wire is wrapped around one leg of the core. The core is made of a material having the magnetization curve shown in Figure 1–10c.

(a) How much current is required to produce 0.012 Wb of flux in the core?
(b) What is the core's relative permeability at that current level?
(c) What is its reluctance?

Solution

(a) The required flux density in the core is

$$B = \frac{\phi}{A} = \frac{1.012 \text{ Wb}}{0.015 \text{ m}^2} = 0.8 \text{ T}$$

From Figure 1–10c, the required magnetizing intensity is

$$H = 115 \text{ A} \cdot \text{turns/m}$$

From Equation (1–20), the magnetomotive force needed to produce this magnetizing intensity is

$$\mathcal{F} = Ni = Hl_c$$
$$= (115 \text{ A} \cdot \text{turns/m})(0.55 \text{ m}) = 63.25 \text{ A} \cdot \text{turns}$$

so the required current is

$$i = \frac{\mathcal{F}}{N} = \frac{63.25 \text{ A} \cdot \text{turns}}{200 \text{ turns}} = 0.316 \text{ A}$$

(b) The core's permeability at this current is

$$\mu = \frac{B}{H} = \frac{0.8 \text{ T}}{115 \text{ A} \cdot \text{turns/m}} = 0.00696 \text{ H/m}$$

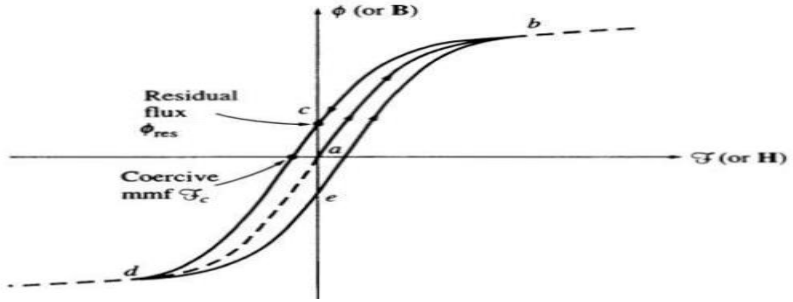
Therefore, the relative permeability is

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.00696 \text{ H/m}}{4\pi \times 10^{-7} \text{ H/m}} = 5540$$

(c) The reluctance of the core is

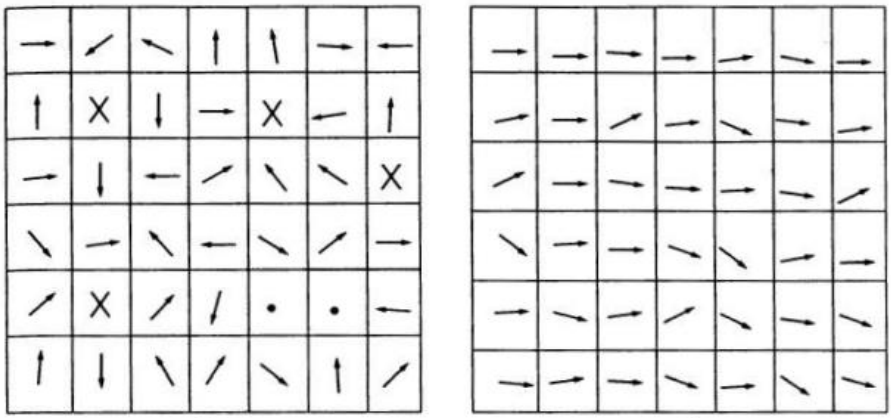
$$\mathcal{R} = \frac{\mathcal{F}}{\phi} = \frac{63.25 \text{ A} \cdot \text{turns}}{0.012 \text{ Wb}} = 5270 \text{ A} \cdot \text{turns/Wb}$$

- **Energy loss in ferromagnetic core hysteresis loss**



- **Hysteresis loop – residual flux**

- **حلقة التخلفية - التدفق المتبقي**



ان القالب يتكون من حقول يسمى كل منها (domain) وهو اصغر جزء في المادة يشكل مغناطيس ، وكل حقل يتكون من جزيئات ... ذرات ... الكترونات) وهذه الكترونات تكون في حركة دائمة ، بحيث تكون الحركة متشابهة في نفس الحقل ، ومختلفة من حقل لأخر ، مما يجعل محصلة التدفق المغناطيسي تساوي الصفر كما هو الحال عند نقطة a في الشكل الذي في صفحة 27

وعند مرور التيار الكهربائي وزيادته تدريجياً فأن الحقول تبدأ بالاصطفاف تدريجياً وبنفس الاتجاه نتيجة تأثير المجال الناشئ من مرور التيار على هذه الحقول والامر الذي يؤدي الى زيادة التدفق بشكل تدريجي كما يظهر من نقطة a الى النقطة b في الشكل الذي في صفحة 27

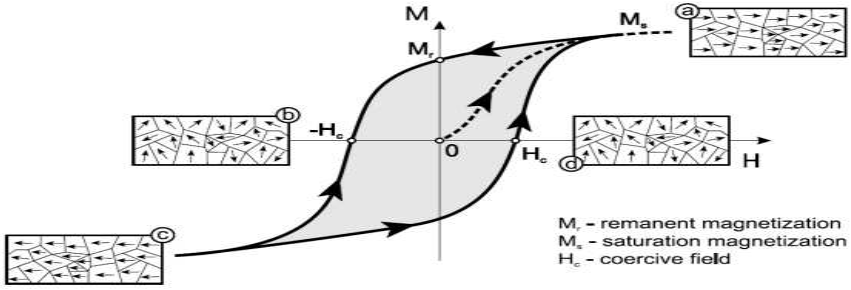
وعند انعكاس التيار فجأة فأن هذا يؤدي الى احداث تضارب مغناطيسي ويؤدي الى احداث خسائر في التدفق المغناطيسي نظراً لضياع جزء من الطاقة في اثناء محاول ال (domain) عكس اتجاهها بما يتناسب مع الاتجاه الجديد للتيار

وعند وصول التيار الى الصفر فأن بعض ال (domain) تبقى محافظة على اتجاهها مما يعني وجود بقايا مغناطيسية (Rezidual flux) كما يظهر عند نقطة c حيث نجد ان b لا يساوي الصفر ولكن i=0 ، وهذه البقايا المغناطيسية تفيد في عمل الالة كما سيتضح عند دراسة المحركات والمولدات

وبالتالي فأن سبب حصول الحلقة الهسترية هو ان التدفق الحاصل في القلب لا يعتمد فقط على مقدار التيار المار في اللمفات وأنما يعتمد على الوضعية السابقة للتدفق داخل هذا القلب

كما ان الحلقة الهستيرية أحد اسباب حدوث الضياعات المغناطيسية (mag. Losses) في المحولات ، ونظراً لاستهلاك جزء من الطاقة في كل مرة يتم فيها عكس اتجاه ال domains

- **Magnetization curve**



- **Faraday's law – induce voltage from a time-varying magnetic field**

- **قانون فاراداي - يستحث الجهد من مجال مغناطيسي متغير بمرور الوقت**

laith abdulsalam awwad

1. Induced voltage magnitude and polarity

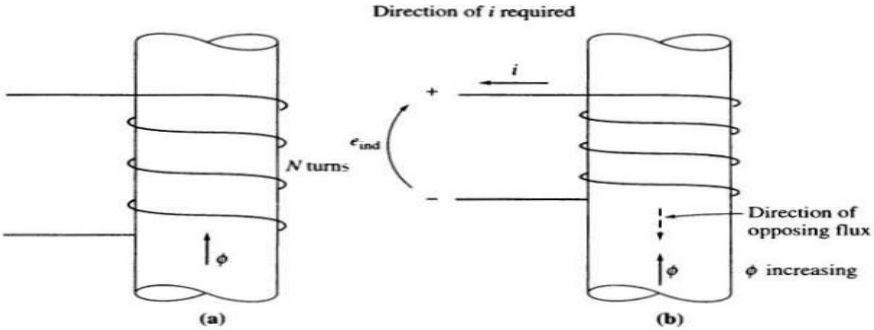
$$e_{\text{ind}} = -\frac{d\phi}{dt}$$

$$e_{\text{ind}} = \sum_{i=1}^{N} e_i$$

$$e_{\text{ind}} = -N\frac{d\phi}{dt}$$

e_{ind} = voltage induced in the coil
N = number of turns of wire in coil
ϕ = flux passing through coil

- **The induced voltage polarity – Lenz's law**



- **Flux and flux linkage**

What is the difference between flux linkage and flux?

Magnetic Flux is "the amount of magnetic field" for want of a better description. Flux linkage is the Flux through a coil multiplied by the number of turns the flux passes through..

التدفق المغناطيسي هو "مقدار المجال المغناطيسي" للحاجة إلى وصف أفضل. ارتباط التدفق هو التدفق عبر ملف مضروبًا في عدد الدورات التي يمر بها التدفق ..

The term in parentheses in Equation (1–40) is called *the flux linkage* λ of the coil, and Faraday's law can be rewritten in terms of flux linkage as

$$e_{\text{ind}} = \frac{d\lambda}{dt} \tag{1–41}$$

where

$$\lambda = \sum_{i=1}^{N} \phi_i \tag{1–42}$$

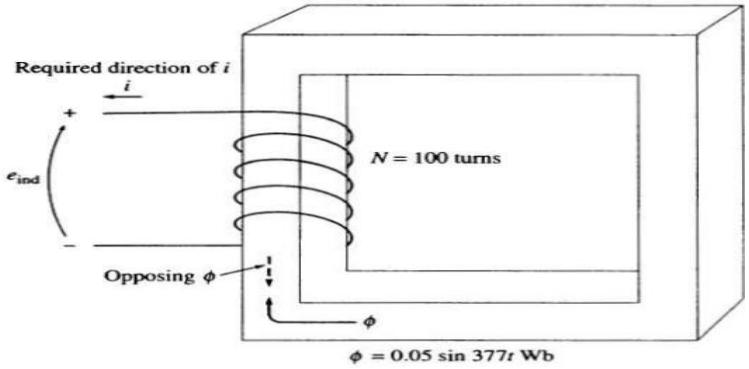
The units of flux linkage are weber-turns.

Example 1–6. Figure 1–15 shows a coil of wire wrapped around an iron core. If the flux in the core is given by the equation

$$\phi = 0.05 \sin 377t \quad \text{Wb}$$

If there are 100 turns on the core, what voltage is produced at the terminals of the coil? Of what polarity is the voltage during the time when flux is *increasing* in the reference direction shown in the figure? Assume that all the magnetic flux stays within the core (i.e., assume that the flux leakage is zero).

The core of Example 1–6. Determination of the voltage polarity at the terminals is shown.



Solution

By the same reasoning as in the discussion on pages 29–30, the direction of the voltage while the flux is increasing in the reference direction must be positive to negative, as shown in Figure 1–15. The *magnitude* of the voltage is given by

$$e_{\text{ind}} = N\frac{d\phi}{dt}$$

$$= (100 \text{ turns})\frac{d}{dt}(0.05 \sin 377t)$$

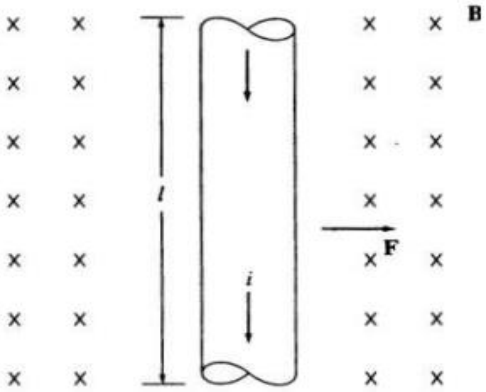
$$= 1885 \cos 377t$$

or alternatively,

$$e_{\text{ind}} = 1885 \sin(377t + 90°) \text{ V}$$

- **Produce an induced force on a wire**
  - **القوى المغناطيسية الناتجة من السلك**

اذا مر تيار كهربائي في سلك واقع في مجال مغناطيسي ثابت فأنه سوف تتولد في هذا السلك قوة دافعة مغناطيسية اثناء الحركة



$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

$$F = ilB \sin \theta$$

i = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

Example 1–7. Figure 1–16 shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T, directed into the page. If the wire is 1.0 m long and carries 0.5 A of current in the direction from the top of the page to the bottom of the page, what are the magnitude and direction of the force induced on the wire?

Solution

The direction of the force is given by the right-hand rule as being to the right. The magnitude is given by

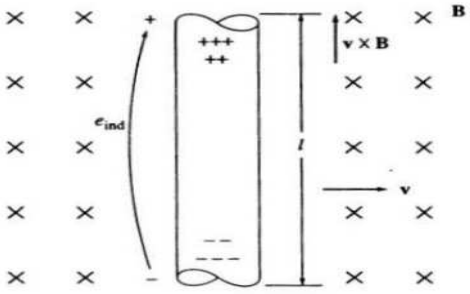
$$F = ilB \sin \theta \qquad (1\text{–}44)$$

$$= (0.5\ \text{A})(1.0\ \text{m})(0.25\ \text{T}) \sin 90° = 0.125\ \text{N}$$

Therefore,

$$\mathbf{F} = 0.125\ \text{N, directed to the right}$$

- **Induced voltage on a conductor**
- **الجهد المستحث على الموصل**



$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

v = velocity of the wire
B = magnetic flux density vector
l = length of conductor in the magnetic field

Example 1–8. Figure 1–17 shows a conductor moving with a velocity of 5.0 m/s to the right in the presence of a magnetic field. The flux density is 0.5 T into the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?

Solution

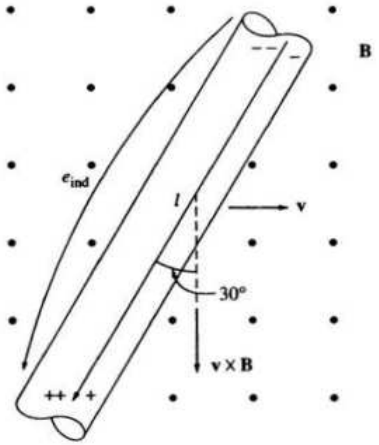
The direction of the quantity $\mathbf{v} \times \mathbf{B}$ in this example is up. Therefore, the voltage on the conductor will be built up positive at the top with respect to the bottom of the wire. The direction of vector $\mathbf{l}$ is up, so that it makes the smallest angle with respect to the vector $\mathbf{v} \times \mathbf{B}$.

Since $\mathbf{v}$ is perpendicular to $\mathbf{B}$ and since $\mathbf{v} \times \mathbf{B}$ is parallel to $\mathbf{l}$, the magnitude of the induced voltage reduces to

$$\begin{aligned} e_{\text{ind}} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45) \\ &= (vB \sin 90°)\, l \cos 0° \\ &= vBl \\ &= (5.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \\ &= 2.5 \text{ V} \end{aligned}$$

Thus the induced voltage is 2.5 V, positive at the top of the wire.

Example 1–9. Figure 1–18 shows a conductor moving with a velocity of 10 m/s to the right in a magnetic field. The flux density is 0.5 T, out of the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?



Solution

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ is down. The wire is not oriented on an up-down line, so choose the direction of l as shown to make the smallest possible angle with the direction of $\mathbf{v} \times \mathbf{B}$. The voltage is positive at the bottom of the wire with respect to the top of the wire. The magnitude of the voltage is

$$\begin{aligned} e_{\text{ind}} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45) \\ &= (vB \sin 90°)\, l \cos 30° \\ &= (10.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \cos 30° \\ &= 4.33 \text{ V} \end{aligned}$$

- **The linear DC machine – a simple example**

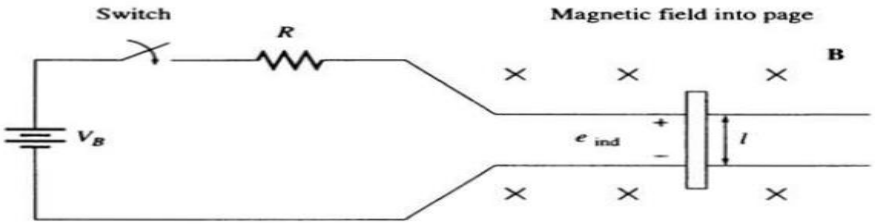


FIGURE 1–19
A linear dc machine. The magnetic field points into the page.



A linear dc machine is about the simplest and easiest-to-understand version of a dc machine, yet it operates according to the same principles and exhibits the same behavior as real generators and motors. It thus serves as a good starting point in the study of machines. A linear dc machine is shown in Figure 1-19. It consists of a battery and a resistance connected through a switch to a pair of smooth, frictionless rails. Along the bed of this "railroad track" is a constant, uniform-density magnetic field directed into the page. A bar of conducting metal is lying across the tracks. How does such a strange device behave? Its behavior can be determined from an application of four basic equations to the machine. These equations are

إن آلة التيار المستمر الخطية هي أبسط وأسهل نسخة للفهم من آلة التيار المستمر ، ومع ذلك فهي تعمل وفقًا لنفس المبادئ وتعرض نفس سلوك المولدات والمحركات الحقيقية. وبالتالي فهي بمثابة نقطة انطلاق جيدة في دراسة الآلات. تظهر آلة التيار المستمر الخطية في الشكل 1-19. يتكون من بطارية ومقاومة متصلة من خلال مفتاح بزوج من القضبان الملساء غير الاحتكاكية. على طول قاع "مسار السكة الحديد" هذا يوجد مجال مغناطيسي ثابت الكثافة وموحد موجه إلى الصفحة. يوجد شريط من المعدن الموصّل عبر المسارات. كيف يتصرف مثل هذا الجهاز الغريب؟ يمكن تحديد سلوكها من خلال تطبيق أربع معادلات أساسية على الآلة. هذه المعادلات

1.The equation for the force on a wire in the presence of a magnetic field:

1. معادلة القوة المؤثرة على سلك في وجود مجال مغناطيسي:

$$\boxed{\mathbf{F} = i(\mathbf{l} \times \mathbf{B})} \qquad (1\text{–}43)$$

where $\mathbf{F}$ = force on wire
i = magnitude of current in wire
$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow
$\mathbf{B}$ = magnetic flux density vector

2. The equation for the voltage induced on a wire moving in a magnetic field:

2. معادلة الجهد المستحث على سلك يتحرك في مجال مغناطيسي:

$$\boxed{e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}} \qquad (1\text{–}45)$$

where e_{ind} = voltage induced in wire
$\mathbf{v}$ = velocity of the wire
$\mathbf{B}$ = magnetic flux density vector
$\mathbf{l}$ = length of conductor in the magnetic field

3. Kirchhoff's voltage law for this machine. From Figure 1-19 this law gives

3. قانون الجهد كيرشوف لهذا الجهاز. من الشكل 1-19 يعطي هذا القانون

$$V_B - iR - e_{\text{ind}} = 0$$

$$\boxed{V_B = e_{\text{ind}} + iR = 0} \qquad (1\text{–}46)$$

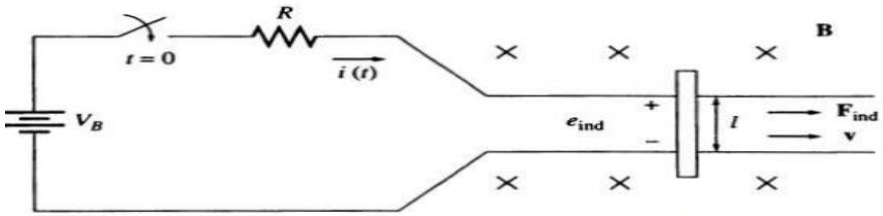
4. Newton's law for the bar across the tracks:

4. قانون نيوتن للشريط عبر المسار:

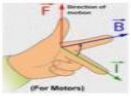
$$\boxed{F_{\text{net}} = ma} \qquad (1\text{–}7)$$

- **Starting the Linear DC Machine**

بدء التشغيل آلالات التيار المستمر الخطية .







1. Current $i = \dfrac{V_B - e_{ind}}{R}$

2. Induced force $F_{ind} = ilB$ to the right

3. Induced voltage $e_{ind} = vBl$ positive upward

$$i\downarrow = \frac{V_B - e_{ind}\uparrow}{R}$$

The result of this action is that eventually the bar will reach a constant steady-state speed where the net force on the bar is zero. This will occur when e_{ind} has risen all the way up to equal the voltage V_B. At that time, the bar will be moving at a speed given by

نتيجة هذا الإجراء هو أن الشريط سيصل في النهاية إلى سرعة ثابتة حيث تكون القوة الكلية على الشريط صفرًا. سيحدث هذا عندما يرتفع e_{ind} على طول الطريق حتى يساوي الجهد V_B. فى ذلك الوقت ، سوف يتحرك الشريط بالسرعة التى قدمها

$$V_B = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl} \qquad (1\text{–}50)$$

The bar will continue to cost along at this no-load at this no-load speed forever unless some external force disturbs it. When the motor is started, the velocity v, induced voltage e_{ind} , current i, and induced force F_{ind} are as sketched in Figure 1-21

سيستمر الشريط في التكلفة عند عدم التحميل عند سرعة عدم التحميل هذه إلى الأبد ما لم تزعجه بعض القوى الخارجية. عند بدء تشغيل المحرك ، تكون السرعة v والجهد المستحث والتيار i والقوة المستحثة كما هو موضح في الشكل 1-21

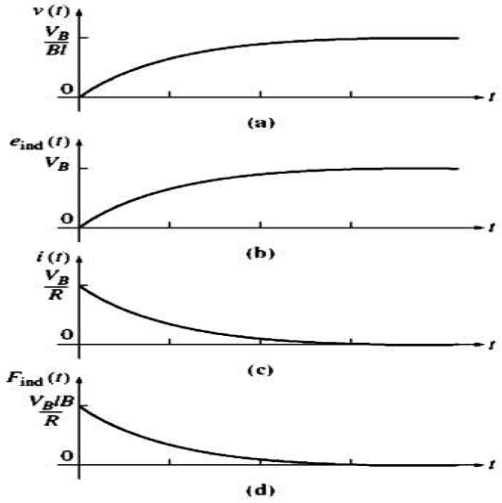


FIGURE 1–21
The linear dc machine on starting. (a) Velocity $v(t)$ as a function of time; (b) induced voltage $e_{ind}(t)$; (c) current $i(t)$; (d) induced force $F_{ind}(t)$.

The summarize, at starting, the linear dc machine behaves as follows:

تلخيص ، عند البدء ، تتصرف آلة التيار المستمر على النحو التالي:

I. Closing the switch produces a current flow $i = V_B / R$.

1. ينتج عن إغلاق المفتاح تدفق تيار

2. The current flow produces a force on the bar given by F = ilB.

2. ينتج التدفق الحالي قوة على الشريط تعطى بواسطة F = ilB.

3. The bar accelerates to the right, producing an induced voltage e_{ind} as it speeds up.

3. يتسارع الشريط إلى اليمين ، وينتج جهدًا مستحثًا أثناء تسارعه.

4. This induced voltage reduces the current flow $i = (V_B - e_{ind}\uparrow)/R$.

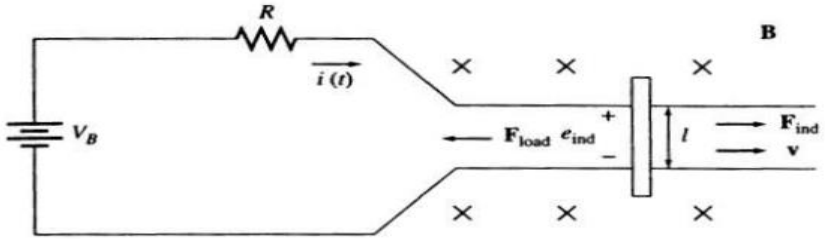
4. هذا الجهد المستحث يقلل من تدفق التيار $i = (V_B - e_{ind}\uparrow)/R$.

5. The induced force is thus decreased ($F = i\uparrow lB$) until eventually F=0. At that point, $e_{ind} = V_B$, i = 0, and the bar moves at a constant no-load speed $V_{ss} = V_b/BL$.

5. وهكذا تنخفض القوة المستحثة ($F = i \uparrow lB$) حتى F = 0 في النهاية. عند هذه النقطة ، $e_{ind} = V_B$، i = 0 ، ويتحرك الشريط بسرعة ثابتة بدون حمل $V_{ss} = V_b / BL$.

- **The Linear DC Machine as a Motor**

- While the load is applied



- The conversion power between mechanical and electrical $P_{conv} = e_{ind}i = F_{ind}v$

Summarize of a dc motor operation

1. A force $\mathbf{F}_{load}$ is applied opposite to the direction of motion, which causes a net force $\mathbf{F}_{net}$ opposite to the direction of motion.
2. The resulting acceleration $a = F_{net}/m$ is negative, so the bar slows down ($v\downarrow$).
3. The voltage $e_{ind} = v\downarrow Bl$ falls, and so $i = (V_B - e_{ind}\downarrow)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a lower speed v.
5. An amount of electric power equal to $e_{ind}i$ is now being converted to mechanical power equal to $F_{ind}v$, and the machine is acting as a motor.

1. يتم تطبيق حمل القوة عكس اتجاه الحركة ، مما يتسبب في صافي القوة F صافي ، عكس اتجاه الحركة.

2. التسارع الناتج $a = F_{net} / m$ سلبي ، لذا يتباطأ الشريط(v↓)

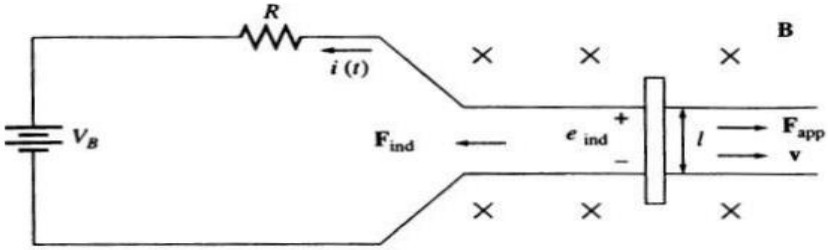
3. الجهد $V_{bl} = e_{ind}$ ينخفض ، وبالتالي فإن $i = (VB - eind↓) / R$ تزداد.

4. القوة المستحثة $F_{ind} = i↑lB$ تزداد حتى أجد $I F_{ind} I = I F_{load} I$ بسرعة أقل v .

5. يتم الآن تحويل مقدار من الطاقة الكهربائية يساوي e_{ind} i إلى طاقة ميكانيكية تساوي V_{ind} f، وma

- **Linear dc generator**

- While the external force is applied on the moving direction

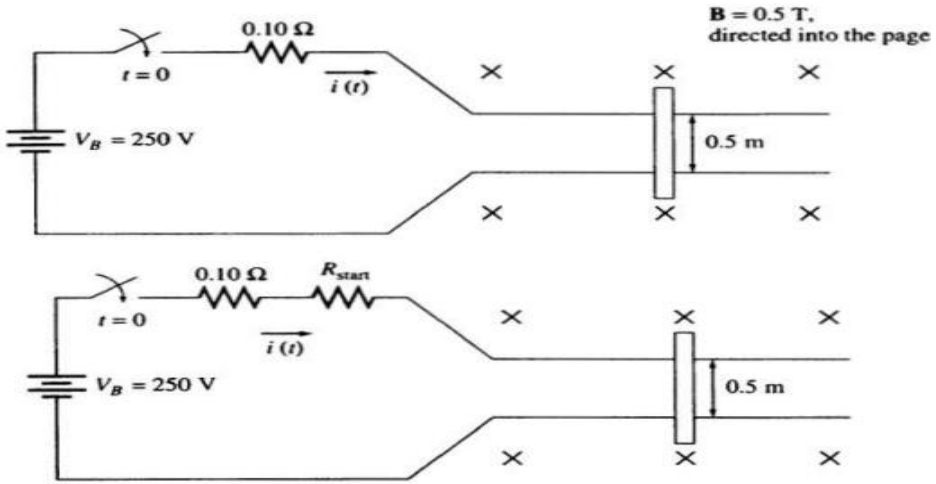


$$P_{conv} = \tau_{ind}\omega$$

Summarize of a dc generator operation

1. A force $\mathbf{F}_{app}$ is applied in the direction of motion; $\mathbf{F}_{net}$ is in the direction of motion.
2. Acceleration $a = F_{net}/m$ is positive, so the bar speeds up ($v\uparrow$).
3. The voltage $e_{ind} = v\uparrow Bl$ increases, and so $i = (e_{ind}\uparrow - V_B)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a higher speed v.
5. An amount of mechanical power equal to $F_{ind}v$ is now being converted to electric power $e_{ind}i$, and the machine is acting as a generator.

- **Starting problem of dc linear machine**



Example 1–10. The linear dc machine shown in Figure 1–27a has a battery voltage of 120 V, an internal resistance of 0.3 Ω, and a magnetic flux density of 0.1 T.

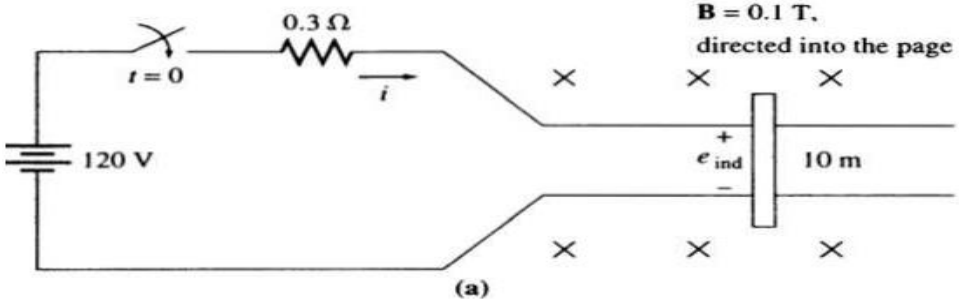
(a) What is this machine's maximum starting current? What is its steady-state velocity at no load?

(b) Suppose that a 30-N force pointing to the right were applied to the bar. What would the steady-state speed be? How much power would the bar be producing or consuming? How much power would the battery be producing or consuming? Explain the difference between these two figures. Is this machine acting as a motor or as a generator?

(c) Now suppose a 30-N force pointing to the left were applied to the bar. What would the new steady-state speed be? Is this machine a motor or a generator now?

(d) Assume that a force pointing to the left is applied to the bar. Calculate speed of the bar as a function of the force for values from 0 N to 50 N in 10-N steps. Plot the velocity of the bar versus the applied force.

(e) Assume that the bar is unloaded and that it suddenly runs into a region where the magnetic field is weakened to 0.08 T. How fast will the bar go now?



Solution

(a) At starting conditions, the velocity of the bar is 0, so $e_{ind} = 0$. Therefore,

$$i = \frac{V_B - e_{ind}}{R} = \frac{120\text{ V} - 0\text{ V}}{0.3\ \Omega} = 400\text{ A}$$

When the machine reaches steady state, $\mathbf{F}_{ind} = 0$ and $i = 0$. Therefore,

$$VB = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl}$$

$$= \frac{120\text{ V}}{(0.1\text{ T})(10\text{ m})} = 120\text{ m/s}$$

(b) Refer to Figure 1–27b. If a 30-N force to the right is applied to the bar, the final steady state will occur when the induced force $\mathbf{F}_{ind}$ is equal and opposite to the applied force $\mathbf{F}_{app}$, so that the net force on the bar is zero:

$$F_{app} = F_{ind} = ilB$$

Therefore,

$$i = \frac{F_{ind}}{lB} = \frac{30\text{ N}}{(10\text{m})(0.1\text{ T})}$$

$$= 30\text{ A} \quad \text{flowing up through the bar}$$

The induced voltage e_{ind} on the bar must be

$$e_{ind} = V_B + iR$$

$$= 120\text{ V} + (30\text{A})(0.3\ \Omega) = 129\text{ V}$$

and the final steady-state speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{129\text{ V}}{(0.1\text{ T})(10\text{ m})} = 129\text{ m/s}$$

The bar is *producing* $P = (129\text{ V})(30\text{ A}) = 3870\text{ W}$ of power, and the battery is *consuming* $P = (120\text{ V})(30\text{ A}) = 3600\text{ W}$. The difference between these two numbers is the 270 W of losses in the resistor. This machine is acting as a *generator.*

(c) Refer to Figure 1–25c. This time, the force is applied to the left, and the induced force is to the right. At steady state,

$$F_{app} = F_{ind} = ilB$$

$$i = \frac{F_{ind}}{lB} = \frac{30\text{ N}}{(10\text{ m})(0.1\text{ T})}$$

$$= 30\text{ A} \quad \text{flowing down through the bar}$$

The induced voltage e_{ind} on the bar must be

$$e_{ind} = V_B - iR$$

$$= 120\text{ V} - (30\text{ A})(0.3\ \Omega) = 111\text{ V}$$

and the final speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{111\text{ V}}{(0.1\text{ T})(10\text{ m})} = 111\text{ m/s}$$

This machine is now acting as a *motor*, converting electric energy from the battery into mechanical energy of motion on the bar.

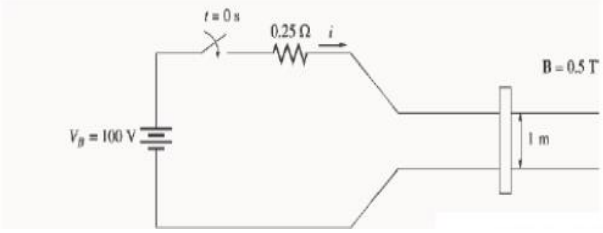
(e) If the bar is initially unloaded, then $e_{ind} = V_B$. If the bar suddenly hits a region of weaker magnetic field, a transient will occur. Once the transient is over, though, e_{ind} will again equal V_B.

1-21. A linear machine has a magnetic flux density of 0.5 T directed into the page, a resistance of 0.25 Ω, a bar length l = 1.0 m, and a battery voltage of 100 V.

(a) What is the initial force on the bar at starting? What is the initial current flow?

(b) What is the no-load steady-state speed of the bar?

(c) If the bar is loaded with a force of 25 N opposite to the direction of motion, what is the new steady-state speed? What is the efficiency of the machine under these circumstances?



SOLUTION

(a) The current in the bar at starting is

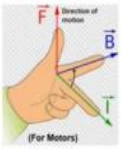
$$i = \frac{V_B}{R} = \frac{100\ \text{V}}{0.25\ \Omega} = 400\ \text{A}$$

Therefore, the force on the bar at starting is

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B}) = (400\ \text{A})(1\ \text{m})(0.5\ \text{T}) = 200\ \text{N, to the right}$$



Left Hand Rule (for Motor)



(b) The no-load steady-state speed of this bar can be found from the equation

$$V_B = e_{\text{ind}} = vBl$$

$$v = \frac{V_B}{Bl} = \frac{100\ \text{V}}{(0.5\ \text{T})(1\ \text{m})} = 200\ \text{m/s}$$

(c) With a load of 25 N opposite to the direction of motion, the steady-state current flow in the bar will be given by

$$F_{\text{app}} = F_{\text{ind}} = ilB$$

$$i = \frac{F_{\text{app}}}{Bl} = \frac{25\ \text{N}}{(0.5\ \text{T})(1\ \text{m})} = 50\ \text{A}$$

The induced voltage in the bar will be

$$e_{\text{ind}} = V_B - iR = 100\ \text{V} - (50\ \text{A})(0.25\ \Omega) = 87.5\ \text{V}$$

and the velocity of the bar will be

$$v = \frac{V_B}{Bl} = \frac{87.5\ \text{V}}{(0.5\ \text{T})(1\ \text{m})} = 175\ \text{m/s}$$

The *input* power to the linear machine under these conditions is

$$P_{\text{in}} = V_B i = (100\ \text{V})(50\ \text{A}) = 5000\ \text{W}$$

The *output* power from the linear machine under these conditions is

$$P_{\text{out}} = V_B i = (87.5\ \text{V})(50\ \text{A}) = 4375\ \text{W}$$

Therefore, the efficiency of the machine under these conditions is

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{4375\ \text{W}}{5000\ \text{W}} \times 100\% = 87.5\%$$

1-22. A linear machine has the following characteristics:

$B = 0.33\ \text{T}$ into page $\qquad R = 0.50\ \Omega$

$l = 0.5\ \text{m} \qquad V_B = 120\ \text{V}$

(a) If this bar has a load of 10 N attached to it opposite to the direction of motion, what is the steady-state speed of the bar?

(b) If the bar runs off into a region where the flux density falls to 0.30 T, what happens to the bar? What is its final steady-state speed?

(c) Suppose V_B is now decreased to 80 V with everything else remaining as in part (b). What is the new steady-state speed of the bar?

(d) From the results for parts (b) and (c), what are two methods of controlling the speed of a linear machine (or a real dc motor)?

SOLUTION

(a) With a load of 20 N opposite to the direction of motion, the steady-state current flow in the bar will be given by

$$F_{app} = F_{ind} = ilB$$

$$i = \frac{F_{app}}{Bl} = \frac{10\text{ N}}{(0.33\text{ T})(0.5\text{ m})} = 60.5\text{ A}$$

The induced voltage in the bar will be

$$e_{ind} = V_B - iR = 120\text{ V} - (60.5\text{ A})(0.50\ \Omega) = 89.75\text{ V}$$

and the velocity of the bar will be

$$v = \frac{e_{ind}}{Bl} = \frac{89.75\text{ V}}{(0.33\text{ T})(0.5\text{ m})} = 544\text{ m/s}$$

(b) If the flux density drops to 0.30 T while the load on the bar remains the same, there will be a speed transient until $F_{app} = F_{ind} = 10$ N again. The new steady state current will be

$$F_{app} = F_{ind} = ilB$$

$$i = \frac{F_{app}}{Bl} = \frac{10\text{ N}}{(0.30\text{ T})(0.5\text{ m})} = 66.7\text{ A}$$

The induced voltage in the bar will be

$$e_{ind} = V_B - iR = 120\text{ V} - (66.7\text{ A})(0.50\ \Omega) = 86.65\text{ V}$$

and the velocity of the bar will be

$$v = \frac{e_{ind}}{Bl} = \frac{86.65\text{ V}}{(0.30\text{ T})(0.5\text{ m})} = 577\text{ m/s}$$

(c) If the battery voltage is decreased to 80 V while the load on the bar remains the same, there will be a speed transient until $F_{app} = F_{ind} = 10$ N again. The new steady state current will be

$$F_{app} = F_{ind} = ilB$$

$$i = \frac{F_{app}}{Bl} = \frac{10\text{ N}}{(0.30\text{ T})(0.5\text{ m})} = 66.7\text{ A}$$

The induced voltage in the bar will be

$$e_{ind} = V_B - iR = 80\text{ V} - (66.7\text{ A})(0.50\ \Omega) = 46.65\text{ V}$$

and the velocity of the bar will be

$$v = \frac{e_{ind}}{Bl} = \frac{46.65\text{ V}}{(0.30\text{ T})(0.5\text{ m})} = 311\text{ m/s}$$

(d) From the results of the two previous parts, we can see that there are two ways to control the speed of a linear dc machine. *Reducing* the flux density B of the machine *increases* the steady-state speed, and *reducing* the battery voltage V_B *decreases* the stead-state speed of the machine. Both of these speed control methods work for real dc machines as well as for linear machines.

نتهى الشابتر ملاحظة بسيطة من ضروري حل اسئلة البروبليم مهمة جداً لفهم المادة لكي تتمكن من المادة اكثر واكثر وبالتوفيق للجميع

Chapter 2 Transformers

المحولات الكهربائية أجهزه ثابتة (غير دواره) تستطيع رفع أو خفض الجهد والتيار المتردد بوسطة الحث الكهرومغناطيسي . وهذا يسمح بنقل القدرة بكفاءة ، وبالتالي فأن المحولات هي المعدة الاساسية التي لايمكن الاستغناء عنها لنقل القدرة وتوزيعها

The transformer ideally changes one ac voltage level to another voltage level without affecting the actual power supplied.

The transformer can be used in distribution system for efficiency issues.

The step-up transformer decreases the line current and decreases the power loss on power line.

The transmission/distribution system with transformer can keep high efficiency

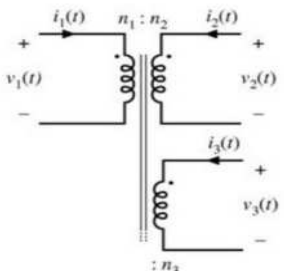
- **Types and construction of transformers**

• The transformer winding connected to the power source is called the primary winding or input winding

The winding connected to the loads is called the secondary winding or output winding

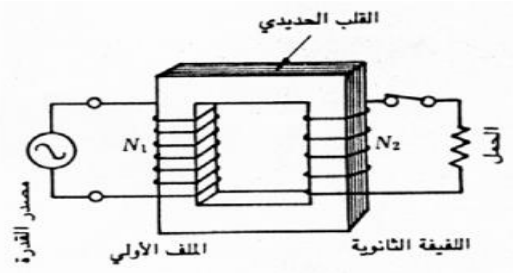
.... tertiary winding

Multiple winding transformer

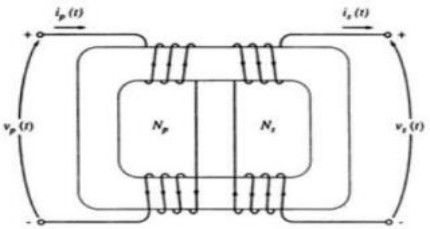


يتكون المحول من الملفات الابتدائية والملفات الثانوية الملفات الابتدائية تكون متصلة مع المصدر والملفات الثانوية مع الحمل

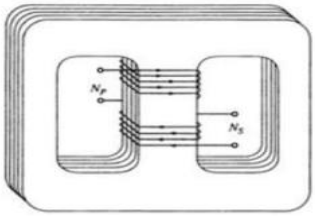
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أنواع الملفات تبعاً لطريقة تركيب لفات الاسلاك



• Core-form: consists a simple rectangular laminated piece of steel with the transformer winding wrapped around two sides of the rectangle



• Shell-form: consists three legs laminated core with winding wrapped around the center leg

•الشكل الأساسي: يتكون من قطعة مستطيلة بسيطة من الصلب مصفح مع لف المحول حول جانبي المستطيل

• شكل شل: يتكون من ثلاثة أرجل مغلفة الأساسية مع لف ملفوفة حول الساق المركزية

شرح عن القلب (core) لعمل قلب المحول تستخدم مادة ذات كثافة فيض مغناطيسي عال وفقد حديدي صغير ويتركب القلب من صفائح (رقيقة) من الصلب السليكوني المشكلة على هيئة شرائط ترص لتشكل قلباً ذا طبقات ، ويجمع القلب .

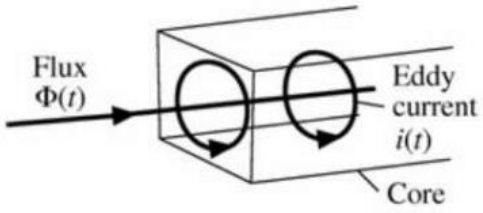
- **Core material and eddy current**

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• **The core is constructed of thin laminations electrically isolated from each other in order to minimize the eddy currents**

يتكون القلب من صفائح رقيقة معزولة كهربائيًا عن بعضها البعض لتقليل التيارات الدوامة

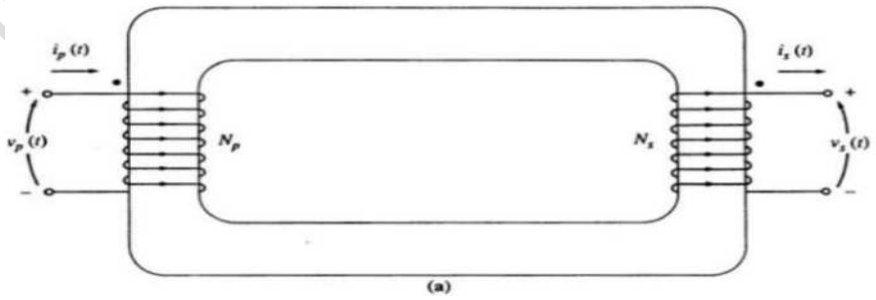
• **Eddy current:**



- **The ideal transformer characteristics**

خصائص المحولات المثالية

لمعرفة العلاقة بين الملفات الثانوية والابتدائية للمحول بصفة عمليه فإننا سوف ندرس هذه العلاقة أولا بالنسبة للمحول المثالي المحول المثالي تكون المفاقيد الحديدية به تساوي صفراً ، و مقاومة الملفات تساوي صفرا (مهملة) وليس به تسرب مغناطيسي . وبالنظر للشكل فإنه يمكن إيجاد مكونات الدائرة على النحو التالي:



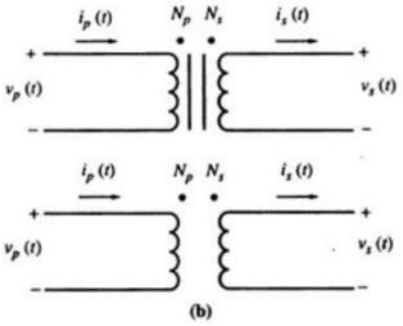
$$\frac{v_P(t)}{v_S(t)} = \frac{N_P}{N_S} = a$$

- Where $a = N_p/N_s$ is the turns ratio
- Energy balance relation

$$N_P i_P(t) = N_S i_S(t) \qquad \frac{i_P(t)}{i_S(t)} = \frac{1}{a}$$

- Phasor relation

$$\frac{\mathbf{V}_P}{\mathbf{V}_S} = a \qquad \frac{\mathbf{I}_P}{\mathbf{I}_S} = \frac{1}{a}$$



- The turns ratio ***a*** only effects the magnitude not the angle

$N_2 > N_1$ step up …… $N_1 > N_2$ step dawn

- **Dot convention in ideal transformer (2 سيركت)**

1. If the primary voltage is positive at the dotted end of the winding with respect to the undotted end, then the secondary voltage will be positive at the dotted end also. Voltage polarities are the same with respect to the dots on each side of the core.

2. If the primary current of the transformer flows into the dotted end of the primary winding, the secondary current will flow out of the dotted end of the secondary winding.

1. إذا كان الجهد الأساسي موجبًا عند النهاية المنقطة للملف فيما يتعلق بالنهاية غير المنقطة ، فسيكون الجهد الثانوي موجبًا عند النهاية المنقطة أيضًا. تكون قطبية الجهد هي نفسها فيما يتعلق بالنقاط الموجودة على كل جانب من جوانب القلب.

2. إذا كان التيار الأساسي للمحول يتدفق إلى النهاية المنقطة للملف الأولي ، فإن التيار الثانوي سوف يتدفق من النهاية المنقطة للملف الثانوي.

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- **Power in an ideal transformer**

The power supplied to the transformer by the primary circuit is given by the equation

$$P_{in} = V_P I_P \cos \theta_P \qquad (2\text{–}6)$$

where θ_p, is the angle between the primary voltage and the primary current. The power supplied by the transformer secondary circuit to its loads is given by the equation

$$P_{out} = V_S I_S \cos \theta_S$$

whereθ_s, is the angle between the secondary voltage and the secondary current. Since voltage and current angles are unaffected by an ideal transformer $\theta_p - \theta_s$= 0. The primary and secondary windings of an ideal transformer have the same power factor. (P_{in}=P_{out})

$$P_{out} = V_S I_S \cos \theta \qquad (2\text{–}8)$$

Applying the turns-ratio equations gives $V_S = V_P/a$ and $I_S = aI_P$, so

$$P_{out} = \frac{V_P}{a}(aI_P) \cos \theta$$

$$P_{out} = V_P I_P \cos \theta = P_{in} \qquad (2\text{–}9)$$

Thus, *the output power of an ideal transformer is equal to its input power.*

The same relationship applies to reactive power Q and apparent power S:

$$Q_{in} = V_P I_P \sin \theta = V_S I_S \sin \theta = Q_{out} \qquad (2\text{–}10)$$

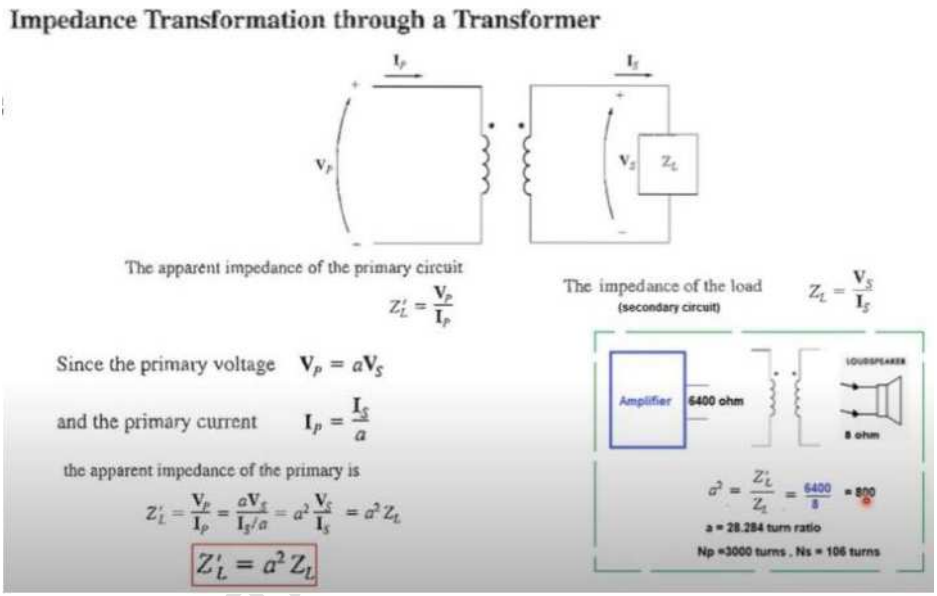
and

$$S_{in} = V_P I_P = V_S I_S = S_{out} \qquad (2\text{–}11)$$

- **Impedance Transformation through a transformer**

نقل الممانعة (المعاوقة) في المحولات

نقل الممانعة من الجانب الثانوي إلى الجانب الابتدائي أو العكس مفيد في تبسيط الدائرة من دائرة كهربائية مزدوجة إلى دائرة مفردة وبالتالي تسهل عملية الحسابات . في الشكل يمكن نقل ممانعة الحمل Z_L الموصلة على الجانب الثانوي إلى الجانب الابتدائي



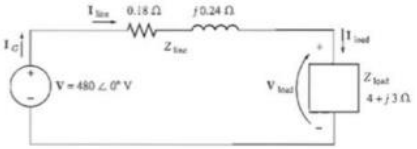
- **Analysis of circuit containing ideal transformers**

تحليل الدوائر التي تحتوي على محولات مثالية

- **All the current and voltage are all referred to one side (primary side)**
- **Note the dot convention for current direction**
- **Impedance transformation**

Example 2–1.

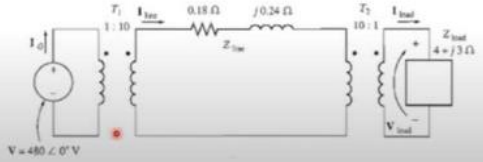
A single-phase power system consists of a 480-V 60-Hz generator supplying a load $Z_{load} = 4 + j3\ \Omega$ through a transmission line of impedance $Z_{line} = 0.18 + j0.24\ \Omega$.



(a) what will the voltage at the load be?

What will the transmission line losses be?

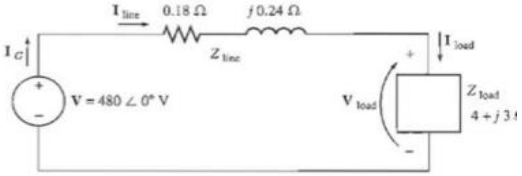
(b) Suppose a 1:10 step-up transformer is placed at the generator end of the transmission line and a 10:1 step-down transformer is placed at the load end of the line (as shown in Figure 2–6b).



(a) what will the voltage at the load be?

What will the transmission line losses be?

Solution:



Here $\mathbf{I}_G = \mathbf{I}_{line} = \mathbf{I}_{load}$

$$\mathbf{I}_{line} = \frac{\mathbf{V}}{Z_{line} + Z_{load}}$$

$$= \frac{480\angle 0^\circ\ \text{V}}{(0.18\ \Omega + j0.24\ \Omega) + (4\ \Omega + j3\ \Omega)}$$

$$= \frac{480\angle 0^\circ}{4.18 + j3.24} = \frac{480\angle 0^\circ}{5.29\angle 37.8^\circ}$$

$$= 90.8\angle -37.8^\circ\ \text{A}$$

Therefore the load voltage is

$$\mathbf{V}_{load} = \mathbf{I}_{line} Z_{load}$$

$$= (90.8\angle -37.8^\circ\ \text{A})(4\ \Omega + j3\ \Omega)$$

$$= (90.8\angle -37.8^\circ\ \text{A})(5\angle 36.9^\circ\ \Omega)$$

$$= 454\angle -0.9^\circ\ \text{V}$$

and the line losses are (power is lost in resistive components only)

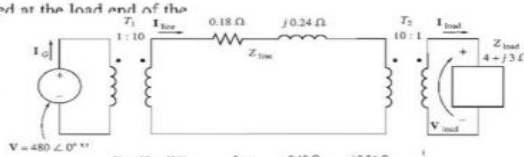
$$P_{loss} = (I_{line})^2 R_{line}$$

$$= (90.8\ \text{A})^2\ (0.18\ \Omega) = 1484\ \text{W}$$

(b) Suppose a 1:10 step-up transformer is placed at the generator end of the transmission line and a 10:1 step-down transformer is placed at the load end of the line (as shown in Figure 2–6b).

What will the load voltage be now?

What will the transmission line losses be now?

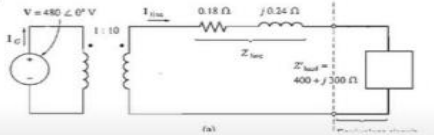


Solution:

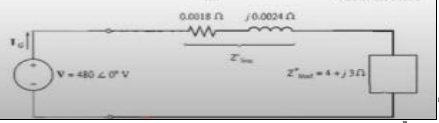
To analyze this system, it is necessary to convert it to a common voltage level.

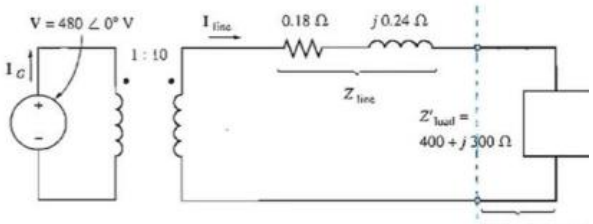
This is done in two steps:

1. Eliminate transformer T_2 by referring the load over to the transmission line's voltage level.



2. Eliminate transformer T_1 by referring the transmission line's elements and the equivalent load at the transmission line's voltage over to the source side.



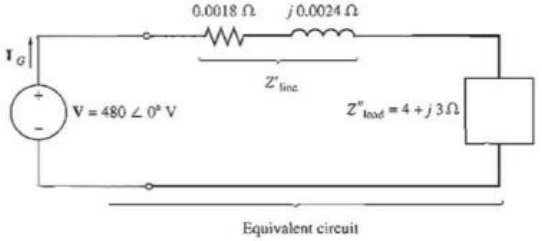


Equivalent circu

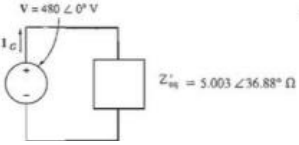
$$Z'_{eq} = a^2 Z_{eq}$$
$$= a^2(Z_{line} + Z'_{load})$$
$$= \left(\frac{1}{10}\right)^2 (0.18\ \Omega + j0.24\ \Omega + 400\ \Omega + j300\ \Omega)$$

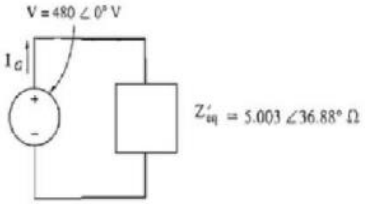
Note: Here 'a' is 1/10

$$= (0.0018\ \Omega + j0.0024\ \Omega + 4\ \Omega + j3\ \Omega)$$
$$= 5.003\ \angle 36.88°\ \Omega$$



Equivalent circuit





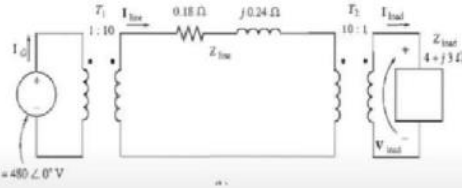
The generator's current is

$$I_G = \frac{480\ \angle 0°\ V}{5.003\ \angle 36.88°\ \Omega} = 95.94\ \angle -36.88°\ A$$

Knowing the current I_G, we can now work back and find I_{line} and I_{load}.

Working back through T_1, we get

$$N_{P1} I_G = N_{S1} I_{line}$$
$$I_{line} = \frac{N_{P1}}{N_{S1}} I_G$$
$$= \frac{1}{10}(95.94\ \angle -36.88°\ A) = 9.594\ \angle -36.88°$$



Working back through T_2 gives

$$N_{P2} I_{line} = N_{S2} I_{load}$$
$$I_{load} = \frac{N_{P2}}{N_{S2}} I_{line}$$
$$= \frac{10}{1}(9.594\ \angle -36.88°\ A) = 95.94\ \angle -36.88°\ A$$

- **Induced voltage and flux linkageλ**

1. The induced voltage $e_{ind} = \frac{d\lambda}{dt}$
2. The flux linkage $\lambda = \sum_{i=1}^{N} \phi_i$
3. Simplified by average flux $\overline{\phi} = \frac{\lambda}{N}$
4. The final induced voltage $e_{ind} = N\frac{d\overline{\phi}}{dt}$

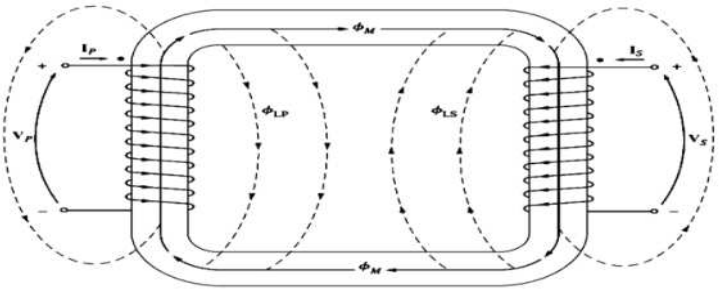


FIGURE 2–10
Mutual and leakage fluxes in a transformer core.

- **Voltage relation علاقة الجهد**

1. Induced voltage on each side $\overline{\phi} = \frac{1}{N_p}\int v_p(t)\,dt$

تنص هذه المعادلة على أن متوسط التدفق في الملف يتناسب مع تكامل الجهد المطبق على الملف ، وثابت التناسب عدد اللفات بالمثل في الملف الأولي 1/ N_p.

هذا التدفق موجود في الملف الأساسي للمحول. ما هو تأثيره على الملف الثانوي للناقل؟ يعتمد التأثير على مقدار التدفق الذي يصل إلى الملف الثانوي. لا يمر كل التدفق الناتج في الملف الأولي أيضًا عبر الملف الثانوي - بعض خطوط التدفق تترك قلب الحديد وتمر عبر الهواء بدلاً من ذلك (انظر الشكل 2-10).

يسمى جزء التدفق الذي يمر عبر أحد ملفات المحول وليس الآخر بتدفق التسرب. وبالتالي يمكن تقسيم التدفق في الملف الأولي للمحول إلى مكونين: التدفق المتبادل ، والذي يظل في القلب ويربط بين كلتا اللفات ، وتدفق تسرب صغير ، يمر عبر الملف الأولي ولكنه يعود عبر الهواء ، متجاوزًا اللف الثانوي:

Primary side flux

$$\bar{\phi}_P = \phi_M + \phi_{LP}$$

$\bar{\phi}_P$ = total average primary flux
ϕ_M = flux component linking both primary and secondary coils
ϕ_{LP} = primary leakage flux

يوجد تقسيم مماثل للتدفق في الملف الثانوي بين التدفق المتبادل وتدفق التسرب الذي يمر عبر الملف الثانوي ولكنه يعود عبر الهواء ، متجاوزًا الملف الأولي:

Secondary side flux

$$\bar{\phi}_S = \phi_M + \phi_{LS}$$

$\bar{\phi}_S$ = total average secondary flux
ϕ_M = flux component linking both primary and secondary coils
ϕ_{LS} = secondary leakage flux

مع تقسيم متوسط التدفق الأولي إلى مكونات متبادلة ومكونات تسرب ، يمكن لقانون فاراداي للدائرة الأولية يمكن تسمية المصطلح الأول من هذا التعبير e_p (t) ، ويمكن تسمية المصطلح الثاني e_{LP} (t).

Induced voltage on primary side

$$v_P(t) = N_P \frac{d\bar{\phi}_P}{dt}$$

$$= N_P \frac{d\phi_M}{dt} + N_P \frac{d\phi_{LP}}{dt}$$

$$v_P(t) = e_P(t) + e_{LP}(t)$$

يمكن أيضًا التعبير عن الجهد على الملف الثانوي للمحول وفقًا لقانون فاراداي كـ

Induced voltage on secondary side

$$v_S(t) = N_S \frac{d\bar{\phi}_S}{dt}$$

$$= N_S \frac{d\phi_M}{dt} + N_S \frac{d\phi_{LS}}{dt}$$

$$= e_S(t) + e_{LS}(t)$$

The primary voltage due to the mutual flux is given by

يتم إعطاء الجهد الأولي بسبب التدفق المتبادل بواسطة

$$e_P(t) = N_P \frac{d\phi_M}{dt}$$

and the secondary voltage due to the mutual flux is given by

ويعطى الجهد الثانوي بسبب التدفق المتبادل بواسطة

$$e_S(t) = N_S \frac{d\phi_M}{dt}$$

Notice from these two relationships that

لاحظ من هاتين العلاقتين أن

$$\frac{e_P(t)}{N_P} = \frac{d\phi_M}{dt} = \frac{e_S(t)}{N_S}$$

Therefore,

$$\boxed{\frac{e_P(t)}{e_S(t)} = \frac{N_P}{N_S} = a}$$

This equation means that the ratio of the primary voltage caused by the mutual flux to the secondary voltage caused by the mutual flux is equal to the turns ratio of the transformer. Since in a well-designed transformer ($\phi_M \gg \phi_{LP}$) and ($\phi_M \gg \phi_{LS}$) the ratio of the total voltage on the primary of a transformer to the total voltage on the secondary of a transformer is approximately

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تعني هذه المعادلة أن نسبة الجهد الأولي الناتج عن التدفق المتبادل إلى الجهد الثانوي الناتج عن التدفق المتبادل تساوي نسبة التحويل (a) للمحول. نظرًا لأنه مصمم جيدًا $(\phi_M \gg \phi_{LP})$ و $(\phi_M \gg \phi_{LS})$ فإن نسبة الجهد الكلي في المرحلة الأولية من المحول إلى إجمالي الجهد في المرحلة الثانوية للمحول هي تقريبًا

$$\frac{v_P(t)}{v_S(t)} = \frac{N_P}{N_S} = a$$

The smaller the leakage fluxes of the transformer are, the closer the total transformer voltage ratio approximates that of the ideal transformer

كلما كانت تدفقات التسرب للمحول أصغر ، كلما اقتربت نسبة الجهد الكلي للمحول من نسبة المحول المثالي

- **The Exact Equivalent Circuit of a Real Transformer**

من الممكن إنشاء دائرة مكافئة تأخذ في الاعتبار جميع العيوب الرئيسية في المحولات الحقيقية. يعتبر كل عيب رئيسي في بدوره ، وتأثيره مدرج في نموذج المحولات.

التأثير الأسهل للنمذجة هو خسائر النحاس. خسائر النحاس هي خسائر مقاومة في اللفات الأولية والثانوية من قلب المحول. يتم نمذجتها عن طريق وضع المقاومة R_p في الدائرة الأولية للمحول والمقاومة R_s في الدائرة الثانوية

As explained , the leakage flux in the primary windings ϕ_{LP} produces a voltageϕ_{LP} given by

كما هو موضح ، فإن تدفق التسرب في اللفات الأولية ϕ_{LP} ينتج جهدًا ϕ_{LP} معطى بواسطة

$$e_{LP}(t) = N_P \frac{d\phi_{LP}}{dt}$$

And the leakage flux in the secondary windings ϕ_{LS} produces a voltage e_{LS} given by

وينتج تدفق التسرب في اللفات الثانوية ϕ_{LS}جهدًا e_{LS} مُعطى بواسطة

$$e_{LS}(t) = N_S \frac{d\phi_{LS}}{dt}$$

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نظرًا لأن معظم مسار تسرب يمر عبر الهواء ، وبما أن الهواء له تردد ثابت أعلى بكثير من تردد القلب ، فإن التدفق يتناسب طرديًا إلى تيار الدائرة الأولية i_p ويكون التدفق ϕ_{LS} متناسبًا طرديًا مع التيار الثانوي i_s

$$\phi_{LP} = (\mathcal{P}N_P)i_P$$
$$\phi_{LS} = (\mathcal{P}N_S)i_S$$

where $\mathcal{P}$ = permeance of flux path
N_P = number of turns on primary coil
N_S = number of turns on secondary coil

Substitute Equations (2–37) into Equations (2–36). The result is

$$e_{LP}(t) = N_P \frac{d}{dt}(\mathcal{P}N_P)i_P = N_P^2 \mathcal{P} \frac{di_P}{dt}$$
$$e_{LS}(t) = N_S \frac{d}{dt}(\mathcal{P}N_S)i_S = N_S^2 \mathcal{P} \frac{di_S}{dt}$$

The constants in these equations can be lumped together. Then

$$\boxed{e_{LP}(t) = L_P \frac{di_P}{dt}}$$

$$\boxed{e_{LS}(t) = L_S \frac{di_S}{dt}}$$

where $L_p = N_S^2 \wp$ is the self-inductance of the primary coil and $L_s = N_S^2 \wp$ is the self-inductance of the secondary coil. Therefore, the leakage flux will be modeled by primary and secondary inductors.

حيث $L_p = N_S^2 \wp$ هو الحث الذاتي للملف الأساسي و $L_s = N_S^2 \wp$ هو الحث الذاتي للملف الثانوي. لذلك ، سيتم نمذجة تدفق التسرب بواسطة محاثات أولية وثانوية.

- THEORY OF OPERATION OF REAL SINGLE-PHASE TRANSFORMERS

المحول الفعلي غير المثالي يختلف عن المحول المثالي في عدة اوجه

1- مقاومة الملف الابتدائي والثانوي ليست مهملة
2- وجود الفيض المغناطيسي المتسرب في الملف الابتدائي والثانوي على التوالي
3- المفاقيد الحديدية غير مهملة
4- نفاذية مادة القلب ليست مالانهاية بما يؤدي الى وجود فيض تبادلي ϕ_M في القلب

لا يمكن بالطبع صنع المحولات المثالية. ما يمكن إنتاجه هو محولات حقيقية - ملفان أو أكثر من الأسلاك ملفوفة حول قلب مغناطيسي حديدي. تقارب خصائص المحولات الحقيقية خصائص المحول المثالي ، ولكن فقط إلى الدرجة العلمية . يتناول هذا القسم سلوك المحولات

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الحقيقية. لفهم عمل المحول حقيقي ، ارجع إلى الشكل8-2. يوضح الشكل8-2 محول يتكون من ملفين من الأسلاك ملفوفين حول قلب المحول. يتم توصيل المحول الأساسي بمصدر طاقة تيار متردد ، والملف الثانوي مفتوح الدائرة. يظهر منحنى التباطؤ للمحول في الشكل 9-2

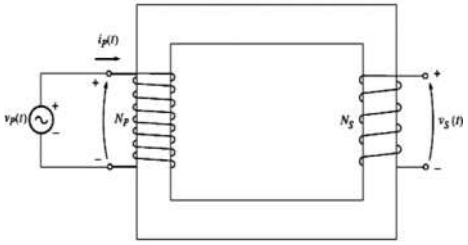


FIGURE 2-8
Sketch of a real transformer with no load attached to its secondary.

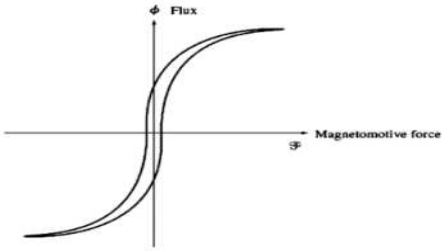


FIGURE 2–9
The hysteresis curve of the transformer.

- **The Magnetization Current in a Real Transformer**

عندما يتم توصيل مصدر طاقة تيار متردد بمحول كما هو موضح في الشكل 8-2 ، يتدفق التيار في دائرته الأولية ، حتى عندما تكون الدائرة الثانوية مفتوحة. هذا التيار هو التيار المطلوب لإنتاج تدفق في قلب مغناطيسي حديدي حقيقي ، كما هو موضح في chapter 1 . ويتكون من مكونين

I. The magnetization current i_M , which is the current required to produce the flux in the transformer core

2. The core-loss current i_H " which is the current required to make up for hysteresis and eddy current losse

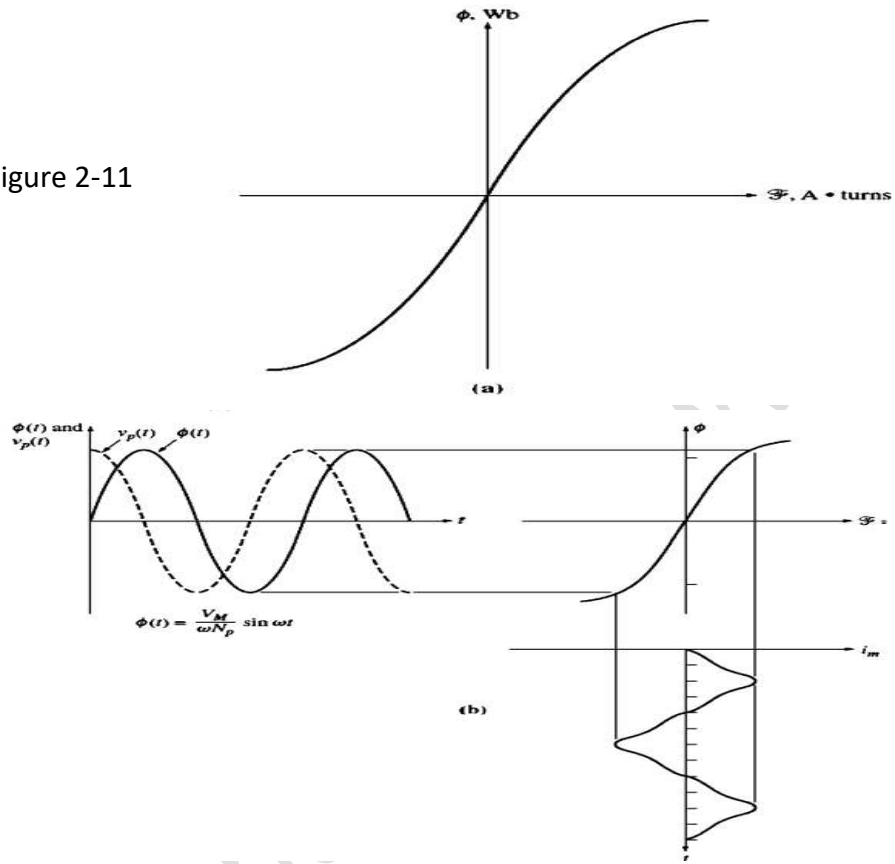
.Iتيار التمغنط i_M ، وهو التيار المطلوب لإنتاج التدفق في قلب المحول

2. تيار الخسارة الأساسية i_H "وهو التيار المطلوب لتعويض التباطؤ وفقدان التيار الدوامي

يوضح الشكل 11-2 منحنى مغنطة لب محول نموذجي. إذا كان التدفق في قلب المحول معروفًا ، فيمكن العثور على حجم تيار المغنطة مباشرة من الشكل 11-2.

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Figure 2-11



I. The magnetization current in the transformer is not sinusoidal. The higher frequency components in the magnetization current are due to magnetic saturation in the transformer core.

2. Once the peak flux reaches the saturation point in the core, a small increase in peak flux requires a very large increase in the peak magnetization current.

3. The fundamental component of the magnetization current lags the voltage applied to the core by 90°.

4. the higher-frequency components in the magnetization current can be quite large compared to the fundamental component. In general,

the further a transformer core is driven into saturation, the larger the harmonic components will become

1- إن تيار المغنطة في المحول ليس جيبيًا. ترجع مكونات التردد العالي في تيار التمغنط إلى التشبع المغناطيسي في قلب المحول .

2- بمجرد وصول ذروة التدفق إلى نقطة التشبع في القلب ، تتطلب الزيادة الصغيرة في ذروة التدفق زيادة كبيرة جدًا في ذروة تيار المغنطة.

3- يتأخر المكون الأساسي لتيار المغنطة في الجهد المطبق على القلب بمقدار 90 درجة.

5- يمكن أن تكون مكونات التردد العالي في تيار المغنطة كبيرة جدًا مقارنة بالمكون الأساسي. بشكل عام ، كلما زاد تشبع قلب المحول ، زادت المكونات التوافقية.

المكون الآخر لتيار بدون حمل في المحول هو التيار المطلوب لتزويد الطاقة لتعويض التباطؤ وخسائر التيار الدوامة في القلب. هذا هو تيار الخسارة الأساسية. افترض أن التدفق في القلب هو جيبي. نظرًا لأن التيارات الدوامة في القلب تتناسب مع $\frac{di}{dt}$ ، فإن التيارات الدوامة تكون أكبر عندما يمر التدفق في القلب عبر 0 Wb. لذلك ، فإن تيار الخسارة الأساسية يكون متأخراً حيث يمر التدفق عبر الصفر. يظهر إجمالي التيار المطلوب لتعويض الخسائر الأساسية في الشكل 12-2

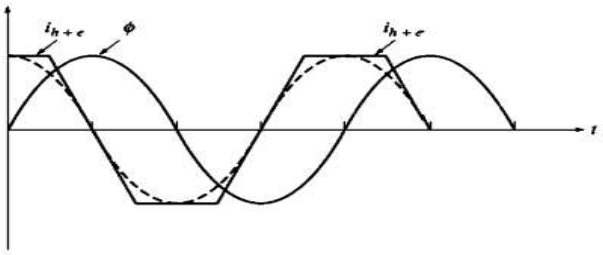


FIGURE 2–12
The core-loss current in a transformer.

Notice the following points about. the core-loss current:

I. The core-loss current is nonlinear because of the nonlinear effects of hysteresis.

2. The fundamental component of the core-loss current is in phase with the voltage applied to the core

1. إن تيار الخسارة الأساسية غير خطي بسبب التأثيرات غير الخطية للتباطؤ.

2. يكون المكون الأساسي لتيار فقد القلب في الطور مع الجهد المطبق على القلب

The total no-load current in the core is called the excitation current of the transformer. It is just the sum of the magnetization current and the core-loss current in the core:

$$i_{ex} = i_m + i_{h+e}$$

The total excitation current in a typical transformer core is shown in Figure 2-13.

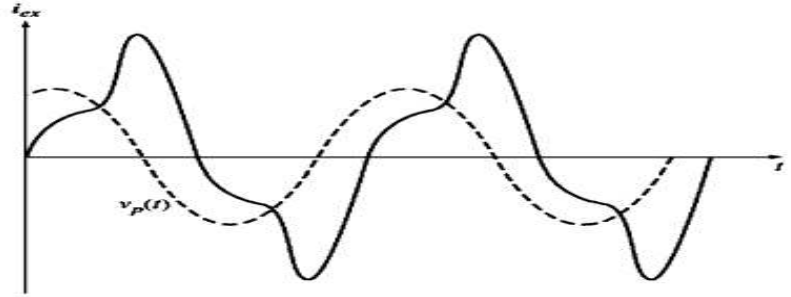


FIGURE 2–13
The total excitation current in a transformer.

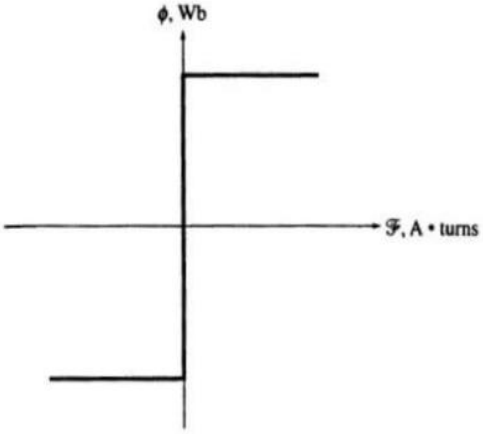
- **The Current Ratio on a Transformer and the Dot Convention**

In the situation shown in Figure 2–14, the primary current produces a positive magnetomotive force $\mathcal{F}_P = N_P i_P$, and the secondary current produces a negative magnetomotive force $\mathcal{F}_S = -N_S i_S$. Therefore, the net magnetomotive force

$$\mathcal{F}_{net} = N_P i_P - N_S i_S \approx 0$$

$$N_P i_P \approx N_S i_S$$

$$\frac{i_P}{i_S} \approx \frac{N_S}{N_P} = \frac{1}{a}$$



- **The assumptions from real to ideal transformer**

1. The core must have no hysteresis or eddy currents.

2. The magnetization curve must have the shape shown in Figure 2-15. Notice that for an unsaturated core the net magnetomotive force $F_{net} = 0$, implying that $N_p i_p = N_s i_s$.

3. The leakage flux in the core must be zero, implying that all the flux in the core couples both windings.

4. The resistance of the transformer windings must be zero.

- **The equivalent circuit of a transformer of model the non-ideal characteristics**

- **الدائرة المكافئة لمحول ذو الخصائص غير المثالية**

1. Copper (I^2R) losses. ***Copper losses are the resistive heating losses in the primary and secondary windings of the transformer. They are proportional to the square of the current in the windings.***

2. Eddy current losses. ***Eddy current losses are resistive heating losses in the core of the transformer. They are proportional to the square of the voltage applied to the transformer.***

3. Hysteresis losses. ***Hysteresis losses are associated with the rearrangement of the magnetic domains in the core during each half-cycle, as explained in Chapter 1. They are a complex, nonlinear function of the voltage applied to the transformer.***

4. Leakage flux. ***The fluxes ϕ_{lp} and ϕ_{ls} which escape the core and pass through only one of the transformer windings are leakage fluxes. These escaped fluxes produce a self-inductance in the primary and secondary coils, and the effects of this inductance must be accounted for.***

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1-خسائر النحاس .(I^2R) خسائر النحاس هي خسائر تنتج حرارة الملفات الأولية والثانوية للمحول. وتتناسب طرديا مع مربع التيار في الملفات.

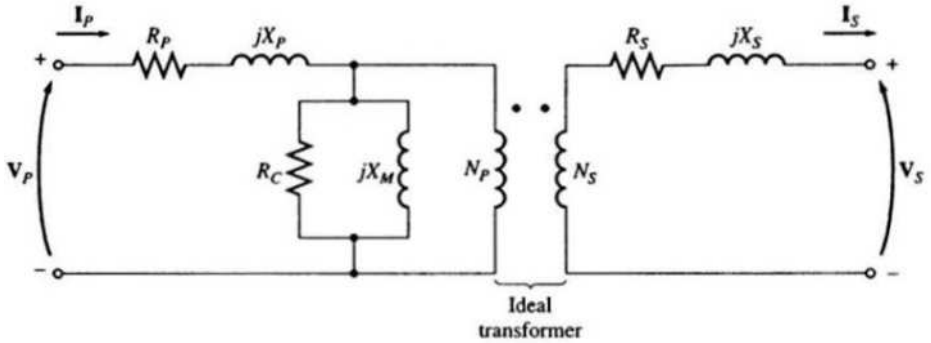
2- خسائر التيارات الدوامية. خسائر التيارات الدوامية هي خسائر تنتج حرارة في قلب المحول. وهي متناسبة مع مربع الجهد المطبق على المحول.

3- خسائر التخلفية. ترتبط خسائر التباطؤ بإعادة ترتيب المجالات المغناطيسية في القلب خلال كل نصف دورة . وهي دالة معقدة غير خطية للجهد المطبق على المحول.

4. تدفق التسرب. هي التي تخرج من القلب وتمر عبر ملف واحد فقط من لفات المحولات هي تدفقات تسرب. تنتج هذه التدفقات المتسربة تحريضًا ذاتيًا في الملفين الأولي والثانوي ، ويجب حساب تأثيرات هذا المحاثة.

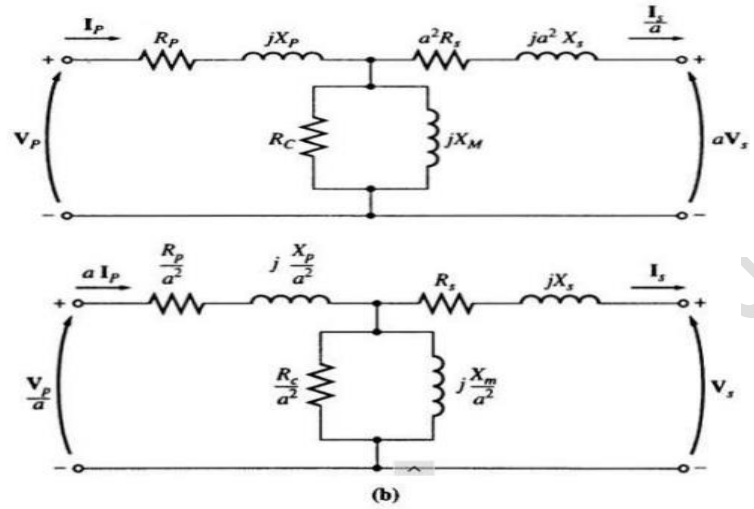
- **Modeling excitation current and copper loss**

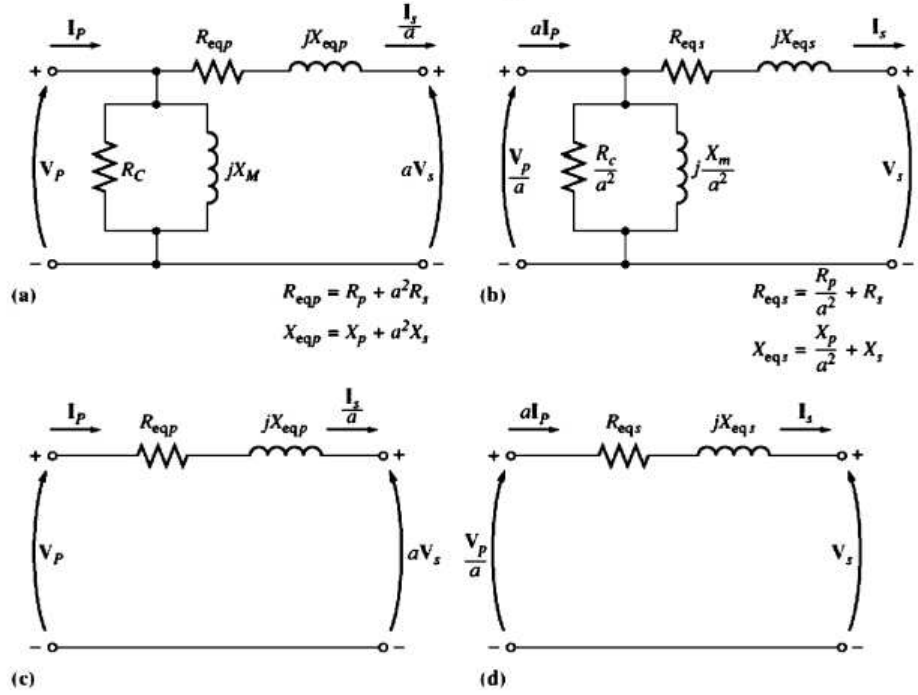
I. The hysteresis and eddy currents is in-phase with input voltage (modeled as a shunt resistor R_c)
II. The magnetization current (modeled as shunt inductor X_m)
III. The copper loss can be modeled as the Series Resistors R_p and R_s



يمكن حذف المحول المثالي بنقل المكونات الجانب الثانوي الى الجانب الابتدائي كما في الشكل a

كما يمكن حذف المحول المثالي بنقل مكونات الجانب الابتدائي الى الجانب الثانوي كما في الشكل b





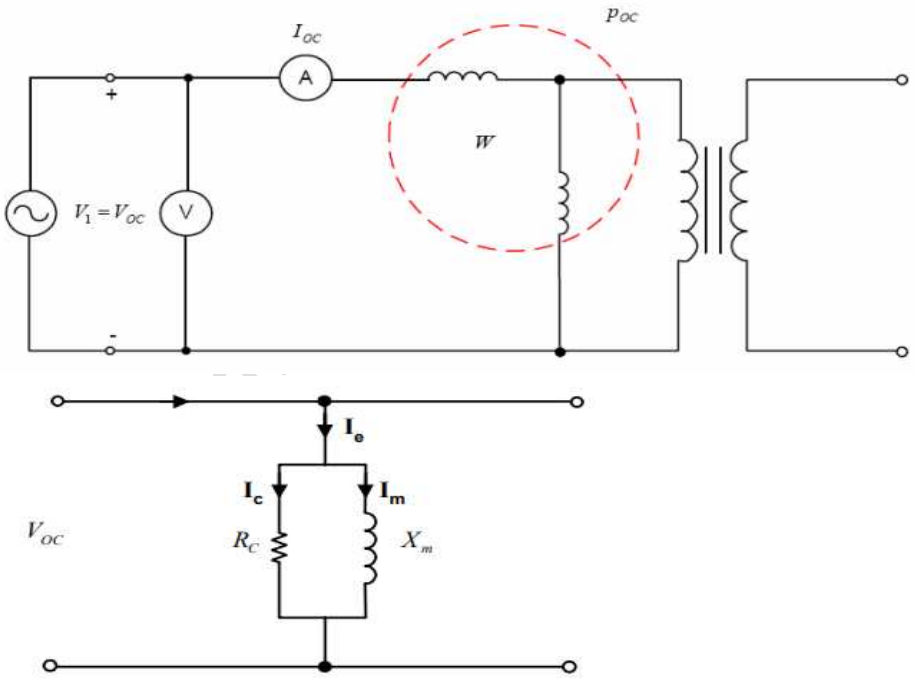
- **Measure the equivalent circuit parameters**

There are two types of measurements used for determination the equivalent circuit parameters

1- **Open circuit test** – used to measure excitation branch
2- **Short circuit test** – used to measure series branch

من الصعب معرفة جميع مكونات الدائرة المكافئة ما عدا لمن قام بتصميم هذه المحولات . عند ذلك فأنه يتوجب ايجاد طرق لمعرفة بعض مكونات الدائرة المكافئة لمعرفة خصائص المحول وحساب هذه المكونات . ويتم ذلك بإجراء اختبارين هما اختبار الدائرة المفتوحة o.c واختبار الدائرة المقصورة s.c

Open circuit test **اختبار الدائرة المفتوحة**



الغرض من هذا الاختبار هو حساب ادميتنس ، ومفاقيد اللاحمل ، وتيار الاستثارة عند اللاحمل ومعامل القدرة عند عدم وجود حمل ، ويتم هذا الاختبار بتوصيل جانب المحول الابتدائي لمصدر الجهد وترك الجانب الثانوي مفتوحاً أو العكس صحيح ، ويتم قياس التيار والقدرة المسحوبة من مصدر الجهد

حيث ان :

P_{oc} - قراءة جهاز قياس القدرة

V_{oc} - قراءة جهاز قياس الفولت

I - قراءة جهاز قياس الامبير

$$P_{oc} = V_{oc} . I_{oc} . \cos\varphi_{oc}$$

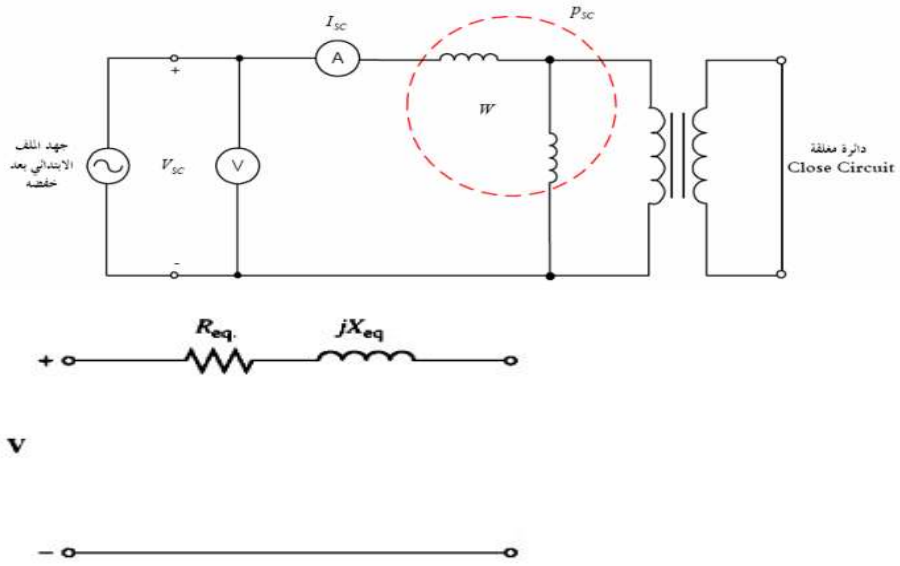
$$\cos\varphi_{oc} = \frac{P_{oc}}{V_{oc} . I_{oc}} = PF_{oc}$$

$$Y_{oc} = \frac{I_{oc}}{V_{oc}} \angle -\varphi_{oc} = \frac{1}{R_c} - j\frac{1}{X_m}$$

$$R_c = \frac{1}{G_{oc}} = \frac{V_{oc}^2}{P_{oc}}$$

$$X_m = \frac{1}{B_{oc}} = \frac{1}{\sqrt{Y_{oc}^2 - G_{oc}^2}}$$

اختبار الدائرة المقصورة Short circuit test



يتم هذا الاختبار بقصر أحدى طرفي الملف وعادة مايكون جانب الجهد المنخفض وتسليط جهد منخفض على جانب الجهد العالي وهو عبارة عن نسبة من الجهد المقنن للمحول على جانبيه الابتدائي ، وعادة ما يكون 2% الى 12% من الجهد المقنن للمحول . ويتم زيادة هذا الجهد حتى يصل التيار المار في الملفات المقصورة الى القيمة المقننة للمحول

مقنن(يعني الاسمي مثال: التيار المقنن للمحول يعني التيار الاسمي للمحول الذي يكون مكتوب على اللوحة الاسمية)

$$P_{sc} = V_{sc}.I_{sc}.\cos\varphi_{sc}$$

$$\cos\varphi_{sc} = \frac{p_{sc}}{V_{sc}.I_{sc}} = PF_{sc}$$

$$Z_{sc} = \frac{V_{sc}}{I_{sc}}\angle\varphi_{sc} = R_{eq} - jX_{eq}$$

ومن هنا يتبين لنا ما يلي

P_{sh} - قراءة جهاز قياس القدرة

V_{sh} - قراءة جهاز قياس الفولت

I_{sh} - قراءة جهاز قياس الاميتر

الحالة الاولى المنسوبة للابتدائي

$$R_{eq} = \frac{p_{sc}}{I_{SC}^2} \qquad R_1 = R_{eq} - R'_2 = \frac{R_{eq}}{2} = a^2R_2$$

$$x_{SC} = \sqrt{Z_{SC}^2 - R_{SC}^2} \qquad X_1 = X_{eq} - X'_2 = \frac{X_{eq}}{2} = a^2X_2$$

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$$R_{eq} = \frac{p_{sc}}{I_{SC}^2} \qquad R_2 = R_{eq} - R'_1 = \frac{R_{eq}}{2} = \frac{R_1}{a^2}$$

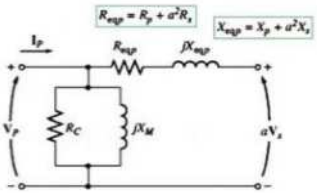
$$x_{SC} = \sqrt{Z_{SC}^2 - R_{SC}^2} \qquad X_2 = X_{eq} - X'_1 = \frac{X_{eq}}{2} = \frac{X_1}{a^2}$$

R_{eq} & X_{eq} can be calculated from series impedance $Z_{SE} = R_{eq} + jX_{eq}$

$$Z_{SE} = \frac{V_{SC}\angle 0^\circ}{I_{SC}\angle -\theta^\circ} = \frac{V_{SC}}{I_{SC}}\angle\theta^\circ$$

$$= \frac{V_{SC}}{I_{SC}}\angle \cos^{-1} \text{PF}$$

Open-circuit test (on primary)	Short-circuit test (on primary)
$V_{OC} = 8000$ V	$V_{SC} = 489$ V
$I_{OC} = 0.214$ A	$I_{SC} = 2.5$ A
$V_{OC} = 400$ W	$P_{SC} = 240$ W



The power factor during the *short-circuit* test is

$$\text{PF} = \cos\theta = \frac{P_{SC}}{V_{SC}I_{SC}}$$

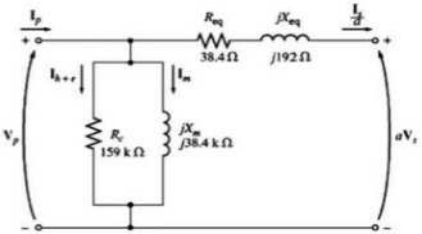
$$= \cos\theta = \frac{240 \text{ W}}{(489 \text{ V})(2.5 \text{ A})} = 0.196 \text{ lagging}$$

The series impedance is given by

$$Z_{SE} = \frac{V_{SC}}{I_{SC}}\angle \cos^{-1} \text{PF}$$

$$= \frac{489 \text{ V}}{2.5 \text{ A}}\angle 78.7^\circ$$

$$= 195.6\angle 78.7^\circ = 38.4 + j192\ \Omega$$



$R_{eq} = 38.4\ \Omega \qquad X_{eq} = 192\ \Omega$

$$= 0.0000063 - j0.0000261 = \frac{1}{R_C} - j\frac{1}{X_M} \qquad X_M = \frac{1}{0.0000261} = 38.4 \text{ k}\Omega$$

- **Per Unit system of Measurement**

Per-Unit System of Measurements. In the case of power system analysis used in electric, the per-unit system is the parameters of different terms that used as the base of the system. Due to this system, the measurements become easy. In such system where numerous transformer has employed the calculations of quantities become easy.

نظام القياسات لكل وحدة. في حالة تحليل نظام الطاقة المستخدم في الكهرباء ، فإن النظام لكل وحدة هو معايير المصطلحات المختلفة التي تستخدم كأساس للنظام. بفضل هذا النظام ، تصبح القياسات سهلة. في مثل هذا النظام حيث استخدم العديد من المحولات تصبح حسابات الكميات سهلة.

The basic configuration of a per-unit system is to get large differences in different parameters to get basic relations. So denotations of components in the network having per unit parameters are easy to handle. This system offered the unit denotation for voltage, current, power, and other electrical terms. Here we cover its different parameters related to the per-unit system

التكوين الأساسي لنظام كل وحدة هو الحصول على اختلافات كبيرة في المعايير المختلفة للحصول على العلاقات الأساسية. لذلك من السهل التعامل مع دلالات المكونات في الشبكة التي تحتوي على معاير لكل وحدة. قدم هذا النظام دلالة الوحدة للجهد والتيار والطاقة والمصطلحات الكهربائية الأخرى. نحن هنا نغطي معايير المختلفة المتعلقة بنظام كل وحدة.

Per Unit System

$$I_{\text{base1}} = \frac{S_{\text{base1}}}{V_{\text{base1}}}$$

$$S_{\text{base1}} = S_{\text{base2}}$$

$$V_{\text{base2}} = \frac{N_2}{N_1} V_{\text{base1}}$$

$$I_{\text{base2}} = \frac{S_{\text{base2}}}{V_{\text{base2}}}$$

$$I_{\text{base1}} = \frac{S_{\text{base2}}}{\frac{N_1}{N_2} V_{\text{base2}}}$$

$$= \frac{N_2}{N_1} I_{\text{base2}}$$

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Any quantity can be expressed on a per- unit basis by the equation

$$Quantity\ per\ unit = \frac{Actual\ value \text{ القيمة الفعلية}}{base\ value\ of\ quantity \text{ القيمة الاساسية للكمية}}$$

Example 2–3.

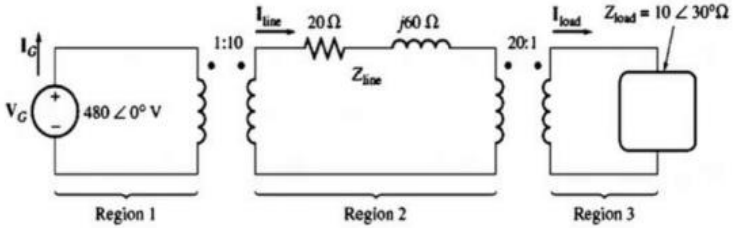
A simple power system is shown in Figure 2–22.

This system contains a 10 kVA , 480-V generator connected to an ideal 1:10 step-up transformer, a transmission line, an ideal 20:1 step-down transformer, and a load.

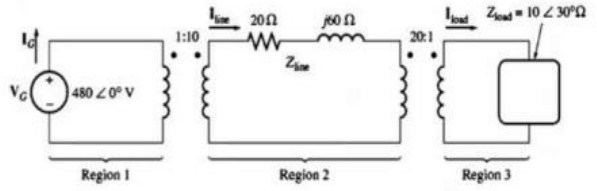
The impedance of the transmission line is $20 + j60\ \Omega$, and the impedance of the load is $10\angle 30°\Omega$.

The base values for this system are chosen to be 480 V and 10 kVA at the generator.

(a) Find the base voltage, current, impedance, and apparent power at every point in the power system.
(b) Convert this system to its per-unit equivalent circuit.
(c) Find the power supplied to the load in this system.
(d) Find the power lost in the transmission line.



(a) Find the base voltage, current, impedance, and apparent power at every point in the power system.



Solution:

(a)

In the generator region, V_{base} = 480 V and S_{base} = 10 kVA,

$$\text{so} \quad I_{base\,1} = \frac{S_{base}}{V_{base\,1}} = \frac{10{,}000\ \text{VA}}{480\ \text{V}} = 20.83\ \text{A}$$

$$Z_{base\,1} = \frac{V_{base\,1}}{I_{base\,1}} = \frac{480\ \text{V}}{20.83\ \text{A}} = 23.04\ \Omega$$

The turns ratio of transformer T_1 is a = 1/10 = 0.1,

in the transmission line region

the base voltage

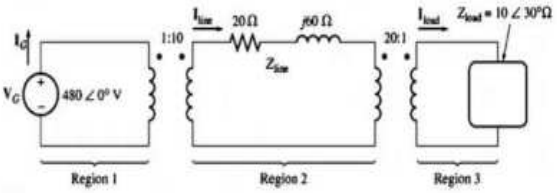
$$V_{base\,2} = \frac{V_{base\,1}}{a} = \frac{480\ \text{V}}{0.1} = 4800\ \text{V}$$

The other base quantities are

$S_{base\,2}$ = 10 kVA (remains constant through out)

$$I_{base\,2} = \frac{10{,}000\ \text{VA}}{4800\ \text{V}} = 2.083\ \text{A}$$

$$Z_{base\,2} = \frac{4800\ \text{V}}{2.083\ \text{A}} = 2304\ \Omega$$



The turns ratio of transformer T_2 is a = 20/1 = 20,

in the load region is

the base voltage $V_{base\,3} = \frac{V_{base\,2}}{a} = \frac{4800\ \text{V}}{20} = 240\ \text{V}$

The other base quantities are

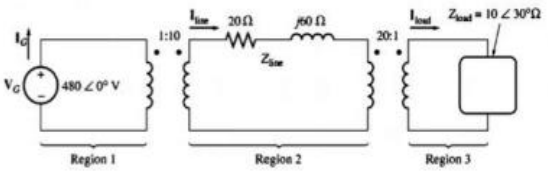
$S_{base\,3}$ = 10 kVA (remains constant through out)

$$I_{base\,3} = \frac{10{,}000\ \text{VA}}{240\ \text{V}} = 41.67\ \text{A}$$

$$Z_{base\,3} = \frac{240\ \text{V}}{41.67\ \text{A}} = 5.76\ \Omega$$

(b) Convert this system to its per-unit equivalent circuit.

generator region, $V_{base} = 480$ V
transmission line region $Z_{base\,2} = 2304\ \Omega$
load region $Z_{base\,3} = 5.76\ \Omega$



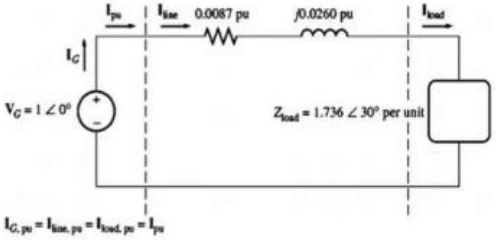
(b) To convert a power system to a per-unit system, each component must be divided by its base value in its region of the system.

The *generator's* per-unit volt- age is

$$V_{G,pu} = \frac{\text{actual value}}{\text{base value}} = \frac{480\angle 0°\ \text{V}}{480\ \text{V}} = 1.0\angle 0°\ \text{pu}$$

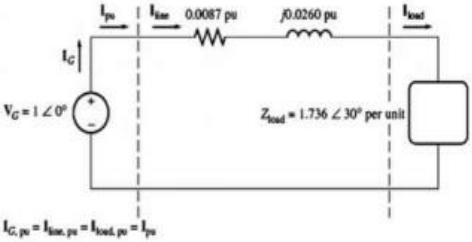
$$Z_{line} = \frac{\text{actual value}}{\text{base value}} = \frac{20 + j60\ \Omega}{2304\ \Omega} = 0.0087 + j0.0260\ \text{pu}$$

$$Z_{load,pu} = \frac{\text{actual value}}{\text{base value}} = \frac{10\angle 30°\ \Omega}{5.76\ \Omega} = 1.736\angle 30°\ \text{pu}$$



(c) Find the power supplied to the load in this system.

The per-unit power of the load is
$P_{load,pu} = I_{pu}^2 R_{pu}$



The current flowing in this per-unit power system is

$$I_{pu} = \frac{V_{pu}}{Z_{tot,pu}}$$

$$= \frac{1\angle 0°}{(0.0087 + j0.0260) + (1.736\angle 30°)}$$

$$= 0.569\angle -30.6°\ \text{pu}$$

Therefore, the per-unit power of the load is

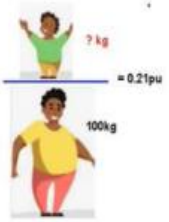
$$P_{load,pu} = I_{pu}^2 R_{pu} = (0.569)^2(1.503) = 0.487$$

$Z_{load,pu} = 1.736\angle 30°$ pu $= 1.5 + j0.868$

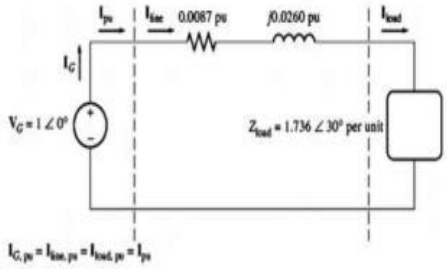
Therefore, the actual power supplied to the load is

$$P_{load} = P_{load,pu} S_{base} = (0.487)(10{,}000\ \text{VA})$$

$$= 4870\ \text{W}$$



(d) Find the power lost in the transmission line.



The per-unit power lost in the transmission line is

$$P_{line,pu} = I_{pu}^2 R_{line,pu} = (0.569)^2(0.0087) = 0.00282$$

the actual power lost in the transmission line is

$$P_{line} = P_{line,pu} S_{base} = (0.00282)(10{,}000 \text{ VA})$$
$$= 28.2 \text{ W}$$

Example 2–4.

Sketch the approximate per-unit equivalent circuit for the transformer in Example 2–2.

Use the transformer's ratings as the system base. Example 2–2. transformer of a 20-kVA, 8000/240-V, 60-Hz

Solution:

To convert to per-unit, the primary circuit base impedance must be found.

$$Z_{base\,1} = \frac{(V_{base\,1})^2}{S_{base\,1}} = \frac{(8000 \text{ V})^2}{20{,}000 \text{ VA}} = 3200\ \Omega$$

On the primary,
$V_{base\,1} = 8000$ V
$S_{base\,1} = 20{,}000$ VA

$S = VxI = Vx\frac{V}{Z} = \frac{V^2}{Z}$

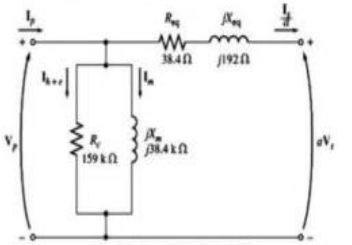
$Z = \frac{V^2}{S}$

Therefore,

$$Z_{SE,pu} = \frac{38.4 + j192\ \Omega}{3200\ \Omega} = 0.012 + j0.06 \text{ pu}$$

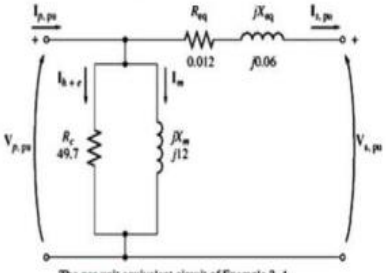
$$R_{C,pu} = \frac{159 \text{ k}\Omega}{3200\ \Omega} = 49.7 \text{ pu}$$

$$Z_{M,pu} = \frac{38.4 \text{ k}\Omega}{3200\ \Omega} = 12 \text{ pu}$$



referred to the primary side

(Final circuit of example 2.2)



The per-unit equivalent circuit of Example 2–4.

The per-unit approximate equivalent circuit, expressed to the transformer's own base.

- **Transformer Voltage Regulation and efficiency**

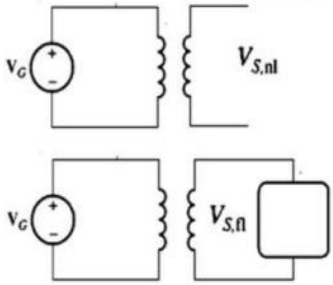
كفاءة المحول وا تنظيم الجهد





Voltage regulation (VR) is a quantity that compares the output voltage of the transformer at no load with the output voltage at full load.

$$\text{VR} = \frac{V_{S,\text{nl}} - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\%$$



Since at no load, $V_S = V_P/a$,
the voltage regulation can also be expressed as

$$\text{VR} = \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\%$$

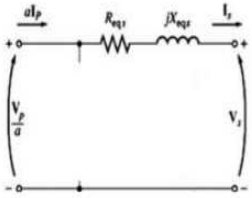
Usually it is a good practice to have as small a voltage regulation as possible.

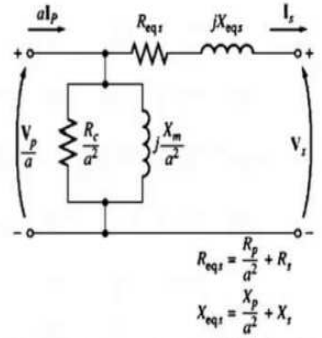
How can the voltage regulation of a transformer be determined? (Next Slide)

How can the voltage regulation of a transformer be determined?

To determine the voltage regulation of a transformer, it is necessary to understand the voltage drops within it.

The effects of the excitation branch on transformer voltage regulation can be ignored, so only the series impedances need be considered.





(b) Approximate transformer models, referred to the secondary side

The easiest way to determine the effect of the impedances and the current phase angles on the transformer voltage regulation is to examine a *phasor diagram*.

In all the following phasor diagrams, the phasor voltage $\mathbf{V}_S$ is assumed to be at an angle of 0°, and all other voltages and currents are compared to that reference.

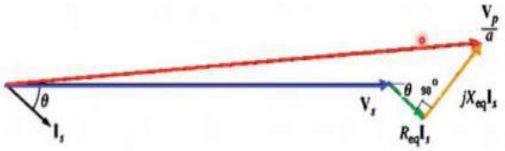
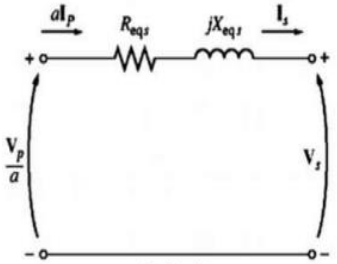


FIG 2-26 Phasor diagram of a transformer operating at a lagging power factor.

By applying Kirchhoff's voltage law to the equivalent circuit in Figure 2–18(d) the primary voltage can be found as

$$-\frac{\mathbf{V}_P}{a} + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S + \mathbf{V}_S = 0$$

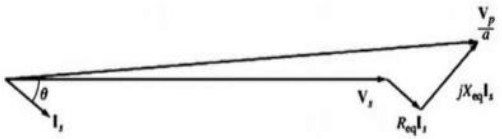
$$\frac{\mathbf{V}_P}{a} = \mathbf{V}_S + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S$$



2–18(d)

A transformer phasor diagram is just a visual representation of this equation.

$$VR = \frac{V_P/a - V_{S,fl}}{V_{S,fl}} \times 100\%$$



It is easy to see that $V_P/a > V_S$ for lagging loads,

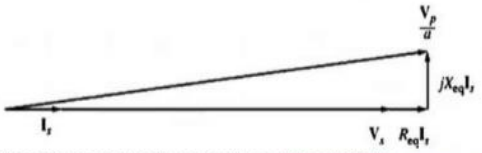
So, VR > 0.

FIGURE 2–26 Phasor diagram of a transformer operating at a lagging power factor.

At unity power factor

$V_P/a > V_S$

so VR > 0.



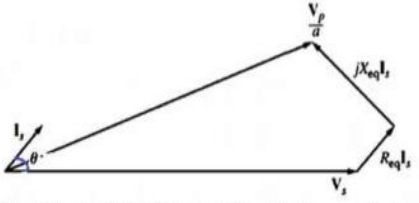
Phasor diagram of a transformer operating at unity power factor. (means θ = 0)

However, this time the voltage regulation is a smaller number than it was with a lagging current.

If the secondary current is leading, the secondary voltage can actually be *higher* than the referred primary voltage.

If this happens, the transformer actually has a *negative* voltage regulation.

Negative voltage regulation means the voltage increases with the load. undesirable condition

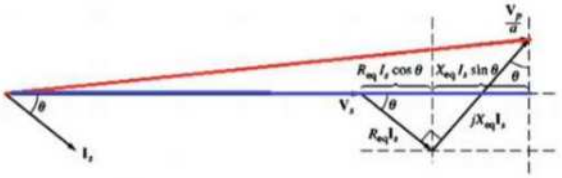


Phasor diagram of a transformer operating at leading power factor.

Derivation of the approximate equation for V_p/a.



$$\frac{V_p}{a} = V_s + R_{eq} I_s + jX_{eq} I_s$$



$$\frac{V_p}{a} \approx V_s + R_{eq} I_s \cos\theta + X_{eq} I_s \sin\theta$$

كفاءة المحول

كفاءة أي جهاز كهربائي كنسبة القدرة الخارجية الى القدرة الداخلة

$$\xi = \frac{P_{out}}{P_{in}} = \frac{P_{out}}{P_{out} + P_{loss}}$$

$$p_{in} = p_{out} + p_{core} + p_{cu}$$

$$p_{out} = V_2 I_2 \cos\phi$$

$$p_{cu} = I_1^2 V_1^2 + I_2^2 V_2^2$$

P_{out} - القدرة المعطاة من المحول

P_{in} - القدرة التي يستهلكها المحول

P_{loss} - القدرة المفقودة

P_{core} – الفواقد الحديدية

P_{cu} - الفواقد النحاسية

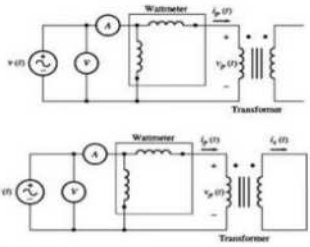
Example 2–5.

A 15-kVA, 2300/230-V transformer is to be tested to determine

(a) its excitation branch components,
(b) its series impedances,
(c) its voltage regulation.

The following test data have been taken from the primary side of the transformer:

Open-circuit test	Short-circuit test
$V_{OC} = 2300$ V	$V_{SC} = 47$ V
$I_{OC} = 0.21$ A	$I_{SC} = 6.0$ A
$P_{OC} = 50$ W	$P_{SC} = 160$ W



(a) Find the equivalent circuit of this transformer referred to the high-voltage side.
(b) Find the equivalent circuit of this transformer referred to the low-voltage side.
(c) Calculate the full-load voltage regulation at 0.8 lagging power factor, 1.0 power factor, and at 0.8 leading power factor.

(e) What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

Abdulsalam Awwad

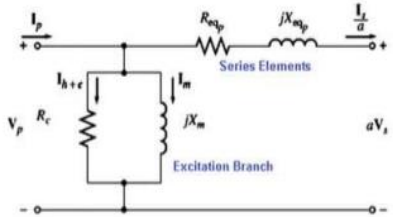
Solution:

(a) Find the equivalent circuit of this transformer referred to the high-voltage side.

2300/230-V transformer

(Primary side)

[this part is exactly same as example 2.2]



(a) The excitation branch values of the transformer equivalent circuit can be calculated from the *open-circuit test* data, and the series elements can be calculated from the *short-circuit test* data.

From the open-circuit test data, the open-circuit impedance angle is

$$\theta_{OC} = \cos^{-1}\frac{P_{OC}}{V_{OC}I_{OC}} \qquad [P = VI\cos\theta]$$

$$= \cos^{-1}\frac{50\text{W}}{(2300\text{ V})(0.21\text{ A})} = 84°$$

Open-circuit test
V_{OC} = 2300 V
I_{OC} = 0.21 A
P_{OC} = 50 W

The excitation admittance is thus

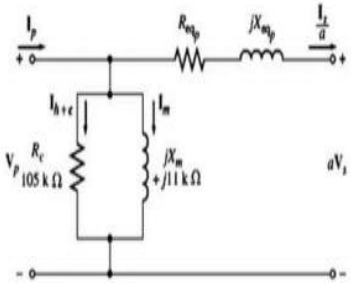
$$Y_E = \frac{I_{OC}}{V_{OC}}\angle -84° = \frac{0.21\text{ A}}{2300\text{ V}}\angle -84°$$

$$= 9.13\times 10^{-5}\angle -84°\,\Omega = 0.0000095 - j0.0000908\,\Omega = \frac{1}{R_C} - j\frac{1}{X_M}$$

The elements of the excitation branch referred to the primary are

$$R_C = \frac{1}{0.0000095} = 105\text{ k}\Omega$$

$$X_M = \frac{1}{0.0000908} = 11\text{ k}\Omega$$



From the short-circuit test data, the short-circuit impedance angle is

$$\theta_{SC} = \cos^{-1}\frac{P_{SC}}{V_{SC}I_{SC}}$$

$$= \cos^{-1}\frac{160\text{ W}}{(47\text{ V})(6\text{ A})} = 55.4°$$

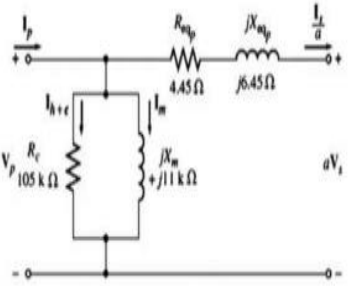
Short-circuit test
V_{SC} = 47 V
I_{SC} = 6.0 A
P_{SC} = 160 W

The equivalent series impedance is thus

$$Z_{SE} = \frac{V_{SC}}{I_{SC}}\angle\theta_{SC}$$

$$= \frac{47\text{ V}}{6\text{ A}}\angle 55.4°\,\Omega$$

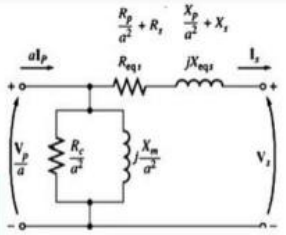
$$= 7.833\angle 55.4° = 4.45 + j6.45$$



The series elements referred to the primary are

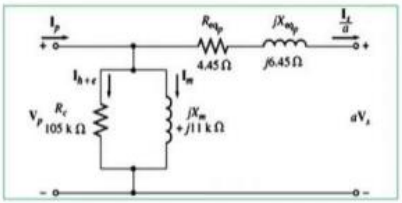
$$R_{eq} = 4.45\,\Omega \qquad X_{eq} = 6.45\,\Omega$$

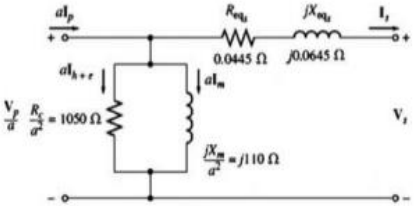
(b) Find the equivalent circuit of this transformer referred to the low-voltage side.



(b) To find the equivalent circuit referred to the low-voltage side, it is simply necessary to divide the impedance by a^2.

Since $a = N_P/N_S = 10$, the resulting values are



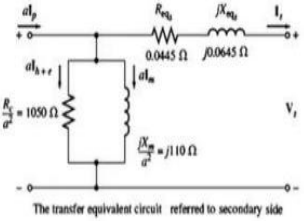


The transfer equivalent circuit referred to secondary side

(c) Calculate the full-load voltage regulation at
0.8 lagging power factor,
1.0 power factor, and at
0.8 leading power factor.

Since at no load, $V_S = V_P/a$, the voltage regulation can also be expressed as

$$\text{VR} = \frac{V_P/a - V_{S,fl}}{V_{S,fl}} \times 100\%$$



The transfer equivalent circuit referred to secondary side

A 15-kVA, 2300/230-V transformer

To calculate V_P/a, use Equation (2–64):

$$\frac{\mathbf{V}_P}{a} = \mathbf{V}_S + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S$$

current $\mathbf{I}_S$ = ?

The full-load current on the secondary side of this transformer is

$$I_{S,\text{rated}} = \frac{S_{\text{rated}}}{V_{S,\text{rated}}} = \frac{15{,}000\ \text{VA}}{230\ \text{V}} = 65.2\ \text{A}$$

At PF = 0.8 lagging,

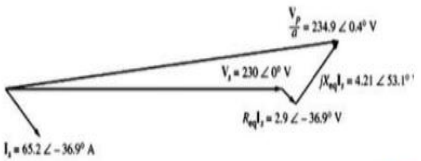
current $\mathbf{I}_S = 65.2\angle -36.9°$ A. $\angle -\cos^{-1}\text{PF}$

Therefore,

$$\frac{\mathbf{V}_P}{a} = 230\angle 0° \text{ V} + (0.0445\ \Omega)(65.2\angle -36.9° \text{ A}) + j(0.0645\ \Omega)(65.2\angle -36.9° \text{ A})$$
$$= 230\angle 0° \text{ V} + 2.90\angle -36.9° \text{ V} + 4.21\angle 53.1° \text{ V}$$
$$= 234.85\angle 0.40° \text{ V}$$

The resulting voltage regulation is

$$\text{VR} = \frac{V_P/a - V_{S,fl}}{V_{S,fl}} \times 100\% = \frac{234.85\ \text{V} - 230\ \text{V}}{230\ \text{V}} \times 100\% = 2.1\%$$



At PF = 1.0, current $\mathbf{I}_S = 65.2 \angle 0° \text{ A}$. Therefore,

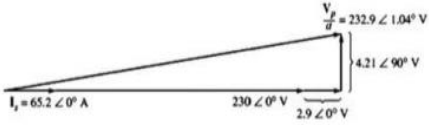
$$\frac{\mathbf{V}_P}{a} = 230 \angle 0° \text{ V} + (0.0445\ \Omega)(65.2 \angle 0° \text{ A}) + j(0.0645\ \Omega)(65.2 \angle 0° \text{ A})$$

$$= 230 \angle 0° \text{ V} + 2.90 \angle 0° \text{ V} + 4.21 \angle 90° \text{ V}$$

$$= 232.94 \angle 1.04° \text{ V}$$

To calculate V_P/a, use Equation (2–64):

$$\frac{\mathbf{V}_P}{a} = \mathbf{V}_S + R_{eq}\mathbf{I}_S + jX_{eq}\mathbf{I}_S$$



The resulting voltage regulation is

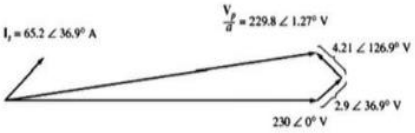
$$\text{VR} = \frac{V_P/a - V_{S,fl}}{V_{S,fl}} \times 100\% = \frac{232.94 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = 1.28\%$$

At PF = 0.8 leading, current $\mathbf{I}_S = 65.2 \angle 36.9° \text{ A}$. Therefore,

$$\frac{\mathbf{V}_P}{a} = 230 \angle 0° \text{ V} + (0.0445\ \Omega)(65.2 \angle 36.9° \text{ A}) + j(0.0645\ \Omega)(65.2 \angle 36.9° \text{ A})$$

$$= 230 \angle 0° \text{ V} + 2.90 \angle 36.9° \text{ V} + 4.21 \angle 126.9° \text{ V}$$

$$= 229.85 \angle 1.27° \text{ V}$$



The resulting voltage regulation is

$$\text{VR} = \frac{V_P/a - V_{S,fl}}{V_{S,fl}} \times 100\% = \frac{229.85 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = -0.062\%$$

(e) What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

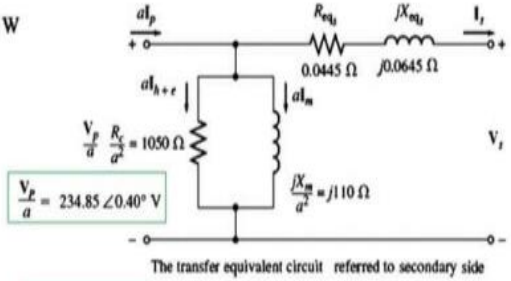
$$\eta = \frac{V_S I_S \cos\theta}{P_{Cu} + P_{core} + V_S I_S \cos\theta} \times 100\%$$

The copper losses are

$$P_{Cu} = (I_S)^2 R_{eq} = (65.2 \text{ A})^2(0.0445\ \Omega) = 189 \text{ W}$$

The core losses are given by

$$P_{core} = \frac{(V_P/a)^2}{R_C} = \frac{(234.85 \text{ V})^2}{1050\ \Omega} = 52.5 \text{ W}$$



The transfer equivalent circuit referred to secondary side

The output power of the transformer at this power factor is

$$P_{out} = V_S I_S \cos\theta$$

$$= (230 \text{ V})(65.2 \text{ A}) \cos 36.9° = 12{,}000 \text{ W}$$

$\mathbf{V}_S = 230 \angle 0° \text{ V}$
$\mathbf{I}_S = 65.2 \angle -36.9° \text{ A}$

Therefore, the efficiency of the transformer at this condition is

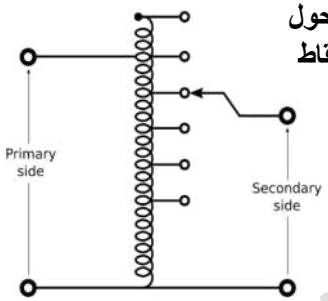
$$\eta = \frac{V_S I_S \cos\theta}{P_{Cu} + P_{core} + V_S I_S \cos\theta} \times 100\% = \frac{12{,}000 \text{ W}}{189 \text{ W} + 52.5 \text{ W} + 12{,}000 \text{ W}} \times 100\%$$

$$= 98.03\%$$

- **THE AUTOTRANSFORMER**

1- **Continuously tune the output voltage magnitude**
2- **The size of auto transformer is smaller than the size of conventional transformer**
3- **Output terminal is not electrical isolation**





المحول الذاتي أو المحول الإلي أو المحول التلقائي، هو محول كهربائي عادي ولكن ذو ملف واحد فقط. الملف له ثلاث نقاط توصيل كهربائية على الأقل. يتم توصيل كل من المصدر الكهربائي والحمل إلى نقطه توصيل خاصه به. بنهاية الملف توجد نقطه توصيل مشتركه بينهما (المصدر والحمل). كل نقطة توصيل يقابلها جهد كهربائي مختلف لكل من المصدر والحمل. يلاحظ ان جزء من نفس الملف يعمل كقاسم مشترك بين الملفات الابتدائية والثانوية.

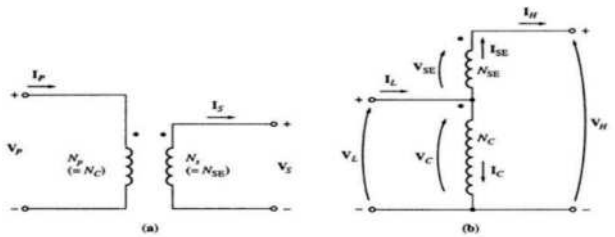
الفرق بين المحول الالي والمحول العادي

يمتاز المحول الالي بصغر الحجم وانها ذو كفاءة عالية (لان الضياعات الكهربائية اقل) ورخص ثمنه مقارنة بالمحول العادي

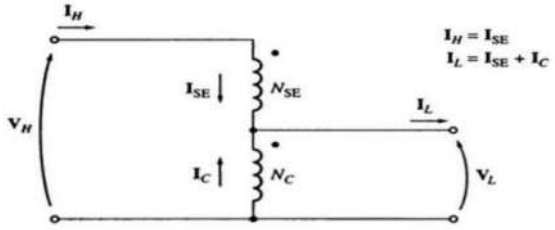
ولكن المحول الالي اكثر حساسية واشد خطورة في التعامل (لوجود اتصال كهربائي)

تصل كفاءة المحول الالي الى 97.7% بينما المحول العادي تصل الى 97%

The step up connection of autotransformer from conventional transformer



The step down connection of autotransformer from conventional transformer



Voltage and current in both coils

- **Coil voltage and current in Nc and Nse still follow the voltage and current relation**
- **The autotransformer terminal voltage and current**

$$\frac{\mathbf{V}_C}{\mathbf{V}_{SE}} = \frac{N_C}{N_{SE}}$$

$$N_C \mathbf{I}_C = N_{SE} \mathbf{I}_{SE}$$

$$\mathbf{V}_L = \mathbf{V}_C$$

$$\mathbf{V}_H = \mathbf{V}_C + \mathbf{V}_{SE}$$

$$\mathbf{I}_L = \mathbf{I}_C + \mathbf{I}_{SE}$$

$$\mathbf{I}_H = \mathbf{I}_{SE}$$

$$\mathbf{V}_H = \mathbf{V}_C + \mathbf{V}_{SE}$$

But $\mathbf{V}_C/\mathbf{V}_{SE} = N_C/N_{SE}$, so

$$\mathbf{V}_H = \mathbf{V}_C + \frac{N_{SE}}{N_C}\mathbf{V}_C$$

Finally, noting that $\mathbf{V}_L = \mathbf{V}_C$, we get

$$\mathbf{V}_H = \mathbf{V}_L + \frac{N_{SE}}{N_C}\mathbf{V}_L$$

$$= \frac{N_{SE} + N_C}{N_C}\mathbf{V}_L$$

or

$$\boxed{\frac{\mathbf{V}_L}{\mathbf{V}_H} = \frac{N_C}{N_{SE} + N_C}}$$

$$\mathbf{I}_L = \mathbf{I}_C + \mathbf{I}_{SE} \tag{2–72}$$

From Equation (2–69), $\mathbf{I}_C = (N_{SE}/N_C)\mathbf{I}_{SE}$, so

$$\mathbf{I}_L = \frac{N_{SE}}{N_C}\mathbf{I}_{SE} + \mathbf{I}_{SE} \tag{2–77}$$

Finally, noting that $\mathbf{I}_H = \mathbf{I}_{SE}$, we find

$$\mathbf{I}_L = \frac{N_{SE}}{N_C}\mathbf{I}_H + \mathbf{I}_H$$

$$= \frac{N_{SE} + N_C}{N_C}\mathbf{I}_H \tag{2–78}$$

or

$$\boxed{\frac{\mathbf{I}_L}{\mathbf{I}_H} = \frac{N_{SE} + N_C}{N_C}} \tag{2–79}$$

- **Apparatus power rating advantage in autotransformer**

- There are two types of rating
  - Power rating on terminals Sio
  - Power rating on windings Sw
- Terminals power rating Sio

$$S_{in} = V_L I_L \quad S_{out} = V_H I_H \qquad S_{in} = S_{out} = S_{IO}$$

- Winding power rating Sw

$$S_W = V_C I_C = V_{SE} I_{SE}$$

- Relation

$$\begin{aligned} S_W &= V_C I_C \\ &= V_L(I_L - I_H) \\ &= V_L I_L - V_L I_H \end{aligned}$$

$$\begin{aligned} S_W &= V_L I_L - V_L I_L \frac{N_C}{N_{SE} + N_C} \\ &= V_L I_L \frac{(N_{SE} + N_C) - N_C}{N_{SE} + N_C} \\ &= S_{IO} \frac{N_{SE}}{N_{SE} + N_C} \end{aligned}$$

$$\boxed{\frac{S_{IO}}{S_W} = \frac{N_{SE} + N_C}{N_{SE}}}$$

Example 2–7. A 100-VA, 120/12–V transformer is to be connected so as to form a step-up autotransformer (see Figure 2–33). A primary voltage of 120 V is applied to the transformer.

(a) What is the secondary voltage of the transformer?

(b) What is its maximum voltampere rating in this mode of operation?

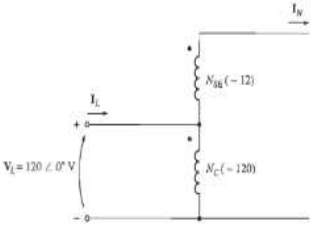
(c) Calculate the rating advantage of this autotransformer connection over the transformer's rating in conventional 120/12–V operation.

To accomplish a step-up transformation with a 120-V primary, the ratio of the turns on the common winding N_C to the turns on the series winding N_{SE} in this transformer must be 120:12 (or 10:1).

(a) This transformer is being used as a step-up transformer. The secondary voltage is V_H, and from Equation (2–76),

$$\mathbf{V}_H = \frac{N_{SE} + N_C}{N_C}\mathbf{V}_L \qquad (2\text{–}76)$$

$$= \frac{12 + 120}{120}\,120\text{ V} = 132\text{ V}$$



(b) The maximum voltampere rating in either winding of this transformer is 100 VA. How much input or output apparent power can this provide? To find out, examine the series winding. The voltage V_{SE} on the winding is 12 V, and the voltampere rating of the winding is 100 VA. Therefore, the *maximum* series winding current is

$$I_{SE,max} = \frac{S_{max}}{V_{SE}} = \frac{100\text{ VA}}{12\text{V}} = 8.33\text{ A}$$

Since I_{SE} is equal to the secondary current I_S (or I_H) and since the secondary voltage $V_S = V_H = 132$ V, the secondary apparent power is

$$S_{out} = V_S I_S = V_H I_H$$

$$= (132\text{ V})(8.33\text{ A}) = 1100\text{ VA} = S_{in}$$

(c) The rating advantage can be calculated from part *(b)* or separately from Equation (2–87). From part *(b)*,

$$\frac{S_{IO}}{S_W} = \frac{1100\text{ VA}}{100\text{ VA}} = 11$$

From Equation (2–87),

$$\frac{S_{IO}}{S_W} = \frac{N_{SE} + N_C}{N_{SE}} \qquad (2\text{–}87)$$

$$= \frac{12 + 120}{12} = \frac{132}{12} = 11$$

Example 2–8. A transformer is rated at 1000 kVA, 12/1.2 kV, 60 Hz when it is operated as a conventional two-winding transformer. Under these conditions, its series resistance and reactance are given as 1 and 8 percent per unit, respectively. This transformer is to be used as a 13.2/12–kV step-down autotransformer in a power distribution system. In the autotransformer connection, (*a*) what is the transformer's rating when used in this manner and (*b*) what is the transformer's series impedance in per-unit?

Solution

(*a*) The N_C/N_{SE} turns ratio must be 12:1.2 or 10:1. The voltage rating of this transformer will be 13.2/12 kV, and the apparent power (voltampere) rating will be

$$S_{IO} = \frac{N_{SE} + N_C}{N_{SE}} S_W$$

$$= \frac{1 + 10}{1} 1000\,\text{kVA} = 11{,}000\,\text{kVA}$$

(*b*) The transformer's impedance in a per-unit system when connected in the conventional manner is

$$Z_{eq} = 0.01 + j0.08\ \text{pu} \qquad \text{separate windings}$$

The apparent power advantage of this autotransformer is 11, so the per-unit impedance of the autotransformer connected as described is

$$Z_{eq} = \frac{0.01 + j0.08}{11}$$

$$= 0.00091 + j0.00727\ \text{pu} \qquad \text{autotransformer}$$

- **Autotransformer summarize**

• When two voltages are fairly close to each other

• The power advantage is very large

• There is a direct physical connection between primary and secondary sides.

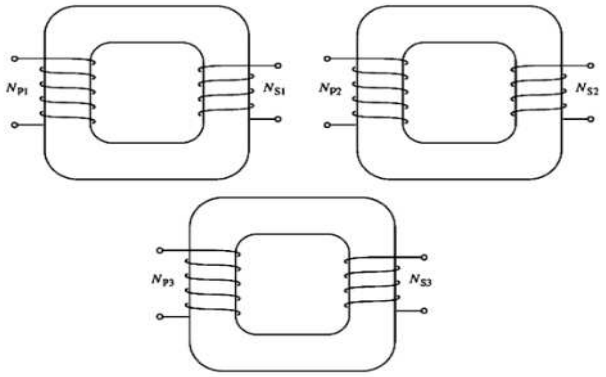
• The autotransformer is a convenient and inexpensive way to tie nearly equal two voltages together

• The electrical isolation of two sides is lost

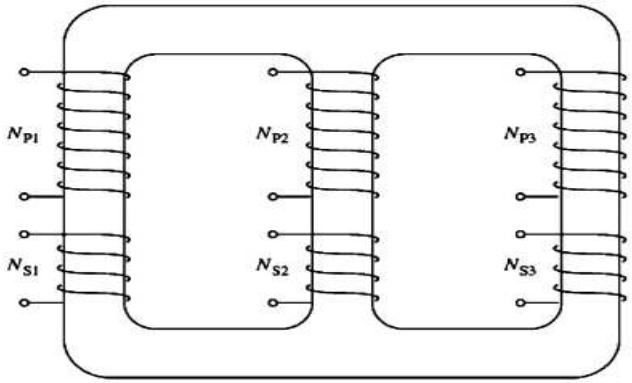
Rim Abdus

في انظمة القدرة ثلاثية الطور من الضروري رفع وخفض الجهدقي مواضع مختلفة من النظام ، وهذا يتم أما عن طريق محولات ذات طور واحد متطابقة (اي لها نفس الموصفات) بحيث يتم توصيل محول واحد لكل طور ، أو اما عن طريق استخدام محول ثلاثي الطور وفي كل حالة فأن لكل طور له ملف ابتدائي وملف ثانوي ، وتلك الملفات اما يتم توصيلها دلتا Δ أو نجمة Y

- **Three-phase transformer – three single – phase transformer banks**

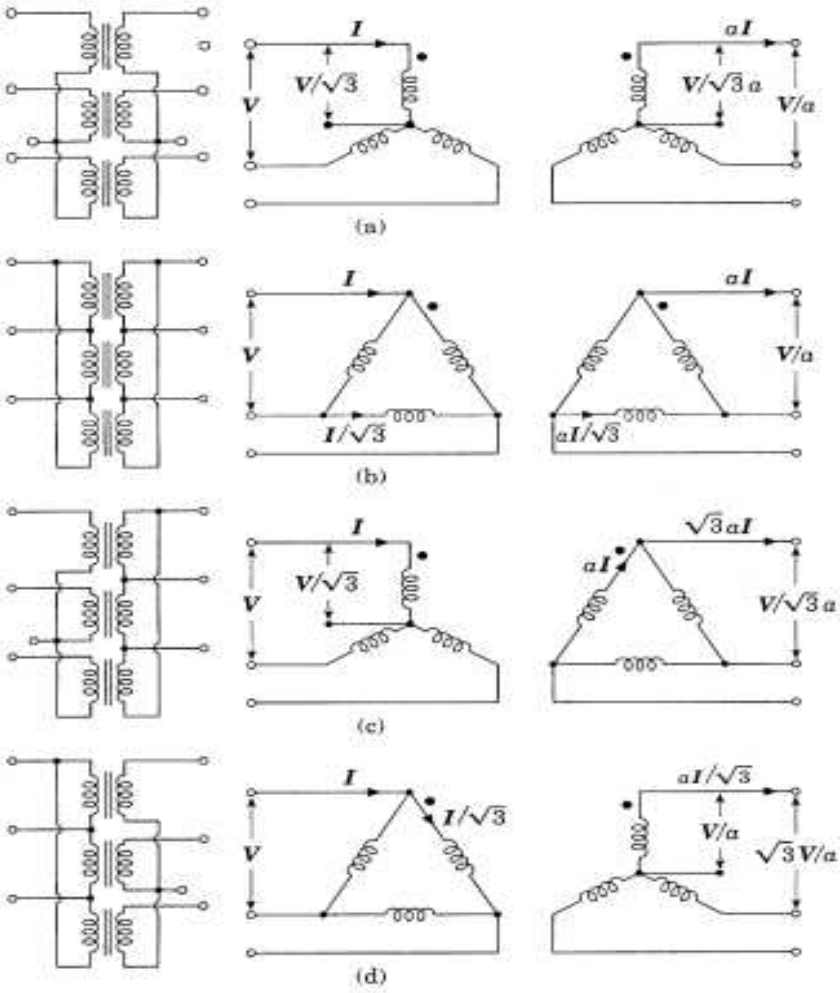


- **Three-phase transformer – one three lags – phase transformer banks**

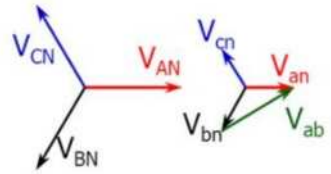


- **Three types of terminal connection**

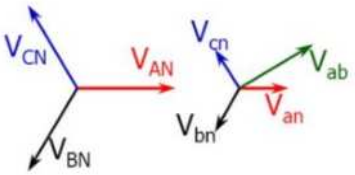
1. Wye–wye (Y–Y)
2. Wye–delta (Y–Δ)
3. Delta–wye (Δ–Y)
4. Delta–delta (Δ–Δ)



(wye-wye connection)



$$V_{ab} = V_{an} - V_{bn} \qquad V_{AN} = aV_{an}$$
$$V_{bc} = V_{bn} - V_{cn} \qquad V_{AB} = aV_{ab}$$
$$V_{ca} = V_{cn} - V_{an}$$



$$V_{ab} = V_{an} - V_{bn}$$
$$V_{bc} = V_{bn} - V_{cn}$$
$$V_{ca} = V_{cn} - V_{an}$$

Solving the unbalance and 3rd harmonic problems in wye-wye connection حل مشاكل عدم الاتزان والتوافقية الثالثة في اتصال ستار ستار

1. Solid ground the neutral of transformers: solve the unbalance problem and support a return path to the 3rd harmonic component.

2. Add a Δ-connected third winding. Since 3rd harmonic components are in-phase in each branch of Δ-connection, 3rd harmonic components will be limited in Δ-connection as the circulating current.

في توصيلة نجمة – نجمة للمحول يتم عمل توصيلة Y عند كل من الجانبين الأولي والثانوي لكل محول . وتيار الاستثارة excitation current لكل محول يحتوي على توافقية الثالثة بالاضافة الى الموجة الاساسية ، وهذه التوافقية الثالثة لها نفس الوجه لكل زاوية وجهية ، وبالتالي فليست هناك دائرة لجريان التوفيقات الثلاث في توصيلة نجمة – نجمة ، ولذا يتشوه الشكل الموجي للجهد وذلك يتسبب في حدوث اضطراب ضجيجي في خطوط الاتصالت المجاورة ، ولهذا السبب فأن توصيل نجمة – نجمة يتسخدم في الحالات الخاصة بالرغم من ان له بعض المزايا مثل سهولة طريقة العزل للمحولات مقارنة بالطرق الاخرى

(Wye – delta connection)

Wye-delta connection summarize

1. No 3rd harmonic component problem:

2. There are phase difference between each sides

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3. In United state, the secondary voltage will lag the primary voltage 30 degrees

توصيلة نجمة – دلتا فيها يتم توصيل نجمة عند الجانب الابتدائي وتوصيل دلتا عند الجانب الثانوي ويكون الجهد الطور V_φعلى الجانب الثانوي مساويا لجهد الخط V_L يستخدم هذا التوصيل بشكل أساسي في محولات الخفض في المحطات الفرعية في نهاية خط النقل. يمكن تأريض نقطة المحايد المتاحة في الجانب الأولي لتجنب التشويه. يمكنه التعامل مع الأحمال الكبيرة غير المتوازنة بشكل جيد ، سيتأخر الجهد الثانوي عن الجهد الأساسي بمقدار 30 درجة

(Delta – wye connection)

• Common connection:

Used on three-wire (delta) to four wire (wye)

Used to isolate ground on wye side from source ground on delta side

توصيلة دلتا – نجمة لها تقريبا نفس مميزات توصيلة نجمة – دلتا ويتخلف جهد الملف الثانوي عن الابتدائي بزاوية 30 هذه المحولات شائعة جدا في المواقع السكنية، التجارية، الصناعية والأماكن عالية الكثافة السكانية لتوفير الطاقة الكهربائية ثلاثية الطور. لكنها غير شائعة في جهود النقل العالية (138 كيلوفولت أو أكثر) بسبب ارتفاع تكلفة العزل بالمقارنة مع توصيلة ستار وتستخدم هذه التوصيلة رفع الجهد

(Delta-Delta Connection)

توصيلة دلتا – دلتا تستخدم بشكل عام للمحولات الكبيرة ذات الجهد المنخفض. وتكون ملفات الجانب الأولي والثانوي متصلة بشكل دلتا. ملائم للأحمال المتزنة والغير متزنة. في حالة وجود التوافقيات الثالثة، فإنها تدور في مسار مغلق وبالتالي لا تظهر في موجة جهد الخرج

TABLE 1: Voltage and Current Ratings of Transformers

Transformer Connection (Primary to Secondary)	Primary				Secondary			
	Line		Phase		Line		Phase	
	Volt.	Current	Volt.	Current	Volt. *	Current	Volt.	Current
Δ-Δ	V	I	V	$\frac{I}{\sqrt{3}}$	$\frac{V}{a}$	aI	$\frac{V}{a}$	$\frac{aI}{\sqrt{3}}$
Y-Y	V	I	$\frac{V}{\sqrt{3}}$	I	$\frac{V}{a}$	aI	$\frac{V}{\sqrt{3}a}$	aI
Y-Δ	V	I	$\frac{V}{\sqrt{3}}$	I	$\frac{V}{\sqrt{3}a}$	$\sqrt{3}aI$	$\frac{V}{\sqrt{3}a}$	aI
Δ-Y	V	I	V	$\frac{I}{\sqrt{3}}$	$\frac{\sqrt{3}V}{a}$	$\frac{aI}{\sqrt{3}}$	$\frac{V}{a}$	$\frac{aI}{\sqrt{3}}$

- **THREE-PHASE TRANSFORMATION USING TWO TRANSFORMERS**

• All techniques that create three-phase power with only two transformers involve a reduction in the power-handling capability of the transformers, but they may be justified by certain economic situations

تتضمن جميع التقنيات التي تنتج طاقة ثلاثية الطور باستخدام محولين فقط انخفاضًا في قدرة المحولات على معالجة الطاقة ، ولكن قد يتم تبريرها من خلال مواقف اقتصادية معينة

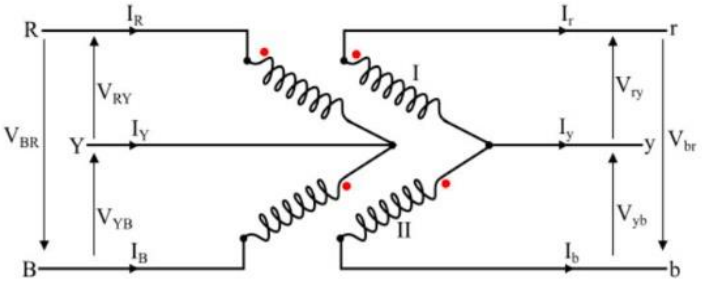
1. The open-Δ (or V–V) connection
2. The open-Y–open-Δ connection
3. The Scott-T connection
4. The three-phase T connection

Open Delta Connection or V-V Connection of Transformers

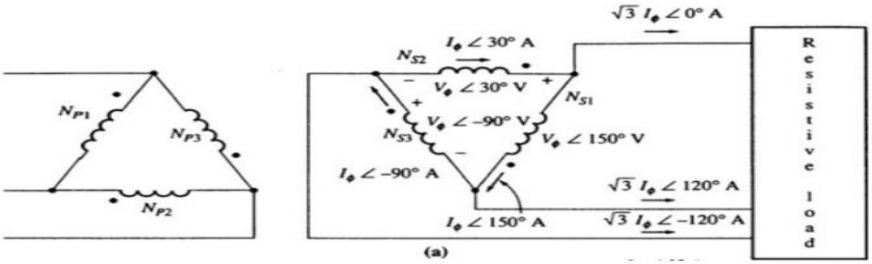
• If this damaged transformer is isolated, the remaining two transformers will function as a 3-phase bank with rating reduced to about 58 % that of the original delta-delta bank. This type of arrangement is known as open-delta or V-V connection

• إذا تم عزل هذا المحول التالف ، فإن المحولين المتبقيين سيعملان كبنك ثلاثي الطور مع تخفيض التصنيف إلى حوالي 58٪ من بنك دلتا دلتا الأصلي. يُعرف هذا النوع من الترتيب باتصال دلتا مفتوح أو اتصال V-V

$$V_{RY} = V_{LP}\angle 0^\circ$$
$$V_{YB} = V_{LP}\angle -120^\circ$$
$$V_{BR} = V_{LP}\angle +120^\circ$$

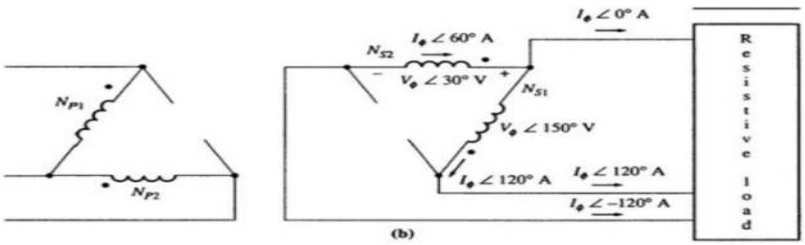


power rating of Δ-Δ connection



$$P = 3V_\phi I_\phi \cos\theta$$
$$= 3V_\phi I_\phi$$

power rating of V-V connection



$$P_1 = 3V_\phi I_\phi \cos(150° - 120°)$$
$$= 3V_\phi I_\phi \cos 30°$$
$$= \frac{\sqrt{3}}{2} V_\phi I_\phi$$

$$P_2 = 3V_\phi I_\phi \cos(30° - 60°)$$
$$= 3V_\phi I_\phi \cos(-30°)$$
$$= \frac{\sqrt{3}}{2} V_\phi I_\phi$$

$$P = \sqrt{3} V_\phi I_\phi$$

comparison of power rating

1. The power rating comparison

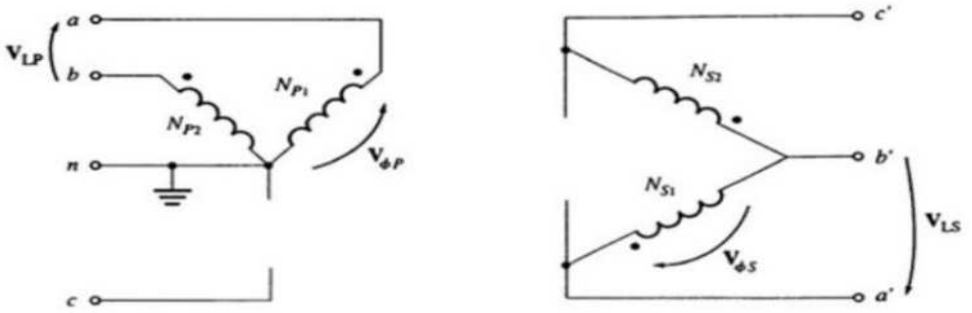
$$\frac{P_{\text{open }\Delta}}{P_{3\text{ phase}}} = \frac{\sqrt{3}V_\phi I_\phi}{3V_\phi I_\phi} = \frac{1}{\sqrt{3}} = 0.577$$

2. Where is the power rating ?
   - The existing reactive power will consume the power rating

$$Q_1 = 3V_\phi I_\phi \sin(150° - 120°)$$
$$= 3V_\phi I_\phi \sin 30°$$
$$= \tfrac{1}{2} V_\phi I_\phi$$

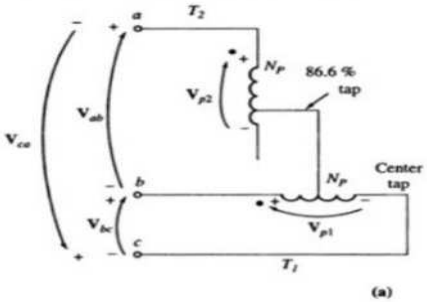
$$Q_2 = 3V_\phi I_\phi \sin(30° - 60°)$$
$$= 3V_\phi I_\phi \sin(-30°)$$
$$= -\tfrac{1}{2} V_\phi I_\phi$$

open Y- open Δ Connection



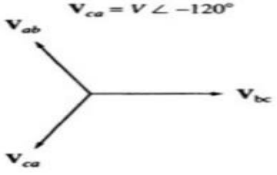
Scott-T connection – railroad applications

1. While in railroad applications, there always need two-phase power system for supporting northbound and southbound rails respectively.

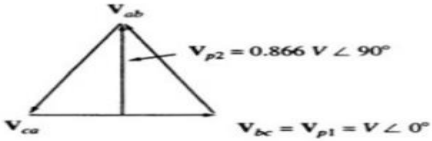


(a)

$V_{ab} = V \angle 120°$
$V_{bc} = V \angle 0°$
$V_{ca} = V \angle -120°$



(b)



(c)

$V_{S2} = \frac{V}{a} \angle 90°$ $\quad a = \frac{N_P}{N_S}$

$V_{S1} = \frac{V}{a} \angle 0°$

(d)

2-3. A 1000-VA 230/115-V transformer has been tested to determine its equivalent circuit. The results of the tests are shown below.

Open-circuit test	Short-circuit test
$V_{OC} = 230$ V	$V_{SC} = 19.1$ V
$I_{OC} = 0.45$ A	$I_{SC} = 8.7$ A
$P_{OC} = 30$ W	$P_{SC} = 42.3$ W

All data given were taken from the primary side of the transformer.

(a) Find the equivalent circuit of this transformer referred to the low-voltage side of the transformer.

(b) Find the transformer's voltage regulation at rated conditions and (1) 0.8 PF lagging, (2) 1.0 PF, (3) 0.8 PF leading.

(c) Determine the transformer's efficiency at rated conditions and 0.8 PF lagging.

SOLUTION

(a) OPEN CIRCUIT TEST:

$$|Y_{EX}| = |G_C - jB_M| = \frac{0.45\text{ A}}{230\text{ V}} = 0.001957$$

$$\theta = \cos^{-1}\frac{P_{OC}}{V_{OC}I_{OC}} = \cos^{-1}\frac{30\text{ W}}{(230\text{ V})(0.45\text{ A})} = 73.15°$$

$$Y_{EX} = G_C - jB_M = 0.001957\angle -73.15° \text{ mho} = 0.000567 - j0.001873 \text{ mho}$$

$$R_C = \frac{1}{G_C} = 1763\,\Omega$$

$$X_M = \frac{1}{B_M} = 534\,\Omega$$

SHORT CIRCUIT TEST:

$$|Z_{EQ}| = |R_{EQ} + jX_{EQ}| = \frac{19.1\text{ V}}{8.7\text{ A}} = 2.2\,\Omega$$

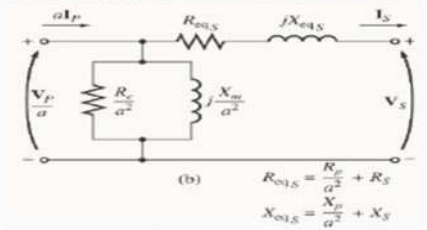
$$\theta = \cos^{-1}\frac{P_{SC}}{V_{SC}I_{SC}} = \cos^{-1}\frac{42.3\text{ W}}{(19.1\text{ V})(8.7\text{ A})} = 75.3°$$

$$Z_{EQ} = R_{EQ} + jX_{EQ} = 2.20\angle 75.3°\,\Omega = 0.558 + j2.128\,\Omega$$

$$R_{EQ} = 0.558\,\Omega$$

$$X_{EQ} = j2.128\,\Omega$$

To convert the equivalent circuit to the secondary side, divide each impedance by the square of the turns ratio ($a = 230/115 = 2$). The resulting equivalent circuit is shown below:



$R_{EQ,S} = 0.140\,\Omega$ $\quad X_{EQ,S} = j0.532\,\Omega$

$R_{C,S} = 441\,\Omega$ $\quad X_{M,S} = 134\,\Omega$

(b) To find the required voltage regulation, we will use the equivalent circuit of the transformer referred to the secondary side. The rated secondary current is

$$I_S = \frac{1000\text{ VA}}{115\text{ V}} = 8.70\text{ A}$$

We will now calculate the primary voltage referred to the secondary side and use the voltage regulation equation for each power factor.

(1) **0.8 PF Lagging:**

$$\mathbf{V}_P' = \mathbf{V}_S + Z_{EQ}\mathbf{I}_S = 115\angle 0^\circ\text{ V} + (0.140 + j0.532\,\Omega)(8.7\angle -36.87^\circ\text{ A})$$

$$\mathbf{V}_P' = 118.8\angle 1.4^\circ\text{ V}$$

$$\text{VR} = \frac{118.8\text{-}115}{115} \times 100\% = 3.3\%$$

(2) **1.0 PF:**

$$\mathbf{V}_P' = \mathbf{V}_S + Z_{EQ}\mathbf{I}_S = 115\angle 0^\circ\text{ V} + (0.140 + j0.532\,\Omega)(8.7\angle 0^\circ\text{ A})$$

$$\mathbf{V}_P' = 116.3\angle 2.28^\circ\text{ V}$$

$$\text{VR} = \frac{116.3\text{-}115}{115} \times 100\% = 1.1\%$$

(3) **0.8 PF Leading:**

$$\mathbf{V}_P' = \mathbf{V}_S + Z_{EQ}\mathbf{I}_S = 115\angle 0^\circ\text{ V} + (0.140 + j0.532\,\Omega)(8.7\angle 36.87^\circ\text{ A})$$

$$\mathbf{V}_P' = 113.3\angle 2.24^\circ\text{ V}$$

$$\text{VR} = \frac{113.3\text{-}115}{115} \times 100\% = -1.5\%$$

(c) At rated conditions and 0.8 PF lagging, the output power of this transformer is

$$P_{OUT} = V_S I_S \cos\theta = (115\text{ V})(8.7\text{ A})(0.8) = 800\text{ W}$$

The copper and core losses of this transformer are

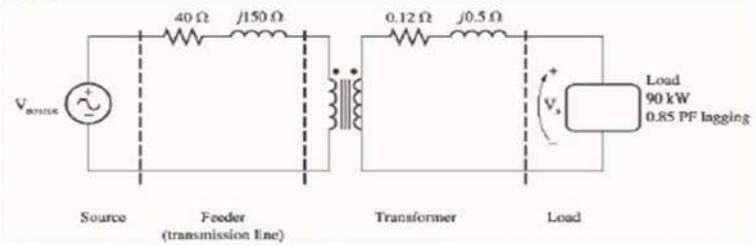
$$P_{CU} = I_S^2\, R_{EQ,S} = (8.7\text{ A})^2\,(0.140\,\Omega) = 10.6\text{ W}$$

$$P_{core} = \frac{\left(V_P'\right)^2}{R_C} = \frac{(118.8\text{ V})^2}{441\,\Omega} = 32.0\text{ W}$$

Therefore the efficiency of this transformer at these conditions is

$$\eta = \frac{P_{OUT}}{P_{OUT} + P_{CU} + P_{core}} \times 100\% = \frac{800\text{ W}}{800\text{ W} + 10.6\text{ W} + 32.0\text{ W}} = 94.9\%$$

2-4. A single-phase power system is shown in Figure P2-1. The power source feeds a 100-kVA 14/2.4-kV transformer through a feeder impedance of 40.0 + *j*150 Ω. The transformer's equivalent series impedance referred to its low-voltage side is 0.12 + *j*0.5 Ω. The load on the transformer is 90 kW at 0.80 PF lagging and 2300 V.



(a) What is the voltage at the power source of the system?

(b) What is the voltage regulation of the transformer?

(c) How efficient is the overall power system?

SOLUTION

To solve this problem, we will refer the circuit to the secondary (low-voltage) side. The feeder's impedance referred to the secondary side is

$$Z_{line}' = \left(\frac{2.4\text{ kV}}{14\text{ kV}}\right)^2 (40\ \Omega + j150\ \Omega) = 1.18 + j4.41\ \Omega$$

The secondary current $\mathbf{I}_S$ is given by

$$I_S = \frac{90\text{ kW}}{(2300\text{ V})(0.85)} = 46.03\text{ A}$$

$$\mathbf{I}_S = 46.03\angle -31.8^\circ\text{ A}$$

(a) The voltage at the power source of this system (referred to the secondary side) is

$$\mathbf{V}_{source}' = \mathbf{V}_S + \mathbf{I}_S Z_{line}' + \mathbf{I}_S Z_{EQ}$$

$$\mathbf{V}_{source}' = 2300\angle 0^\circ\text{ V} + (46.03\angle -31.8^\circ\text{ A})(1.18 + j4.11\ \Omega) + (46.03\angle -31.8^\circ\text{ A})(0.12 + j0.5\ \Omega)$$

$$\mathbf{V}_{source}' = 2467\angle 3.5^\circ\text{ V}$$

Therefore, the voltage at the power source is

$$\mathbf{V}_{source} = (2467\angle 3.5^\circ\text{ V})\frac{14\text{ kV}}{2.4\text{ kV}} = 14.4\angle 3.5^\circ\text{ kV}$$

(b) To find the voltage regulation of the transformer, we must find the voltage at the primary side of the transformer (referred to the secondary side) under full load conditions:

$$\mathbf{V}_P' = \mathbf{V}_S + \mathbf{I}_S Z_{EQ}$$

$$\mathbf{V}_P' = 2300\angle 0^\circ\text{ V} + (46.03\angle -31.8^\circ\text{ A})(0.12 + j0.5\ \Omega) = 2317\angle 0.41^\circ\text{ V}$$

There is a voltage drop of 17 V under these load conditions. Therefore the voltage regulation of the transformer is

$$\text{VR} = \frac{2317 - 2300}{2300} \times 100\% = 0.74\%$$

(c) The overall efficiency of the power system will be the ratio of the output power to the input power. The output power supplied to the load is $P_{OUT} = 90$ kW. The input power supplied by the source is

$$P_{IN} = V_{source}' I_S \cos\theta = (2467\text{ V})(46.03\text{ A})\cos 35.3^\circ = 92.68\text{ kW}$$

Therefore, the efficiency of the power system is

$$\eta = \frac{P_{OUT}}{P_{IN}} \times 100\% = \frac{90\text{ kW}}{92.68\text{ kW}} \times 100\% = 97.1\%$$

2-6. A 15-kVA 8000/230-V distribution transformer has an impedance referred to the primary of $80 + j300\ \Omega$. The components of the excitation branch referred to the primary side are $R_C = 350\ \text{k}\Omega$ and $X_M = 70\ \text{k}\Omega$.

(a) If the primary voltage is 7967 V and the load impedance is $Z_L = 3.2 + j1.5\ \Omega$, what is the secondary voltage of the transformer? What is the voltage regulation of the transformer?

(b) If the load is disconnected and a capacitor of $-j3.5\ \Omega$ is connected in its place, what is the secondary voltage of the transformer? What is its voltage regulation under these conditions?

SOLUTION

(a) The easiest way to solve this problem is to refer all components to the *primary* side of the transformer. The turns ratio is $a = 8000/230 = 34.78$. Thus the load impedance referred to the primary side is

$$Z_L' = (34.78)^2 (3.2 + j1.5\ \Omega) = 3871 + j1815\ \Omega$$

The referred secondary current is

$$\mathbf{I}_S' = \frac{7967\angle 0^\circ\ \text{V}}{(80 + j300\ \Omega) + (3871 + j1815\ \Omega)} = \frac{7967\angle 0^\circ\ \text{V}}{4481\angle 28.2^\circ\ \Omega} = 1.78\angle -28.2^\circ\ \text{A}$$

and the referred secondary voltage is

$$\mathbf{V}_S' = \mathbf{I}_S' Z_L' = (1.78\angle -28.2^\circ\ \text{A})(3871 + j1815\ \Omega) = 7610\angle -3.1^\circ\ \text{V}$$

The actual secondary voltage is thus

$$\mathbf{V}_S = \frac{\mathbf{V}_S'}{a} = \frac{7610\angle -3.1^\circ\ \text{V}}{34.78} = 218.8\angle -3.1^\circ\ \text{V}$$

The voltage regulation is

$$\text{VR} = \frac{7967\text{-}7610}{7610} \times 100\% = 4.7\%$$

(b) The easiest way to solve this problem is to refer all components to the *primary* side of the transformer. The turns ratio is again $a = 34.78$. Thus the load impedance referred to the primary side is

$$Z_L' = (34.78)^2 (-j3.5\ \Omega) = -j4234\ \Omega$$

The referred secondary current is

$$\mathbf{I}_S' = \frac{7967\angle 0^\circ\ \text{V}}{(80 + j300\ \Omega) + (-j4234\ \Omega)} = \frac{7967\angle 0^\circ\ \text{V}}{3935\angle -88.8^\circ\ \Omega} = 2.025\angle 88.8^\circ\ \text{A}$$

and the referred secondary voltage is

$$\mathbf{V}_S' = \mathbf{I}_S' Z_L' = (2.25\angle 88.8^\circ\ \text{A})(-j4234\ \Omega) = 8573\angle -1.2^\circ\ \text{V}$$

The actual secondary voltage is thus

$$\mathbf{V}_S = \frac{\mathbf{V}_S'}{a} = \frac{8573\angle -1.2^\circ\ \text{V}}{34.78} = 246.5\angle -1.2^\circ\ \text{V}$$

The voltage regulation is

$$\text{VR} = \frac{7967 - 8573}{8573} \times 100\% = -7.07\%$$

2-7. A 5000-kVA 230/13.8-kV single-phase power transformer has a per-unit resistance of 1 percent and a per-unit reactance of 5 percent (data taken from the transformer's nameplate). The open-circuit test performed on the low-voltage side of the transformer yielded the following data:

$$V_{OC} = 13.8 \text{ kV} \qquad I_{OC} = 15.1 \text{ A} \qquad P_{OC} = 44.9 \text{ kW}$$

(a) Find the equivalent circuit referred to the low-voltage side of this transformer.

(b) If the voltage on the secondary side is 13.8 kV and the power supplied is 4000 kW at 0.8 PF lagging, find the voltage regulation of the transformer. Find its efficiency.

SOLUTION

(a) The open-circuit test was performed on the low-voltage side of the transformer, so it can be used to directly find the components of the excitation branch relative to the low-voltage side.

$$|Y_{EX}| = |G_C - jB_M| = \frac{15.1 \text{ A}}{13.8 \text{ kV}} = 0.0010942$$

$$\theta = \cos^{-1}\frac{P_{OC}}{V_{OC} I_{OC}} = \cos^{-1}\frac{44.9 \text{ kW}}{(13.8 \text{ kV})(15.1 \text{ A})} = 77.56°$$

$$Y_{EX} = G_C - jB_M = 0.0010942\angle -77.56° \text{ S} = 0.0002358 - j0.0010685 \text{ S}$$

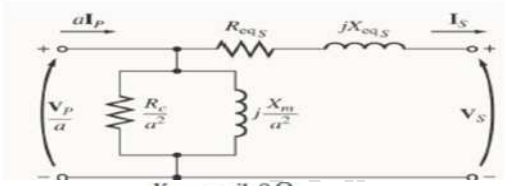
$$R_C = \frac{1}{G_C} = 4240 \ \Omega$$

$$X_M = \frac{1}{B_M} = 936 \ \Omega$$

The base impedance of this transformer referred to the secondary side is

$$Z_{base} = \frac{V_{base}^2}{S_{base}} = \frac{(13.8 \text{ kV})^2}{5000 \text{ kVA}} = 38.09 \ \Omega$$

so $R_{EQ} = (0.01)(38.09 \ \Omega) = 0.38 \ \Omega$ and $X_{EQ} = (0.05)(38.09 \ \Omega) = 1.9 \ \Omega$. The resulting equivalent circuit is shown below:



$$R_{EQ,S} = 0.38 \ \Omega \qquad X_{EQ,S} = j1.9 \ \Omega$$

$$R_{C,S} = 4240 \ \Omega \qquad X_{M,S} = 936 \ \Omega$$

(b) If the load on the secondary side of the transformer is 4000 kW at 0.8 PF lagging and the secondary voltage is 13.8 kV, the secondary current is

$$I_S = \frac{P_{LOAD}}{V_S \text{ PF}} = \frac{4000 \text{ kW}}{(13.8 \text{ kV})(0.8)} = 362.3 \text{ A}$$

$$\mathbf{I}_S = 362.3\angle -36.87° \text{ A}$$

The voltage on the primary side of the transformer (referred to the secondary side) is

$$\mathbf{V}_P' = \mathbf{V}_S + \mathbf{I}_S Z_{EQ}$$

$$\mathbf{V}_P' = 13{,}800\angle 0° \text{ V} + (362.3\angle -36.87° \text{ A})(0.38 + j1.9 \ \Omega) = 14{,}330\angle 1.9° \text{ V}$$

There is a voltage drop of 14 V under these load conditions. Therefore the voltage regulation of the transformer is

$$\text{VR} = \frac{14{,}330 - 13{,}800}{13{,}800} \times 100\% = 3.84\%$$

The transformer copper losses and core losses are

$$P_{CU} = I_S^2 \ R_{EQ,S} = (362.3 \text{ A})^2 (0.38 \ \Omega) = 49.9 \text{ kW}$$

$$P_{core} = \frac{(V_P')^2}{R_C} = \frac{(14{,}330 \text{ V})^2}{4240 \ \Omega} = 48.4 \text{ kW}$$

Therefore the efficiency of this transformer at these conditions is

$$\eta = \frac{P_{OUT}}{P_{OUT} + P_{CU} + P_{core}} \times 100\% = \frac{4000 \text{ kW}}{4000 \text{ kW} + 49.9 \text{ kW} + 48.4 \text{ kW}} = 97.6\%$$

2-11. A 100,000-kVA 230/115-kV Δ-Δ three-phase power transformer has a per-unit resistance of 0.02 pu and a per-unit reactance of 0.055 pu. The excitation branch elements are $R_C = 110$ pu and $X_M = 20$ pu.

(a) If this transformer supplies a load of 80 MVA at 0.85 PF lagging, draw the phasor diagram of one phase of the transformer.

(b) What is the voltage regulation of the transformer bank under these conditions?

(c) Sketch the equivalent circuit referred to the low-voltage side of one phase of this transformer. Calculate all of the transformer impedances referred to the low-voltage side.

SOLUTION

(a) The transformer supplies a load of 80 MVA at 0.85 PF lagging. Therefore, the secondary line current of the transformer is

$$I_{LS} = \frac{S}{\sqrt{3}V_{LS}} = \frac{80{,}000{,}000\text{ VA}}{\sqrt{3}(115{,}000\text{ V})} = 402\text{ A}$$

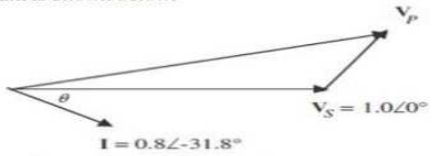
The base value of the secondary line current is

$$I_{LS,base} = \frac{S_{base}}{\sqrt{3}V_{LS,base}} = \frac{100{,}000{,}000\text{ VA}}{\sqrt{3}(115{,}000\text{ V})} = 502\text{ A}$$

so the per-unit secondary current is

$$\mathbf{I}_{LS,pu} = \frac{I_{LS}}{I_{LS,pu}} = \frac{402\text{ A}}{502\text{ A}}\angle\cos^{-1}(0.85) = 0.8\angle -31.8°$$

The per-unit phasor diagram is shown below:



(b) The per-unit primary voltage of this transformer is

$$\mathbf{V}_P = \mathbf{V}_S + \mathbf{I}\,Z_{EQ} = 1.0\angle 0° + (0.8\angle -31.8°)(0.02 + j0.055) = 1.037\angle 1.6°$$

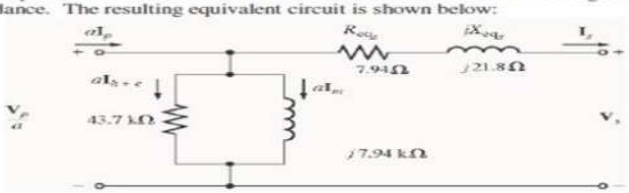
and the voltage regulation is

$$\text{VR} = \frac{1.037 - 1.0}{1.0} \times 100\% = 3.7\%$$

(c) The base impedance of the transformer referred to the low-voltage side is:

$$Z_{base} = \frac{3\,V_{\phi,base}^2}{S_{base}} = \frac{3(115\text{ kV})^2}{100\text{ MVA}} = 397\ \Omega$$

Each per-unit impedance is converted to actual ohms referred to the low-voltage side by multiplying it by this base impedance. The resulting equivalent circuit is shown below:



$R_{EQ,S} = (0.02)(397\ \Omega) = 7.94\ \Omega$
$X_{EQ,S} = (0.055)(397\ \Omega) = 21.8\ \Omega$
$R_C = (110)(397\ \Omega) = 43.7\ \text{k}\Omega$
$X_M = (20)(397\ \Omega) = 7.94\ \text{k}\Omega$

Note how easy it was to solve this problem in per-unit, compared with Problem 2-10 above.

2-12. An autotransformer is used to connect a 13.2-kV distribution line to a 13.8-kV distribution line. It must be capable of handling 2000 kVA. There are three phases, connected Y-Y with their neutrals solidly grounded.

(a) What must the N_C / N_{SE} turns ratio be to accomplish this connection?

(b) How much apparent power must the windings of each autotransformer handle?

(c) If one of the autotransformers were reconnected as an ordinary transformer, what would its ratings be?

SOLUTION

(a) The transformer is connected Y-Y, so the primary and secondary phase voltages are the line voltages divided by $\sqrt{3}$. The turns ratio of each autotransformer is given by

$$\frac{V_H}{V_L} = \frac{N_C + N_{SE}}{N_C} = \frac{13.8\ \text{kV}/\sqrt{3}}{13.2\ \text{kV}/\sqrt{3}}$$

$$13.2\,N_C + 13.2\,N_{SE} = 13.8\,N_C$$

$$13.2\,N_{SE} = 0.6\,N_C$$

Therefore, $N_C / N_{SE} = 22$.

(b) The power advantage of this autotransformer is

$$\frac{S_{IO}}{S_W} = \frac{N_C + N_{SE}}{N_C} = \frac{N_C + 22N_C}{N_C} = 23$$

so 1/22 of the power in each transformer goes through the windings. Since 1/3 of the total power is associated with each phase, the windings in each autotransformer must handle

$$S_W = \frac{2000\ \text{kVA}}{(3)(22)} = 30.3\ \text{kVA}$$

(c) The voltages across each phase of the autotransformer are $13.8/\sqrt{3} = 7967$ V and $13.2/\sqrt{3} = 7621$ V. The voltage across the common winding (N_C) is 7621 kV, and the voltage across the series winding (N_{SE}) is 7967 kV – 7621 kV = 346 V. Therefore, a single phase of the autotransformer connected as an ordinary transformer would be rated at 7621/346 V and 30.3 kVA.

انتهى الشابتر ملاحظة بسيطة من ضروري حل اسئلة البروبليم مهمة جداً لفهم المادة لكي تتمكن من المادة اكثر واكثر وبالتوفيق للجميع